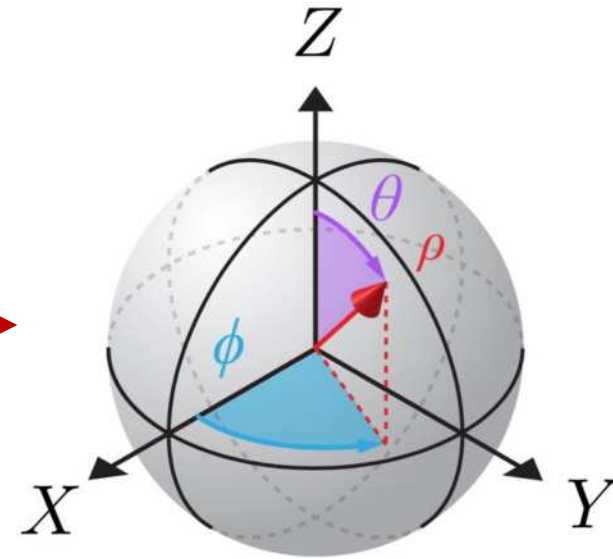
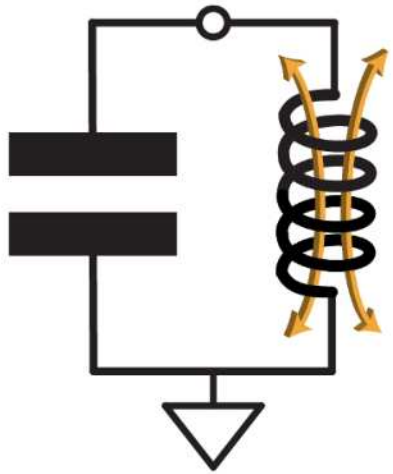


Superconducting Qubits I:

Making Your First Qubit From an Oscillator

Introduction to **Circuit**
Quantum Electrodynamics (cQED)

Zlatko K. Minev



IBM Quantum

IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev



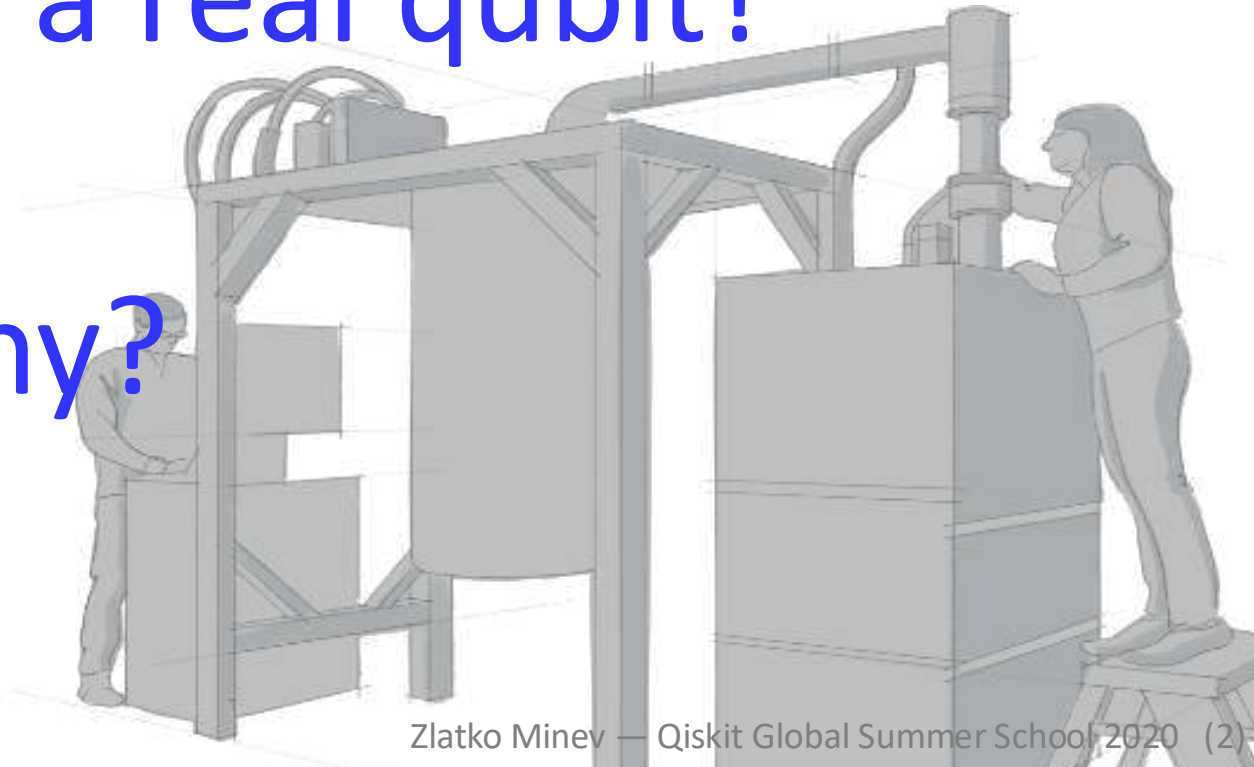
zlatko-minev.com

*Image copyright:
ZKM unless otherwise noted*

What is a real qubit?

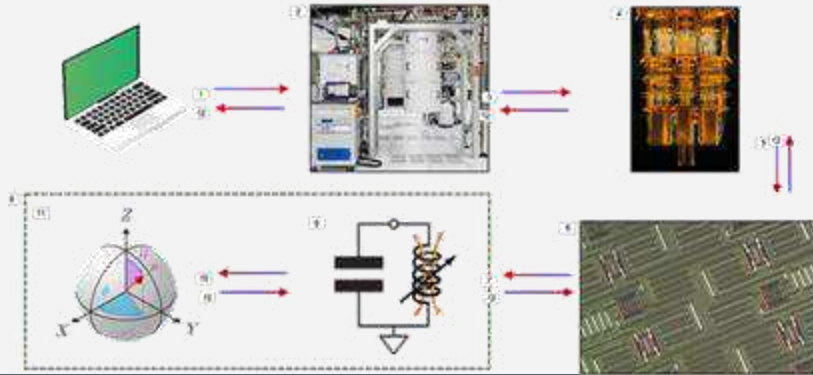
How can you design, control, and measure a real qubit?

Why?

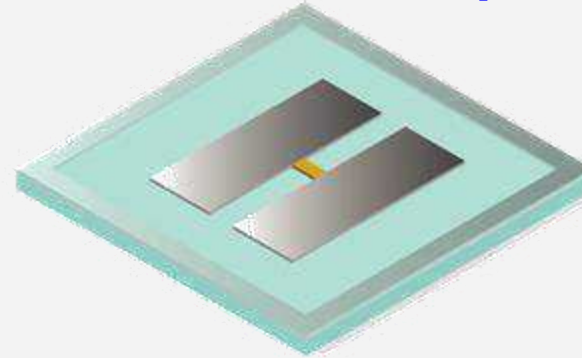


On the road ahead

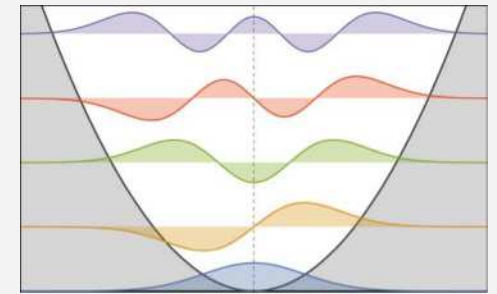
Qubit in the cloud



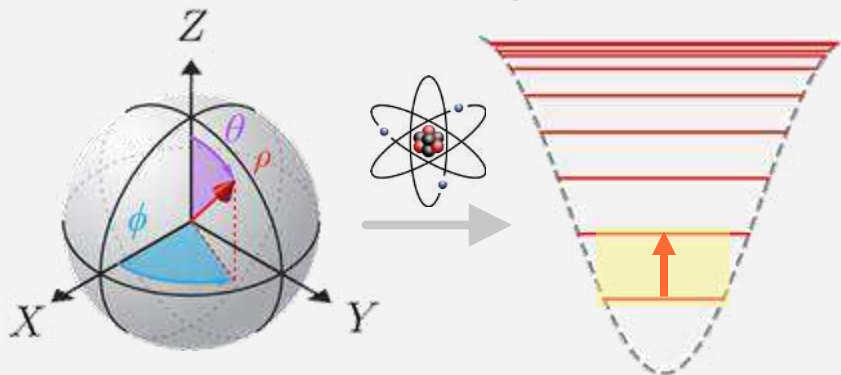
cQED: Transmon qubit



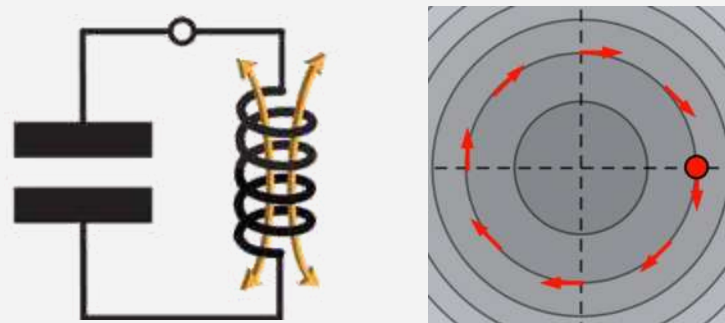
Unveiling the quantum oscillator



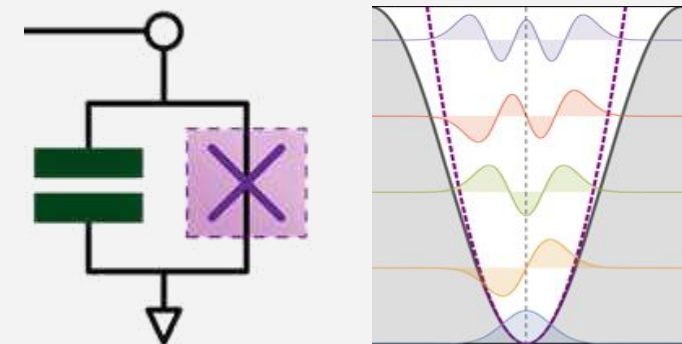
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



This Lecture

Introductory and skill reaffirming

Don't need to know much going in, but we will go far

Examples: simplest, most practical examples

Step by step

Ask questions!

Tightly integrated lab work by Dr. Nick Bronn and Co.!



Advanced
material



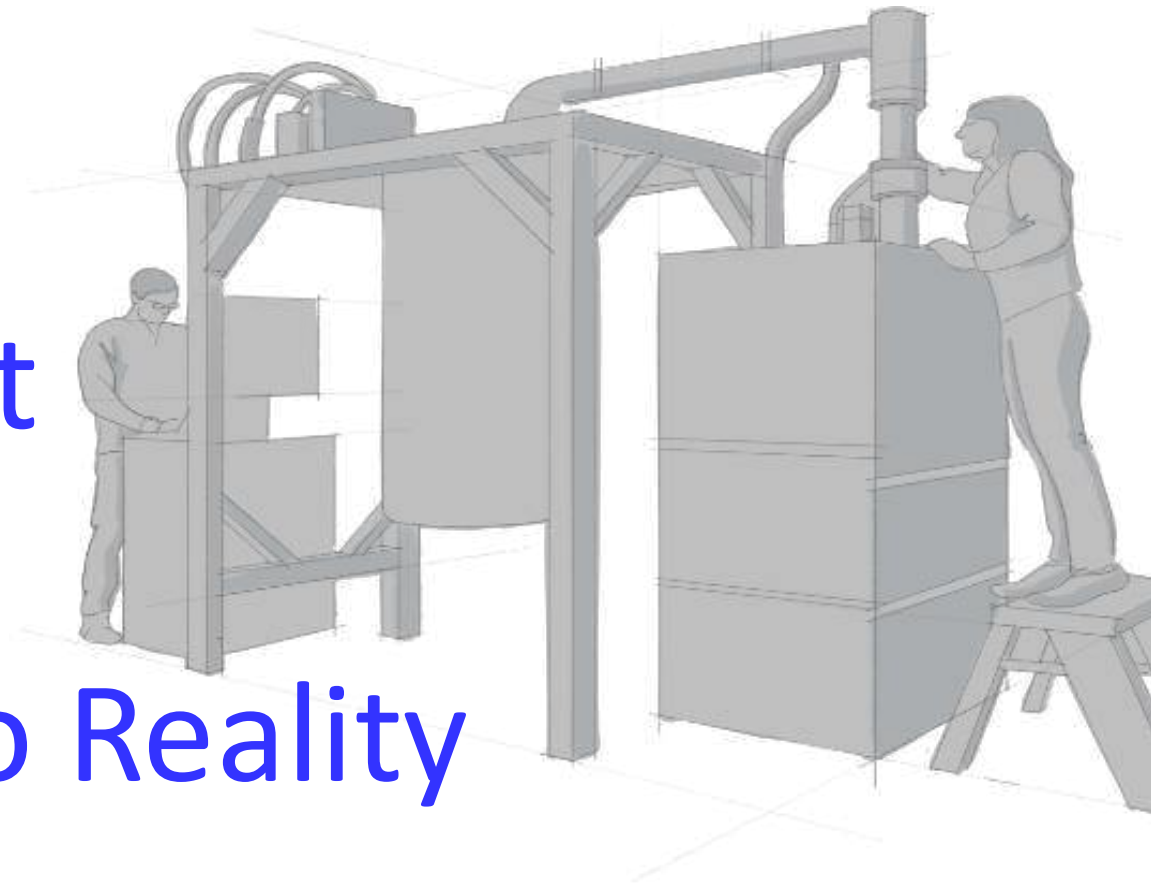
Avoid firehose of information



Thanks to Fred Moxley for gif reference.

Qubit

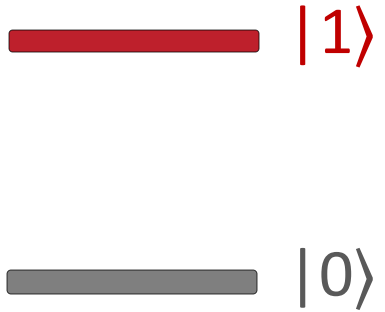
From Idea to Reality



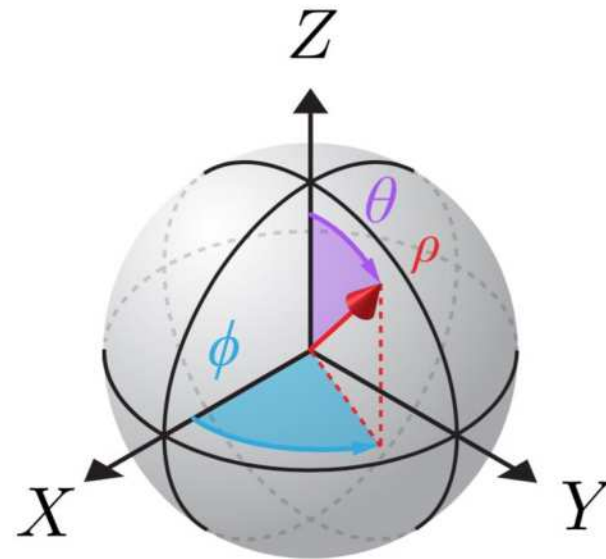
THE BIG PICTURE
before calculations

Qubit: idea

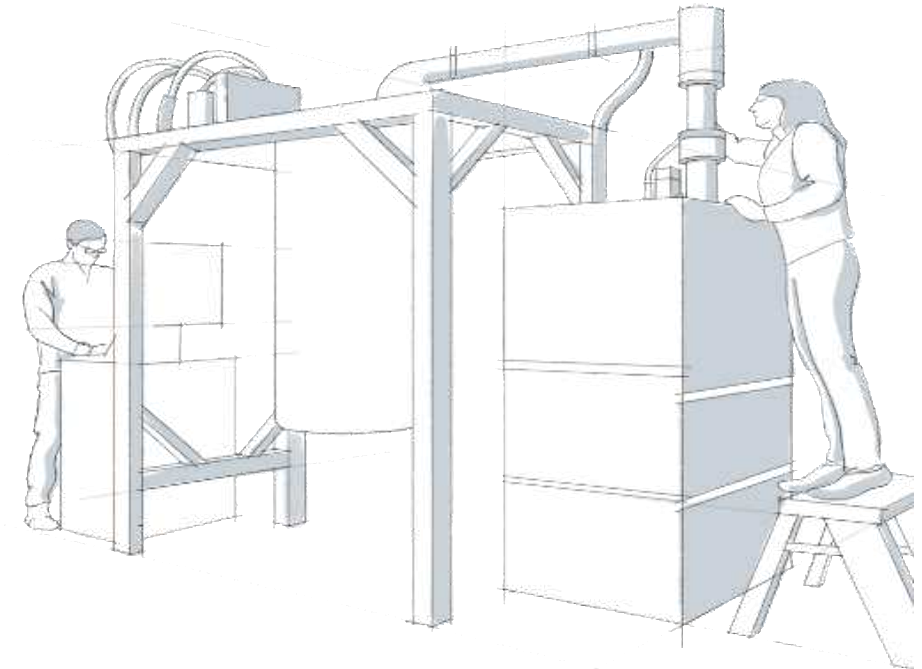
Energy levels



Hilbert space

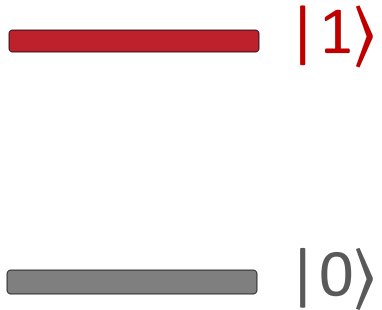


Quantum cloud?

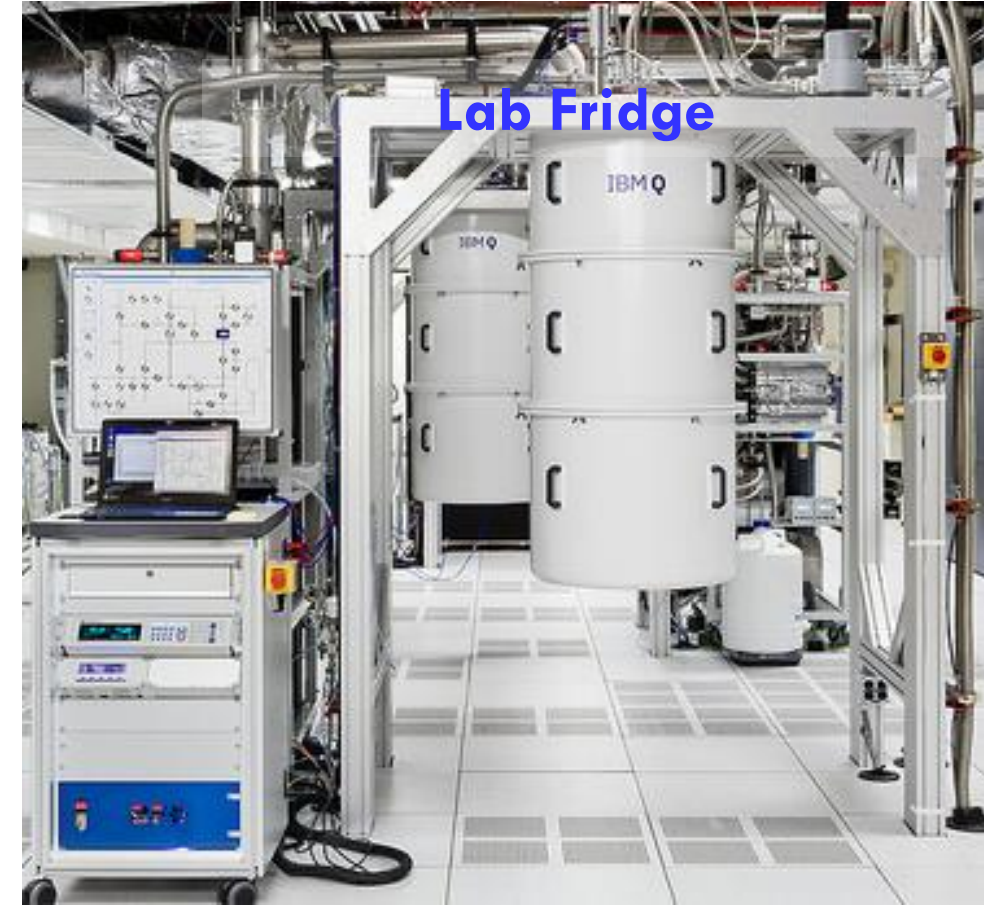
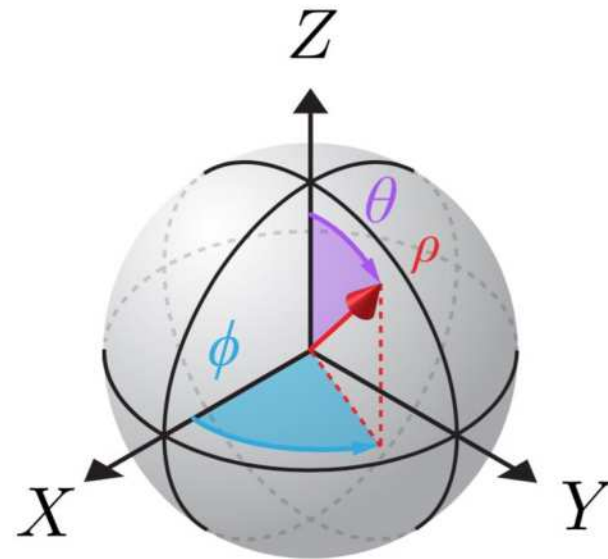


Qubit: idea and reality

Energy levels



Hilbert space



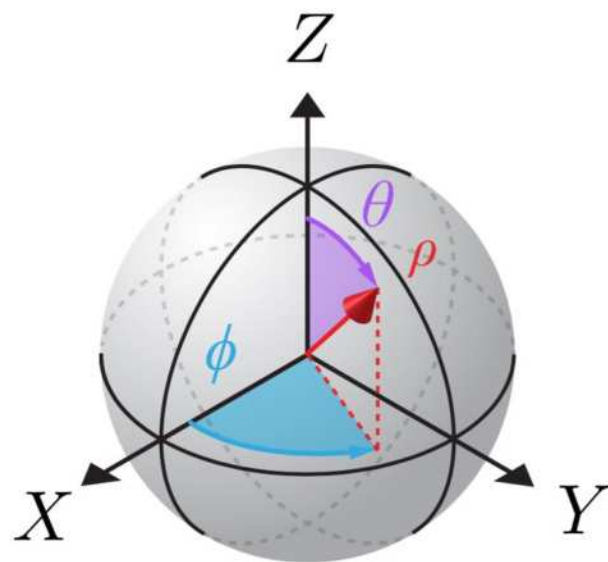
Qubit: idea and reality

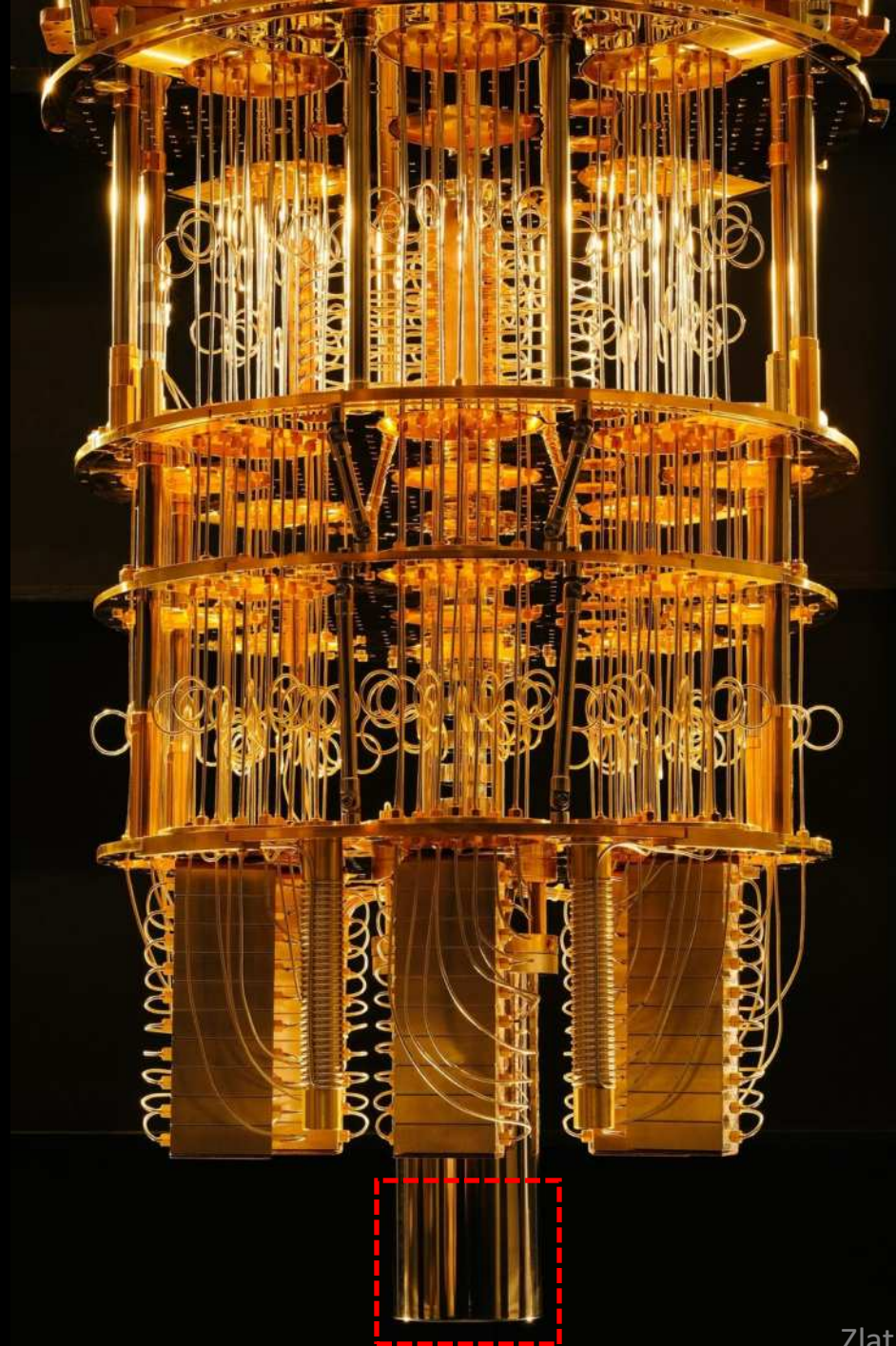
Energy levels

— $|1\rangle$

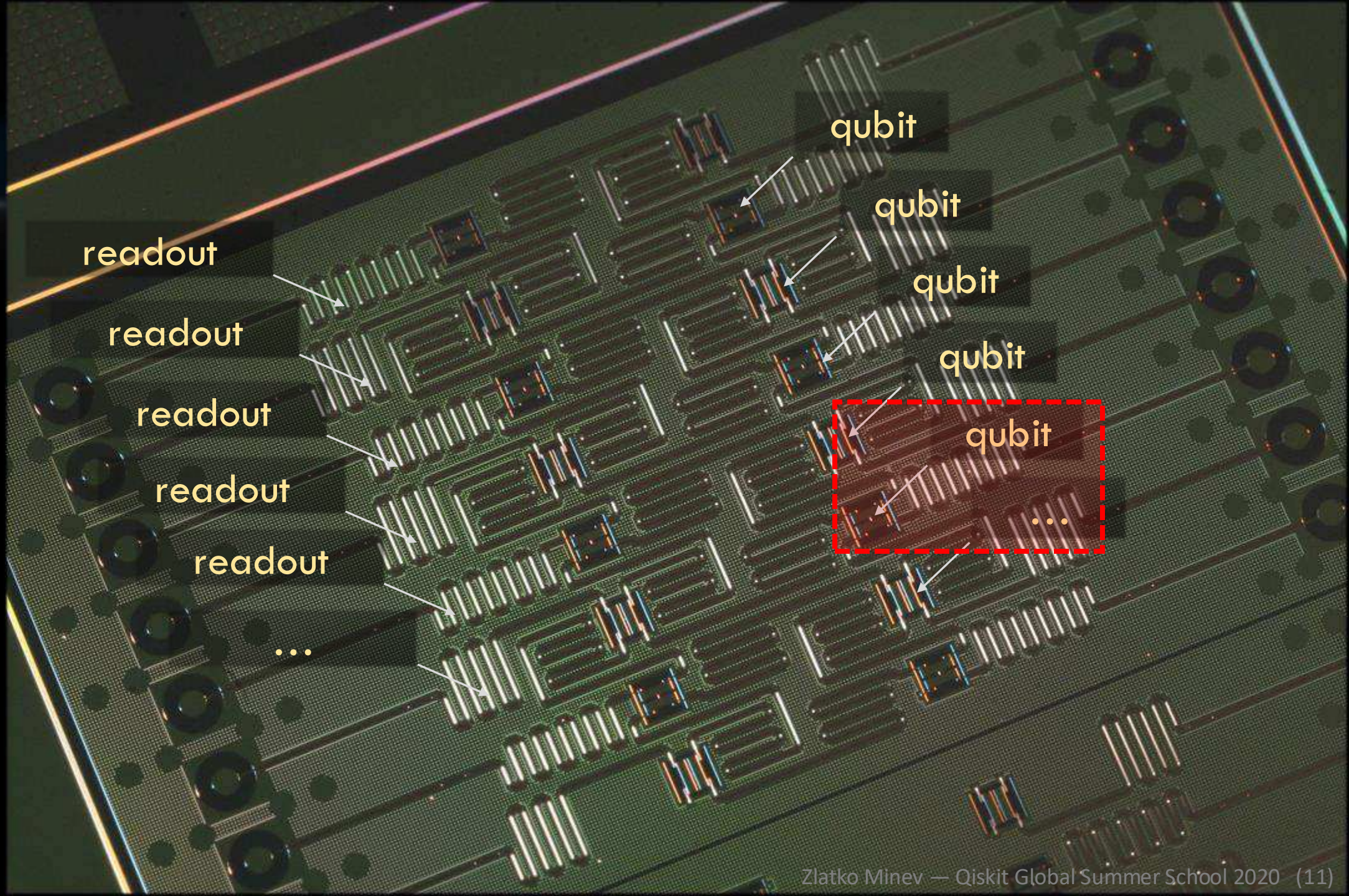
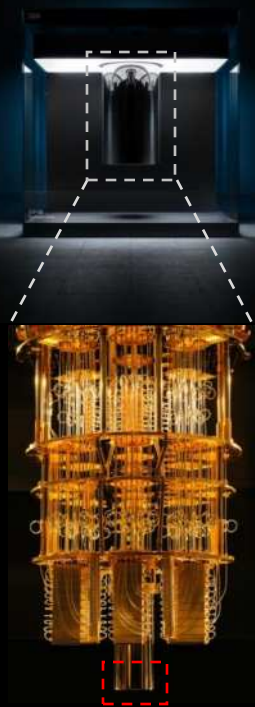
— $|0\rangle$

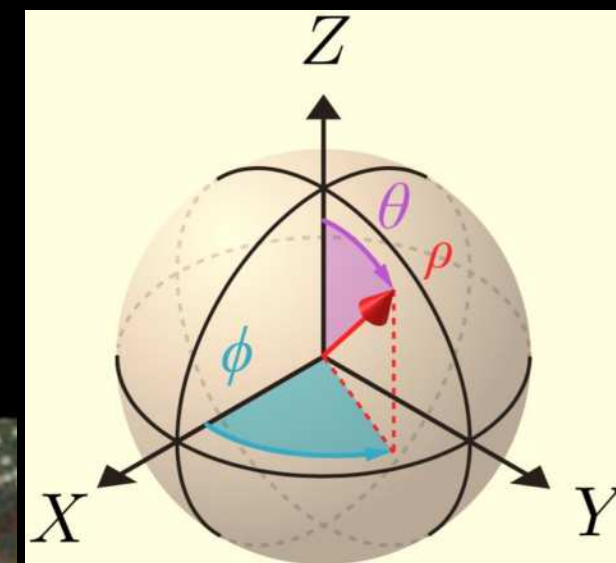
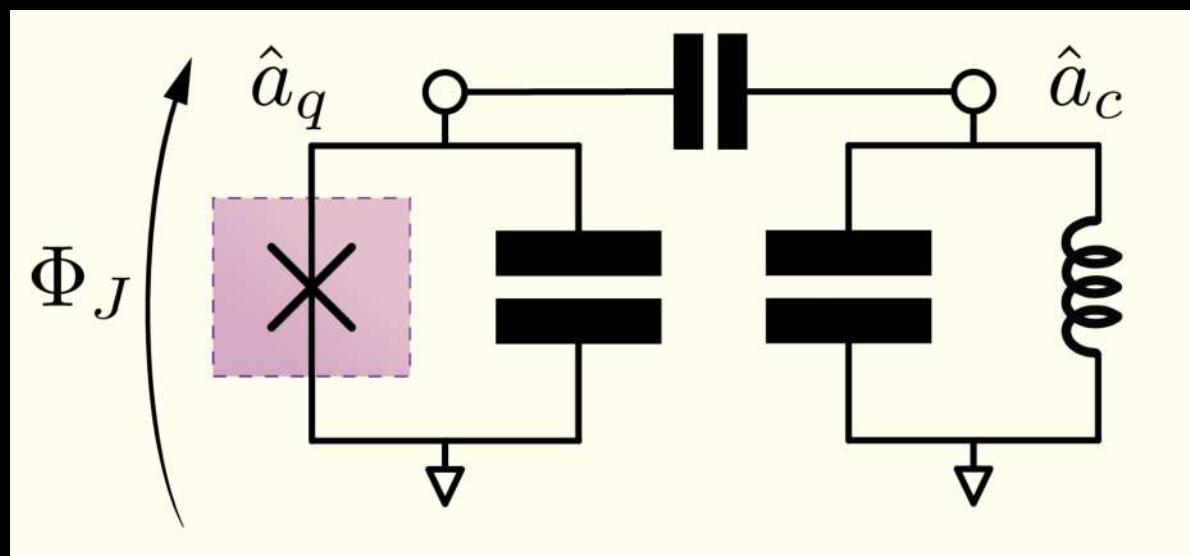
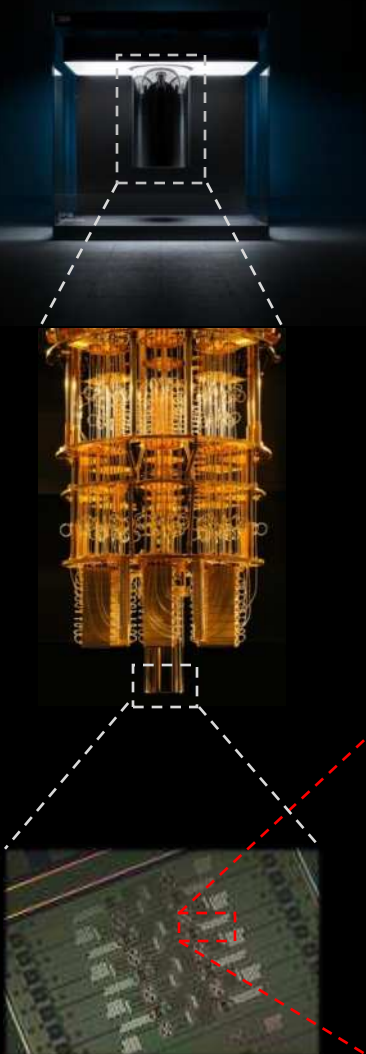
Hilbert space



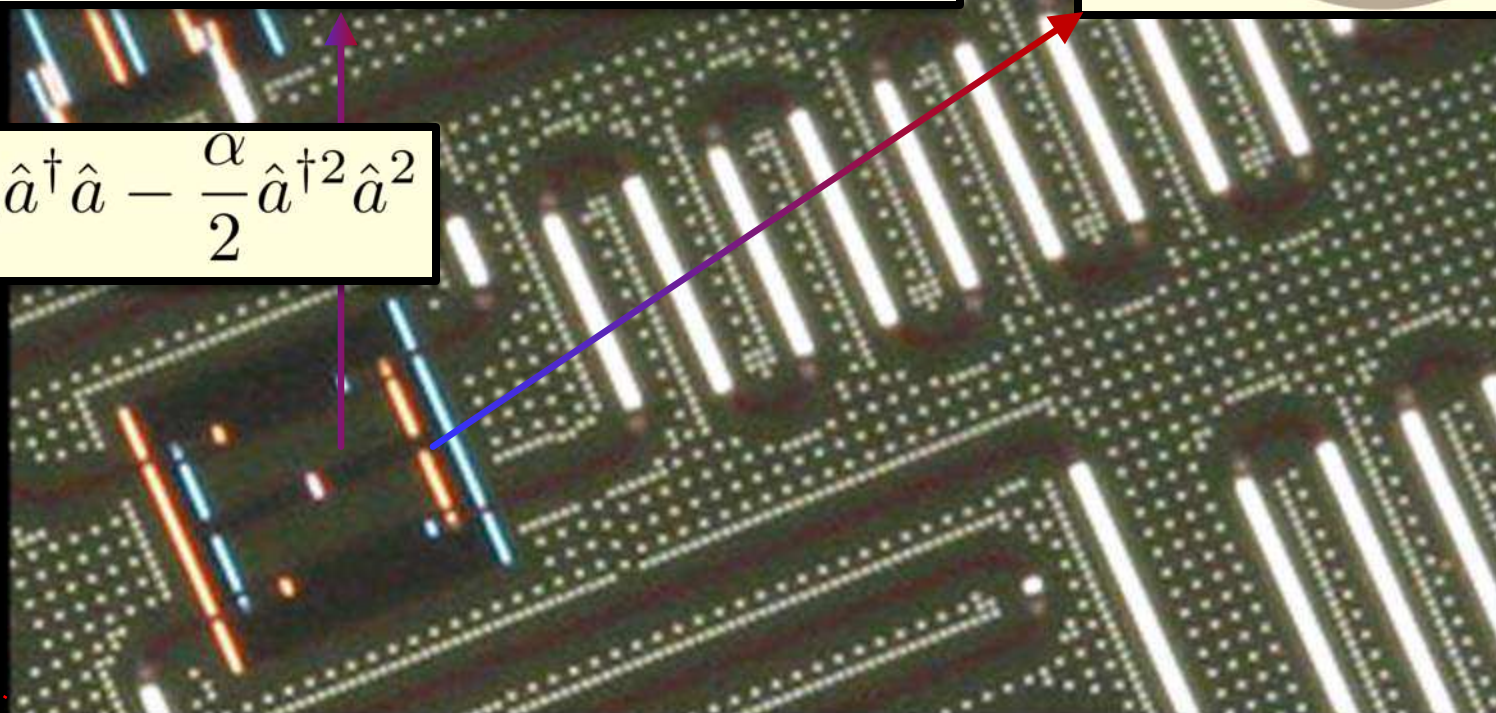


Operation at
15 mK (-273.13 °C)

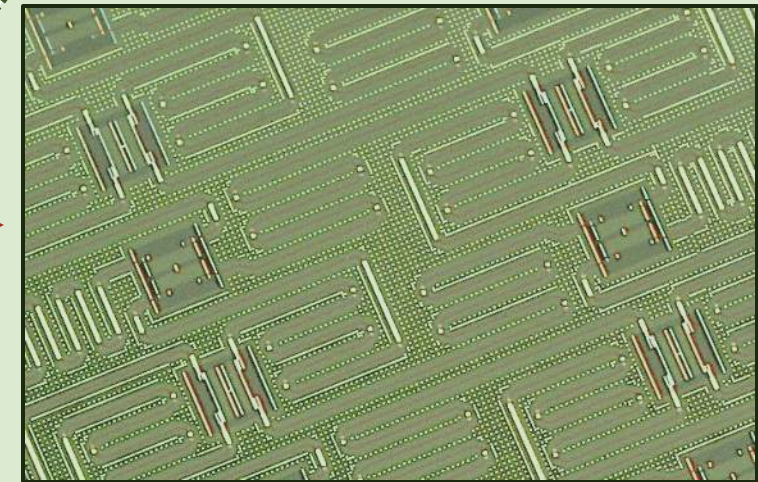
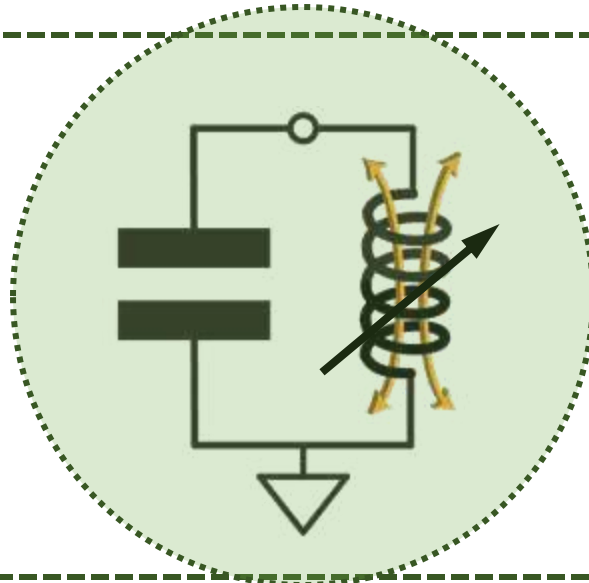
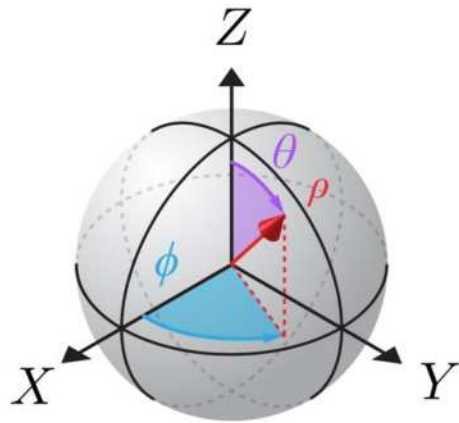
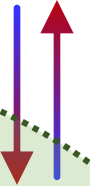
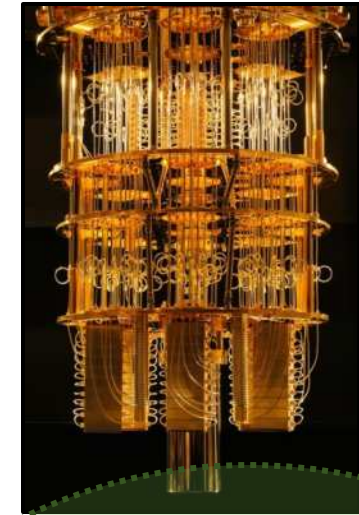


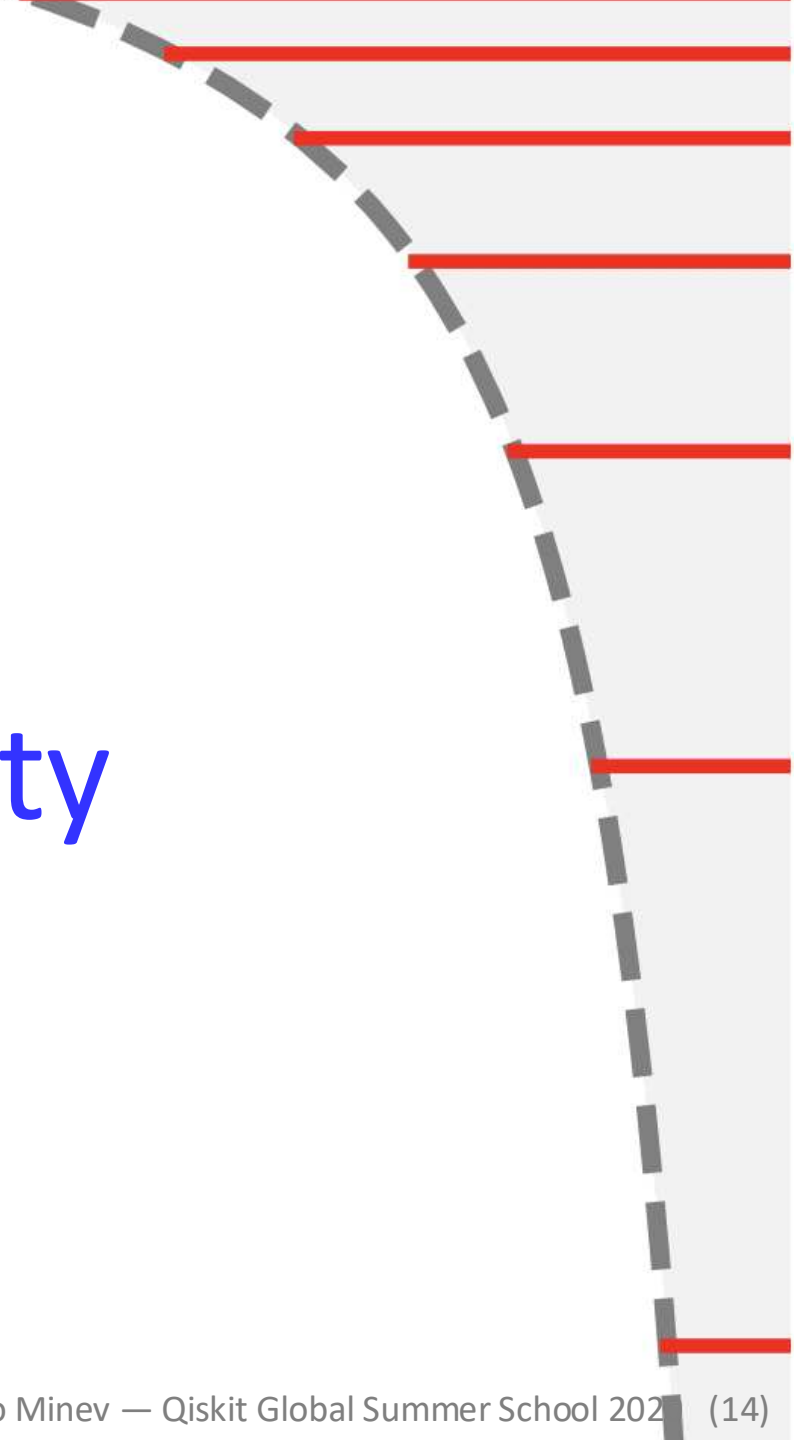


$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$



cQED qubit in the cloud: Summary of flow





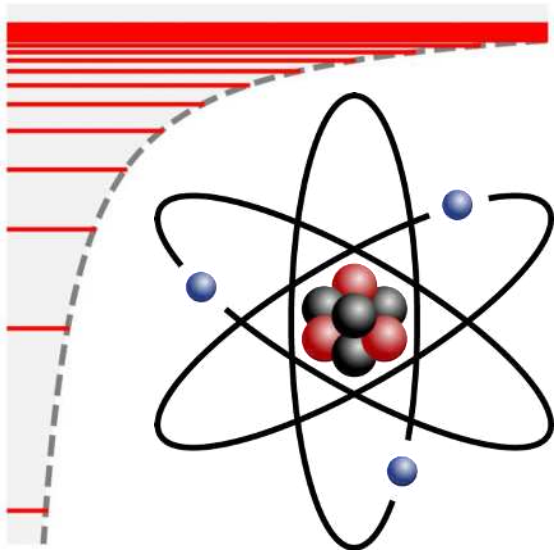
Qubit

From Idea to Reality

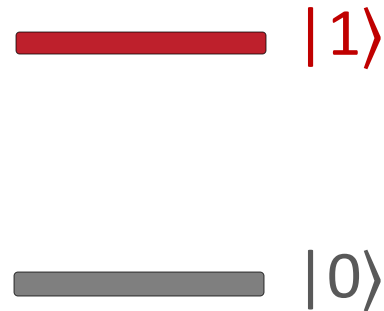
Concepts

From qubit representation to reality

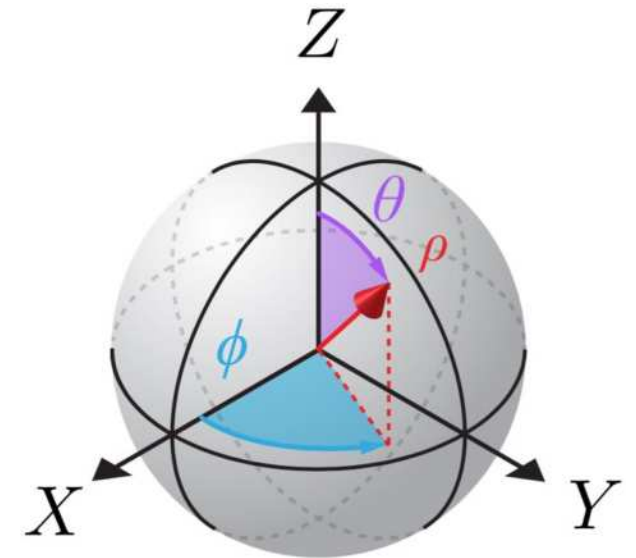
Realization



Energy levels



Hilbert space*



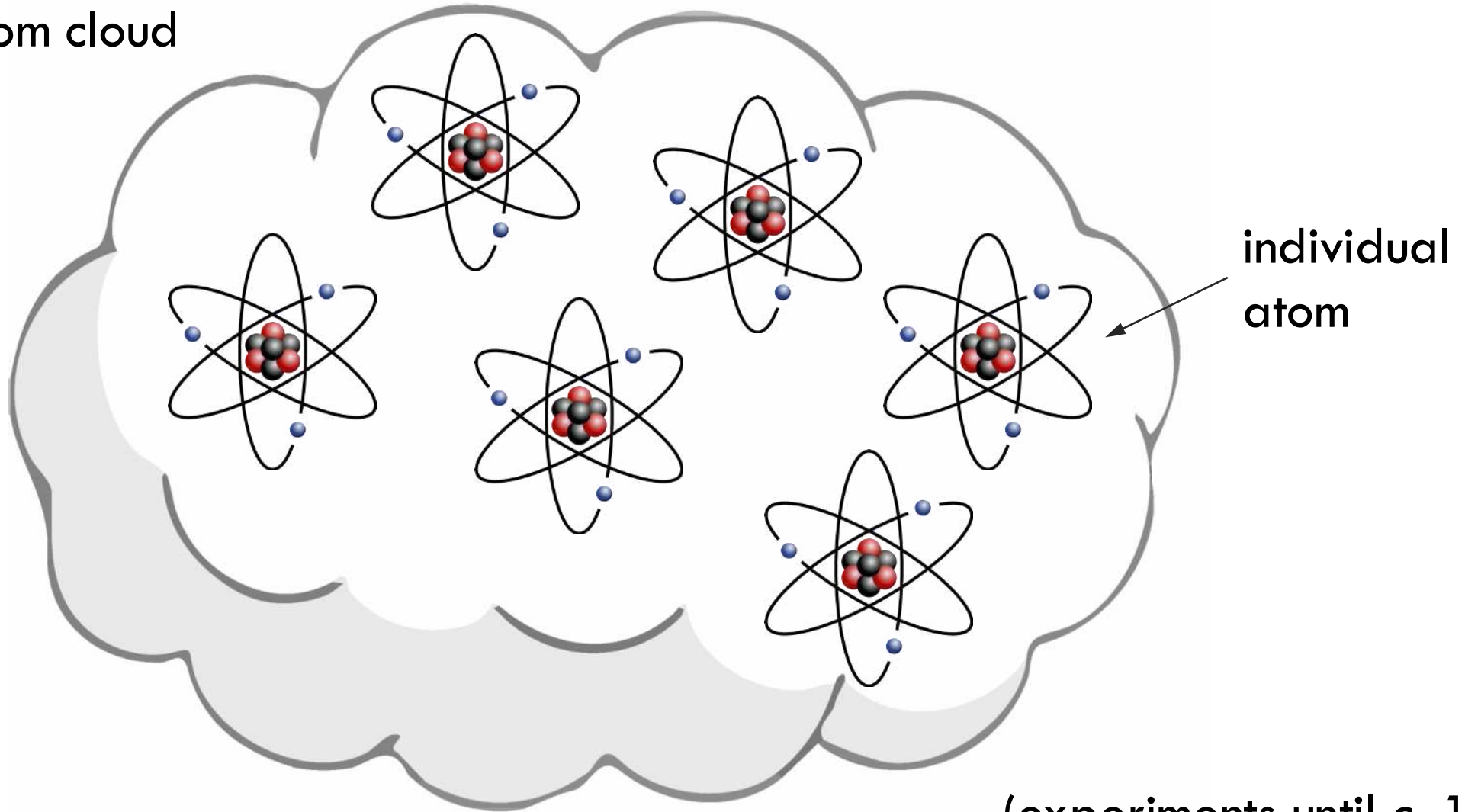
* Bloch sphere is a mere geometrical representation of $SO(3)$, but the density matrix ρ is in $SU(2)$, a double cover of $SO(3)$.

* A density matrix operator lives not in the Hilbert space H but in the Liouville space $H \otimes H$.

Images: Mineev, arXiv:1902.10355; atom art: Indoleces.

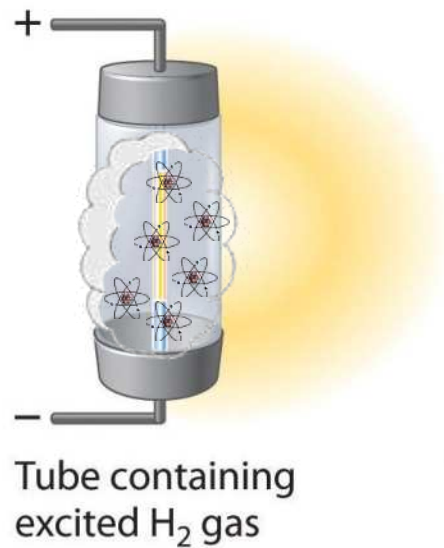
Origins of quantum

e.g., atom cloud

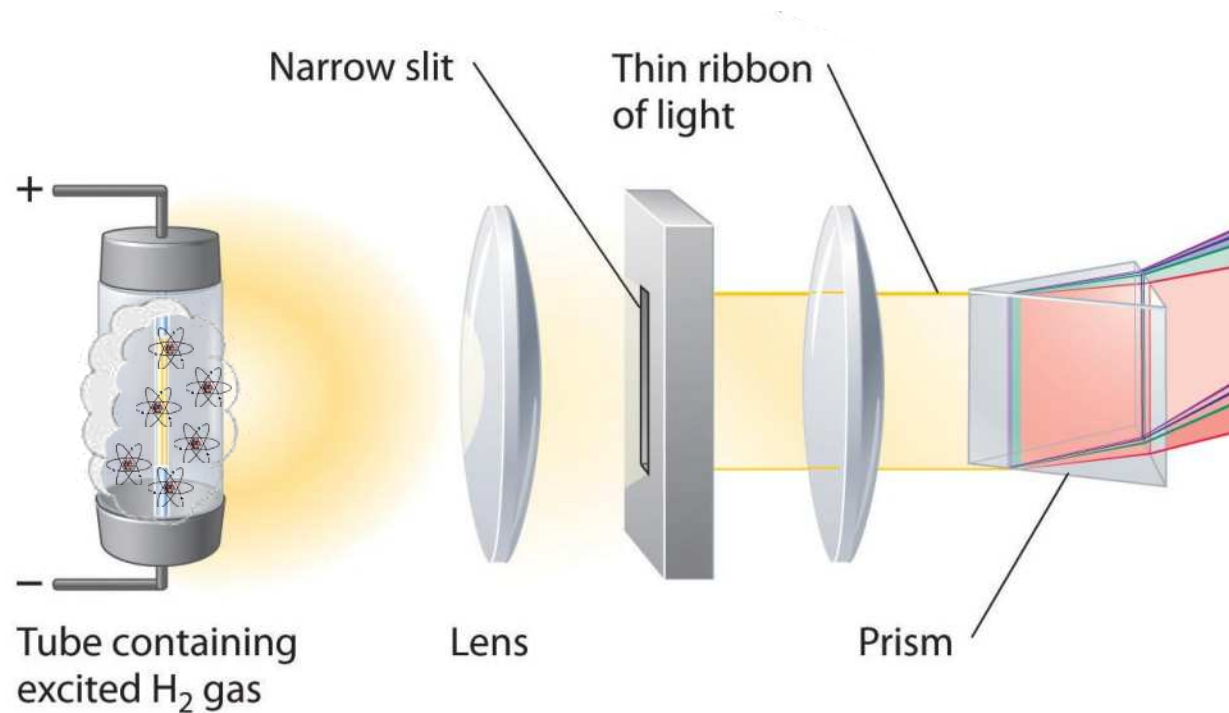


(experiments until c. 1980)

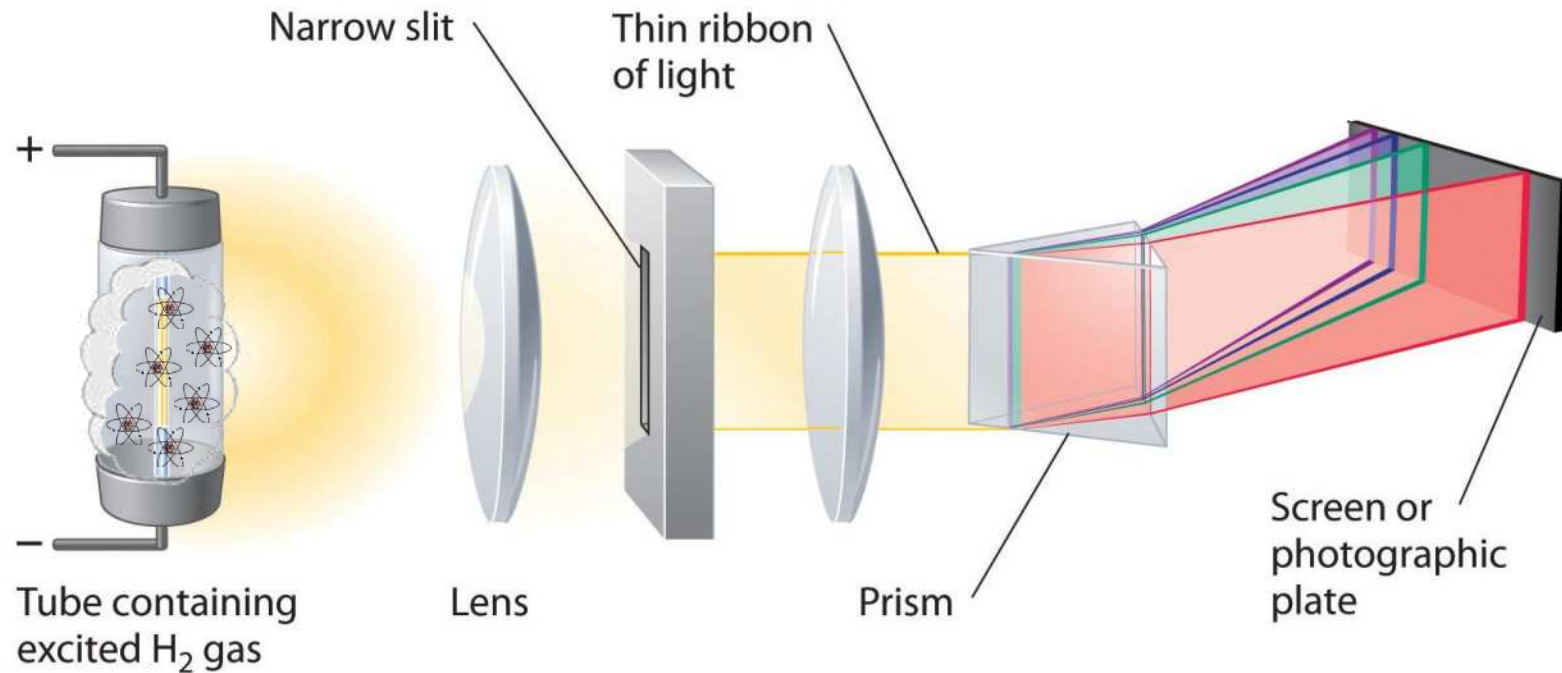
Atomic emission of light



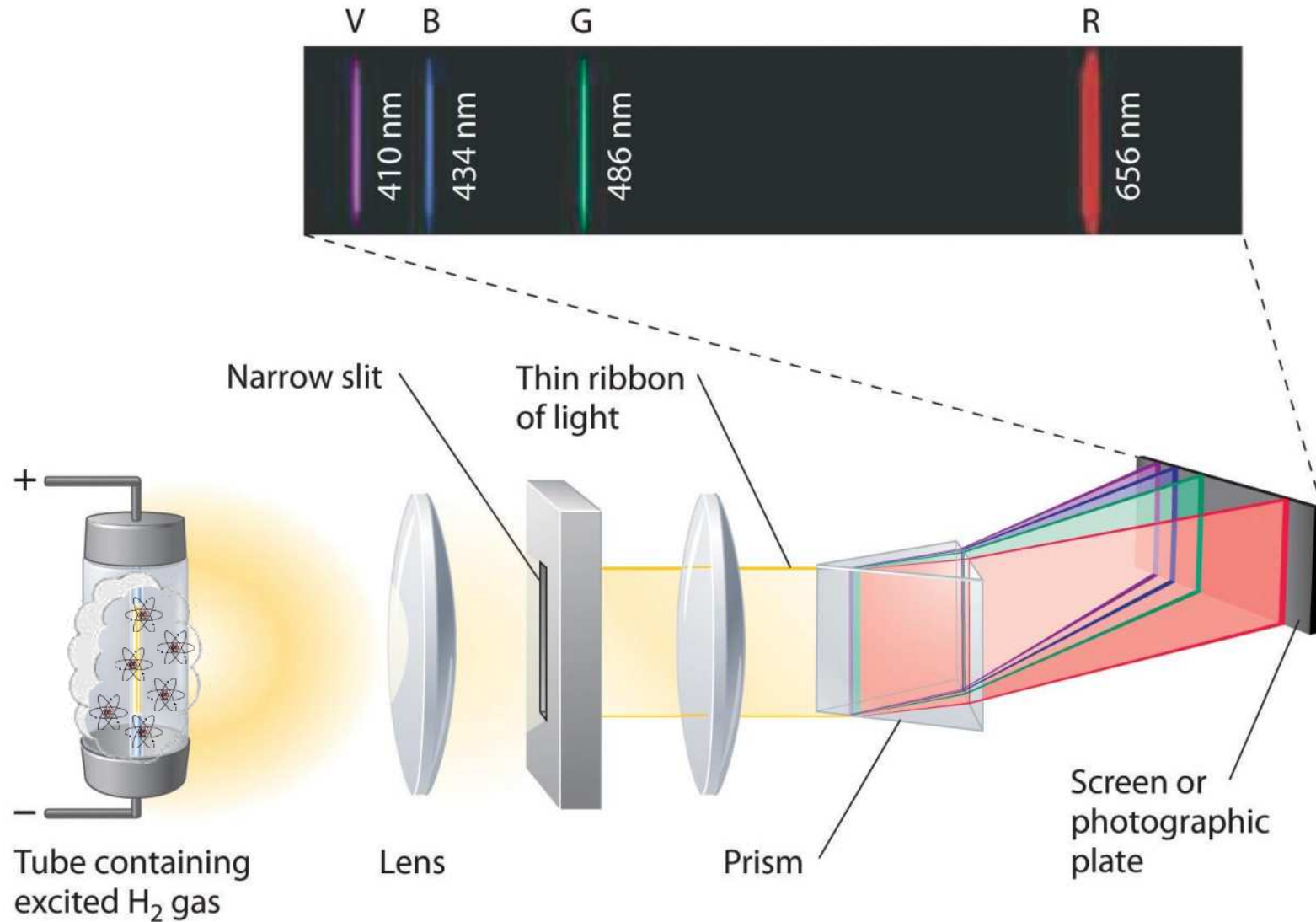
Atomic emission of light



Atomic emission of light



Atomic emission of light



Quantized levels

Atomic emission spectra

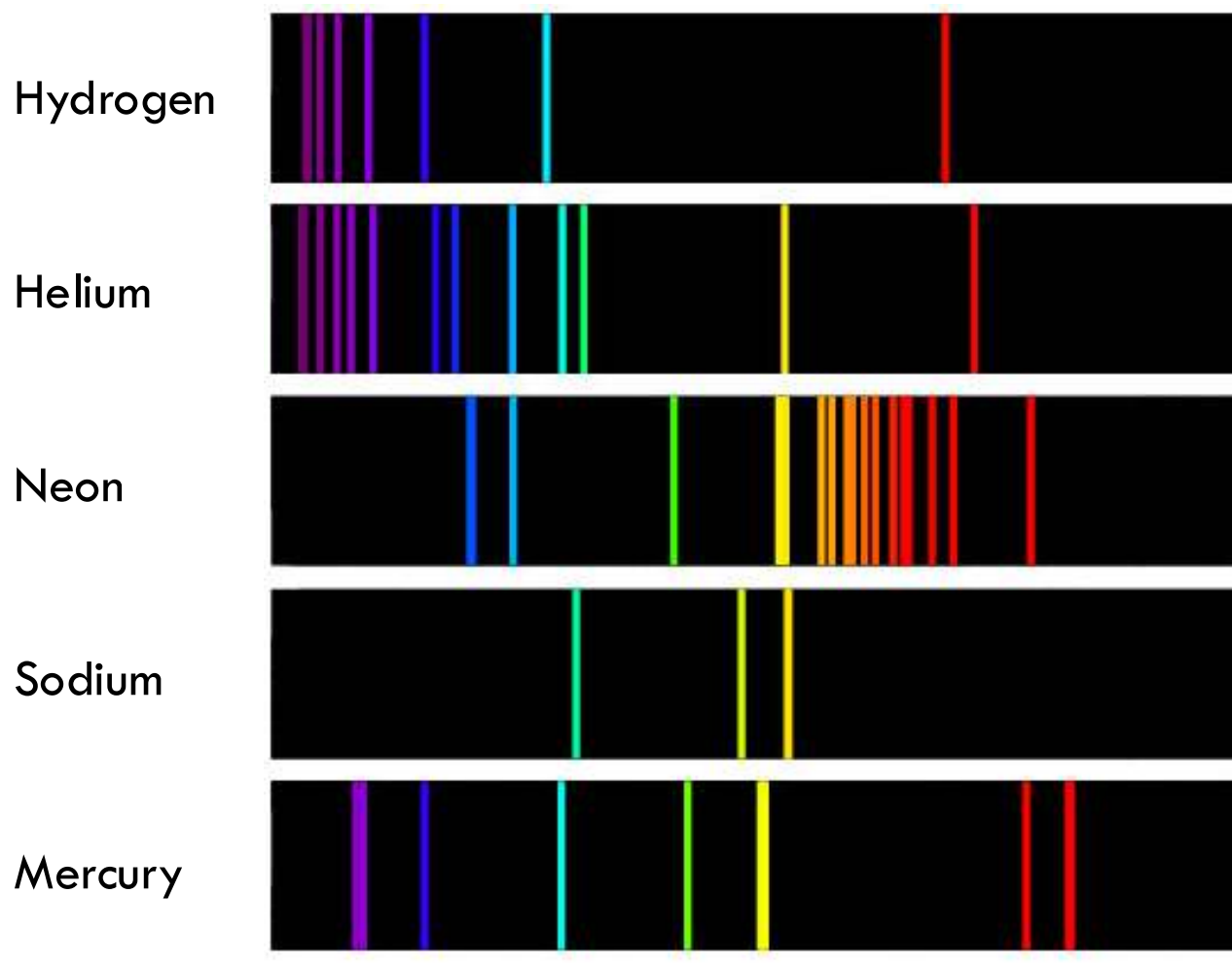
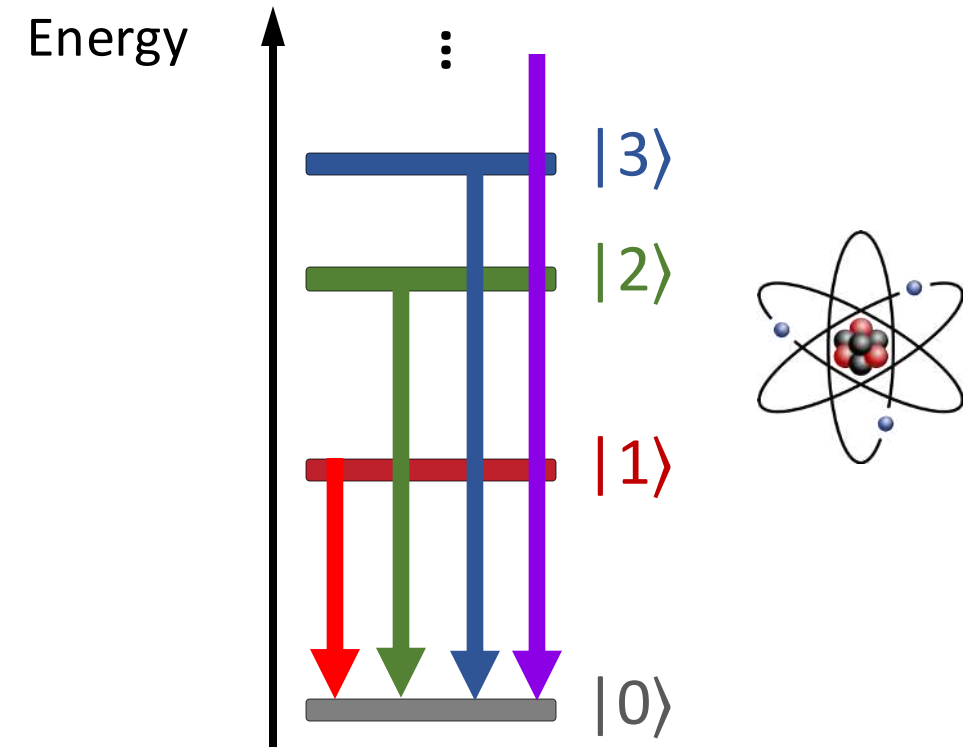


Image credit: NMSU, N. Vogt

Atoms are quantum: discrete intrinsic energy levels*



Quantization of energy

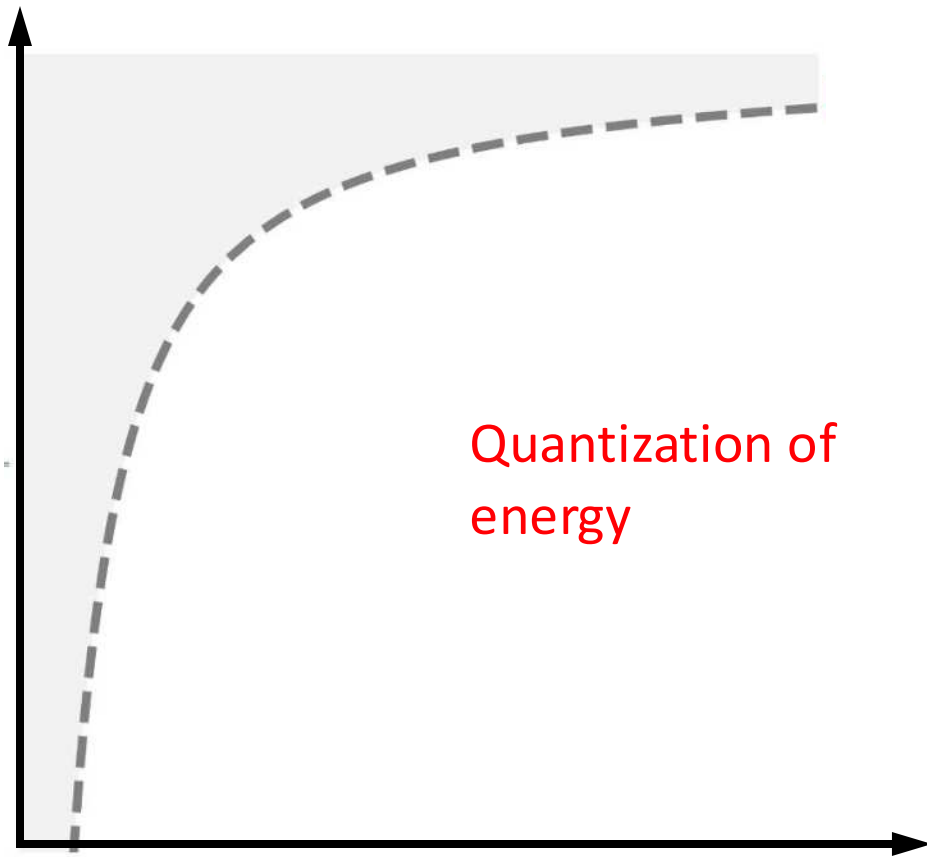
* The notion of an energy level was proposed by Bohr in 1913.
Zlatko Minev — Qiskit Global Summer School 2020 (21)

The light of atoms



Atomic energy levels and transitions

Electron potential-energy landscape



Radial distance from nucleus

Qubit from Atom

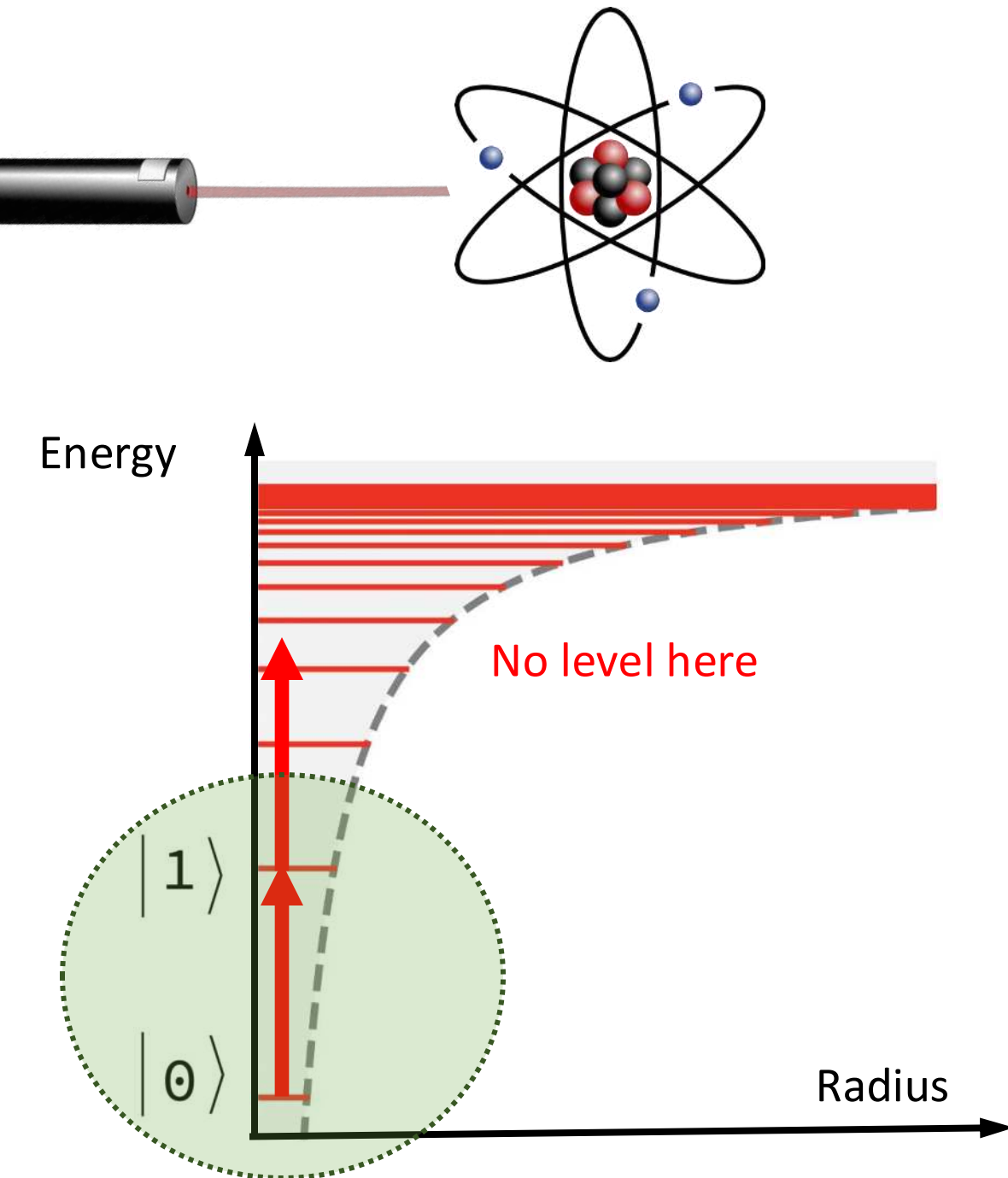
**Anharmonic degree of
freedom and spin**

**Isolated from environment and
thermal bath**

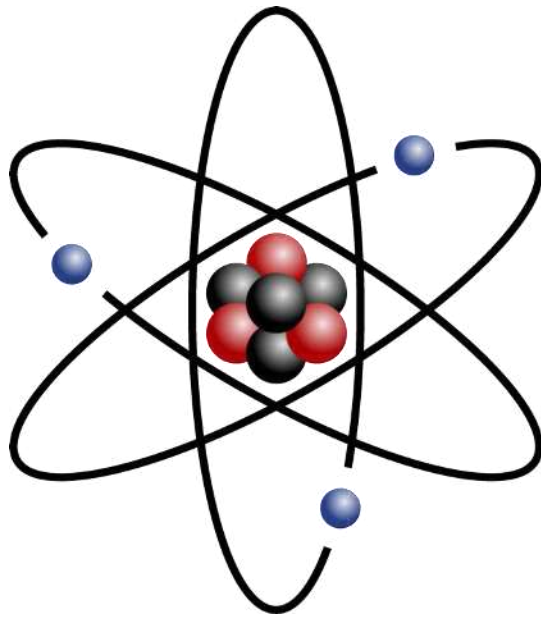
Low-loss

**Level diagram allows for qubit-
specific control and readout**

**There are always more than
two levels**

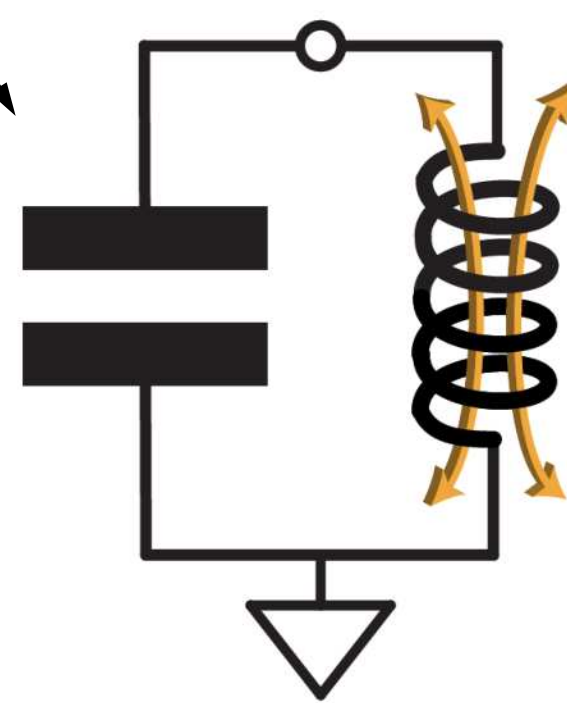


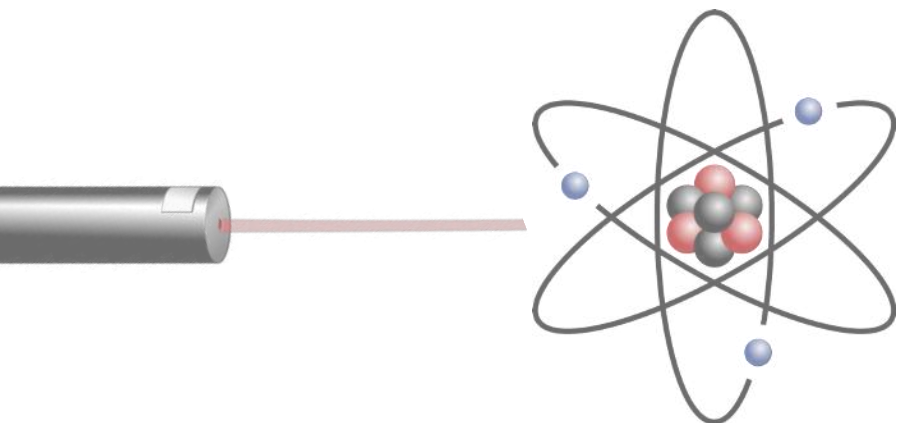
Artificial atoms



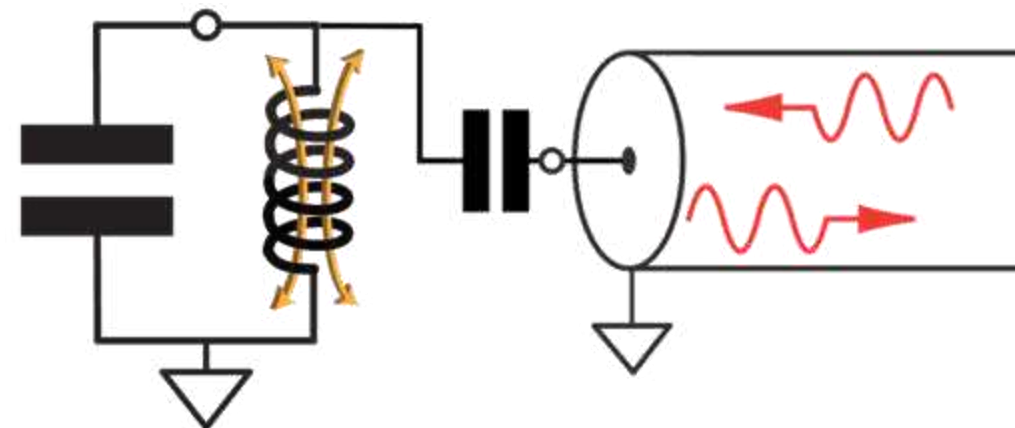
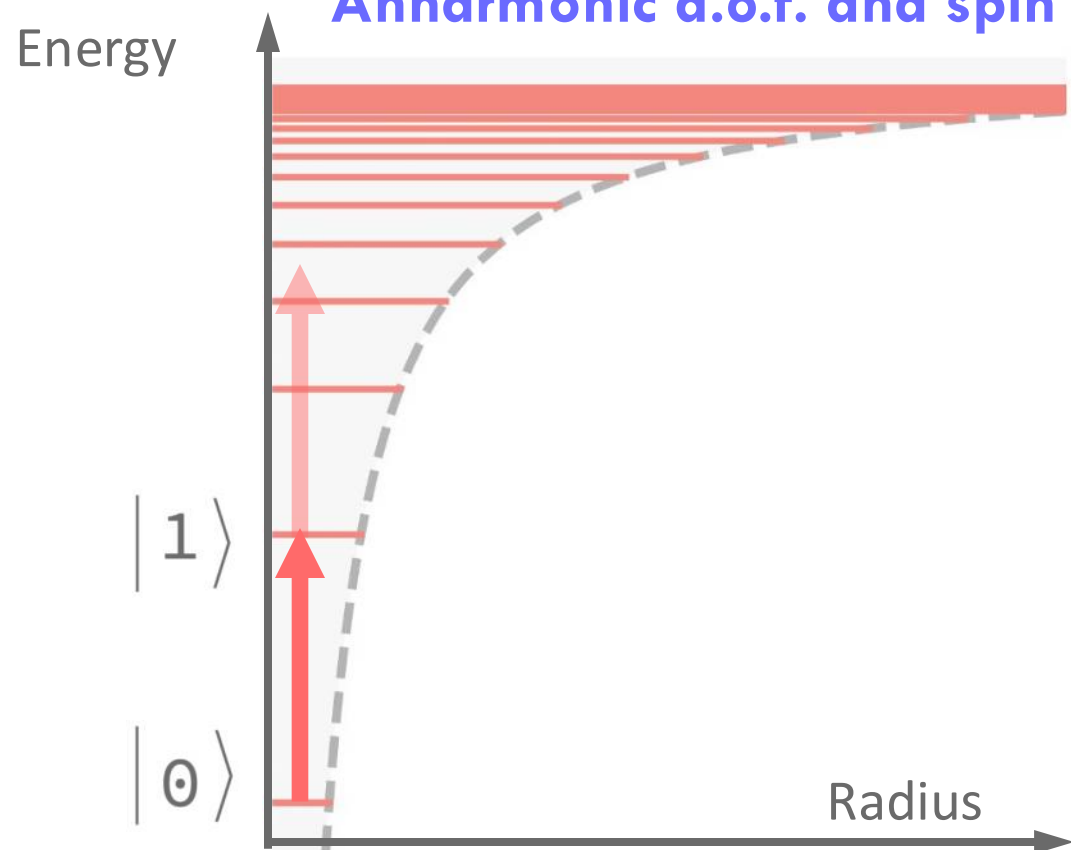
capacitor

inductor

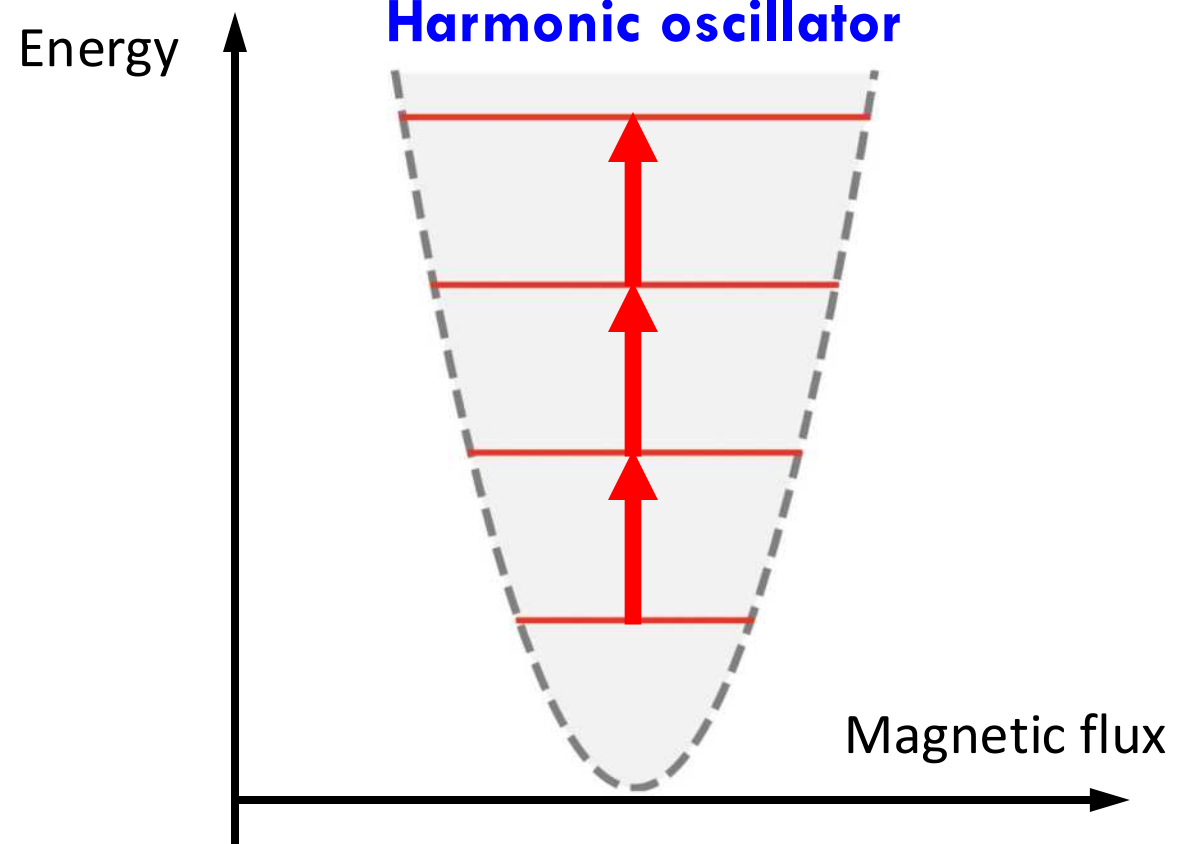


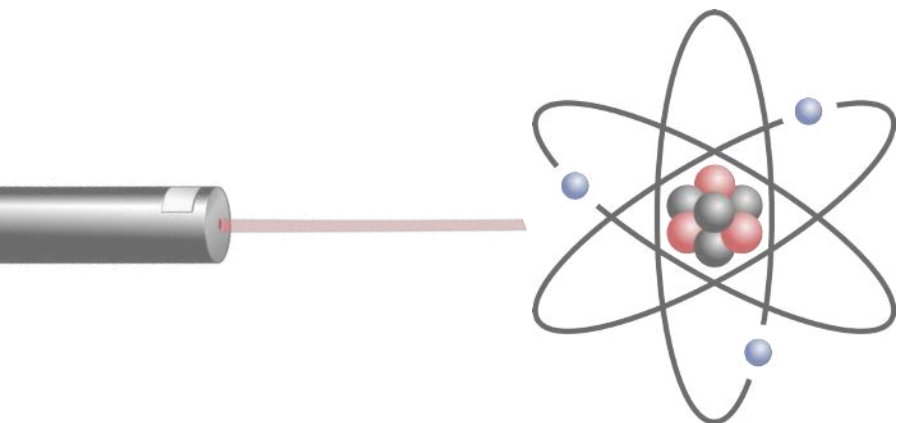


Anharmonic d.o.f. and spin

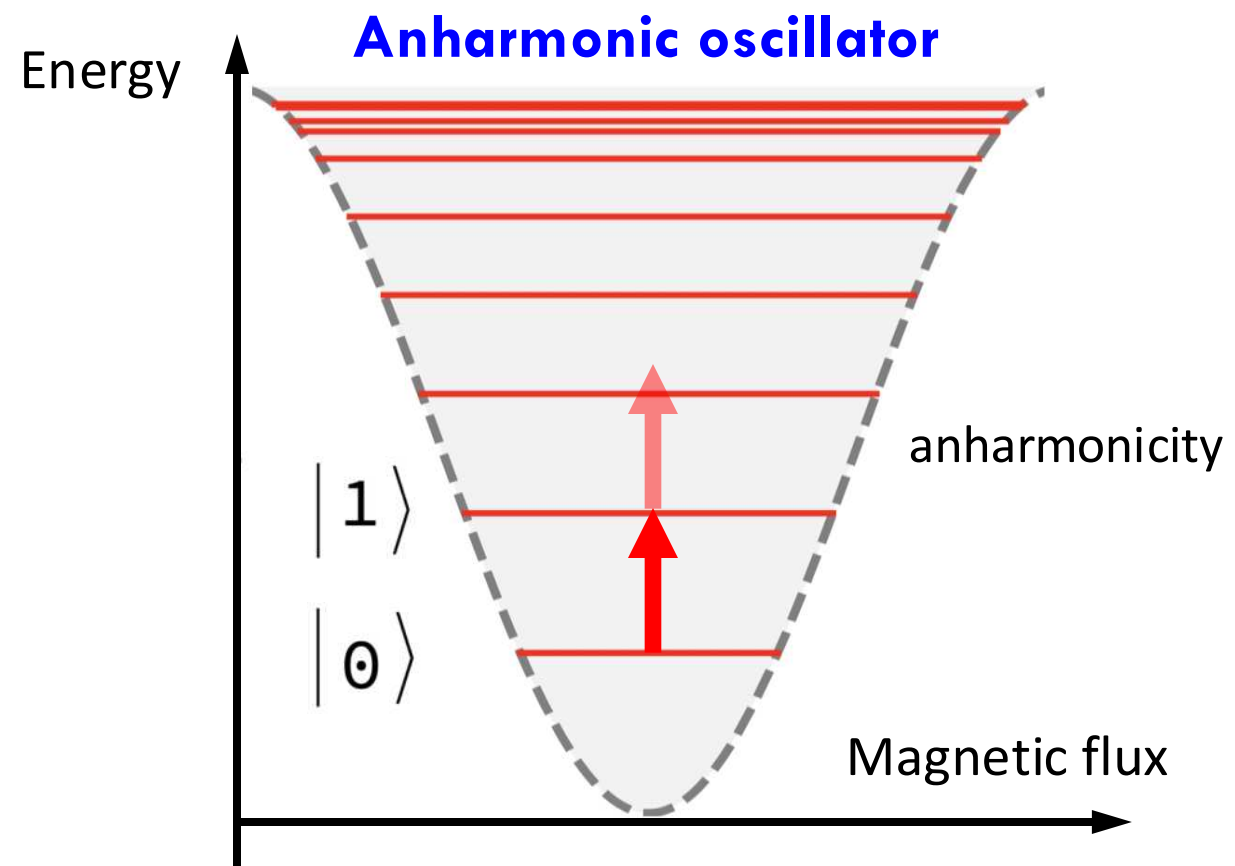
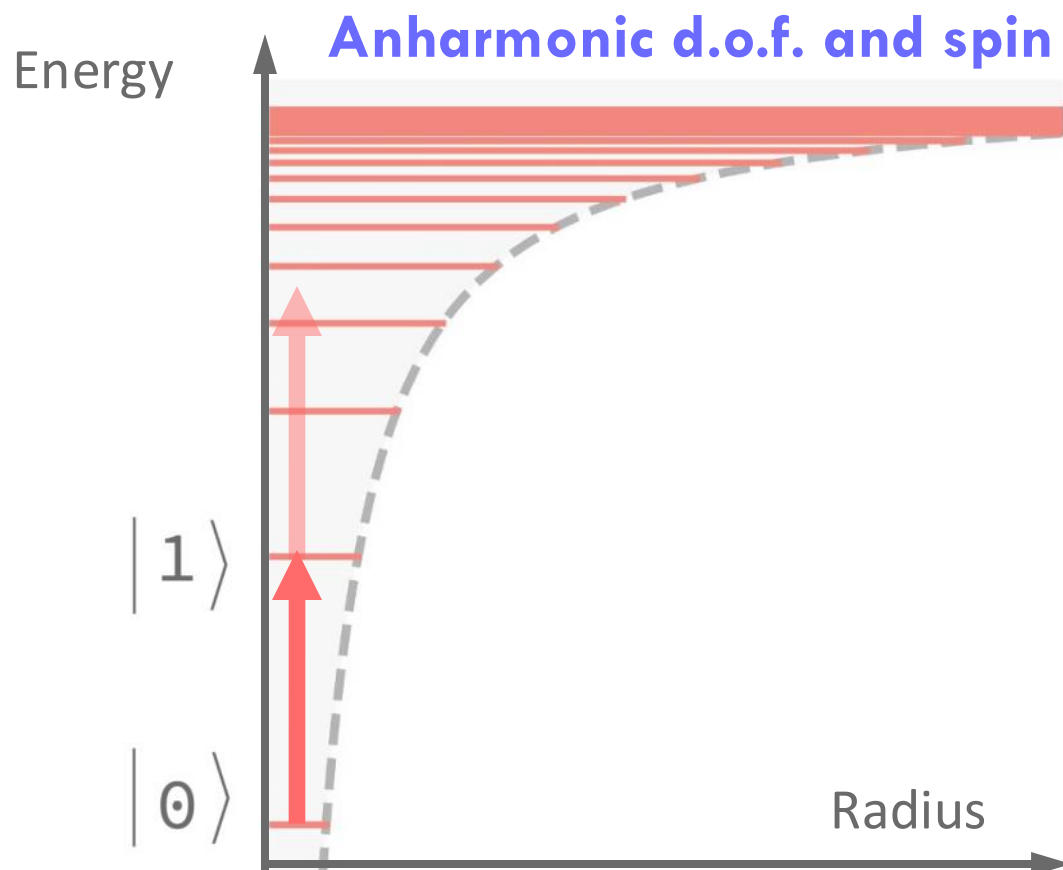
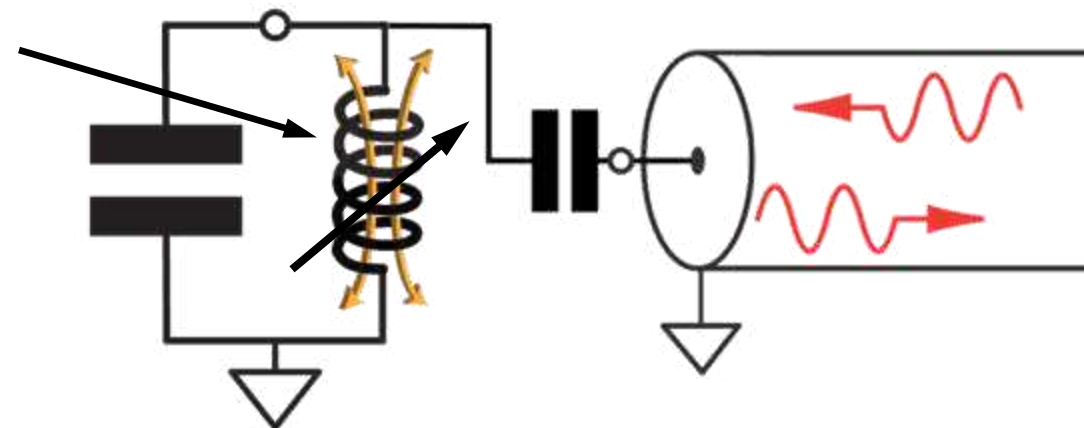


Harmonic oscillator



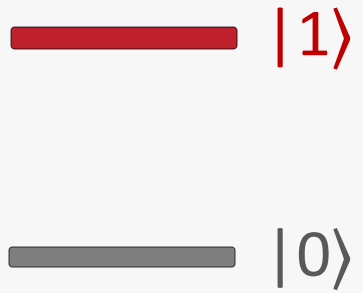


non-linear
inductor

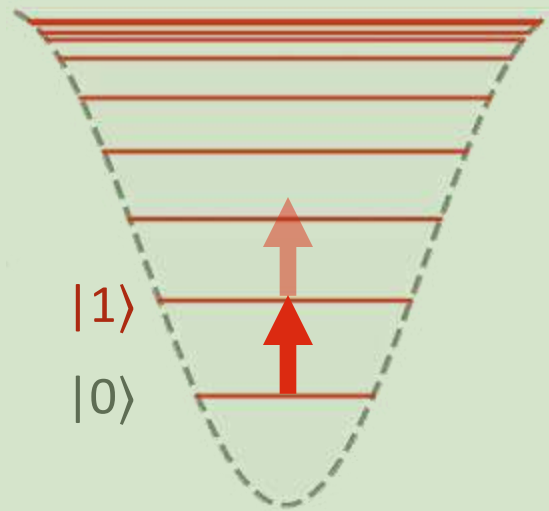


Big-picture connections

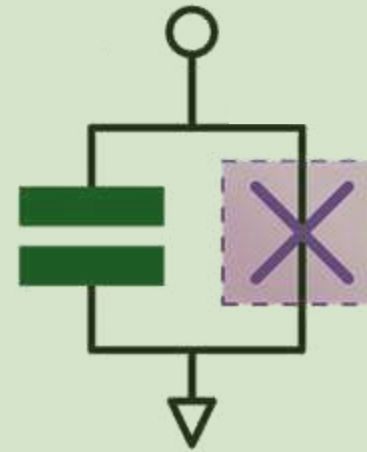
**Idealization of
qubit**



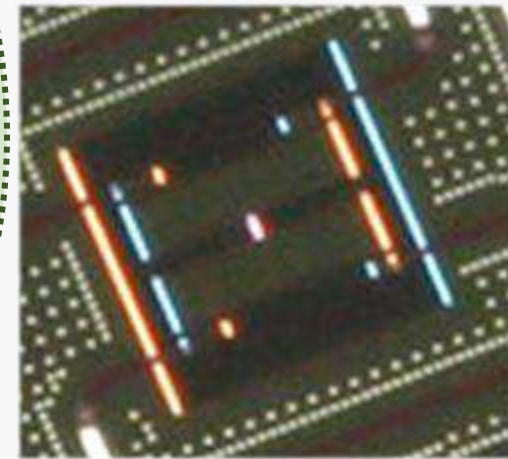
**Anharmonic
oscillator**



**Physical circuit
model**



Physical layout



Idealization



Physical reality

Circuit Quantum Electrodynamics (cQED)

Macro
overview

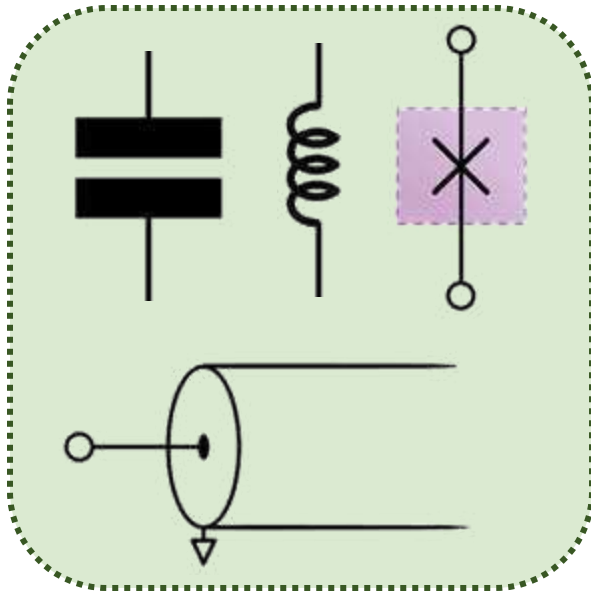
There are two kinds of physicists:

Those who believe all of physics is *spins*.

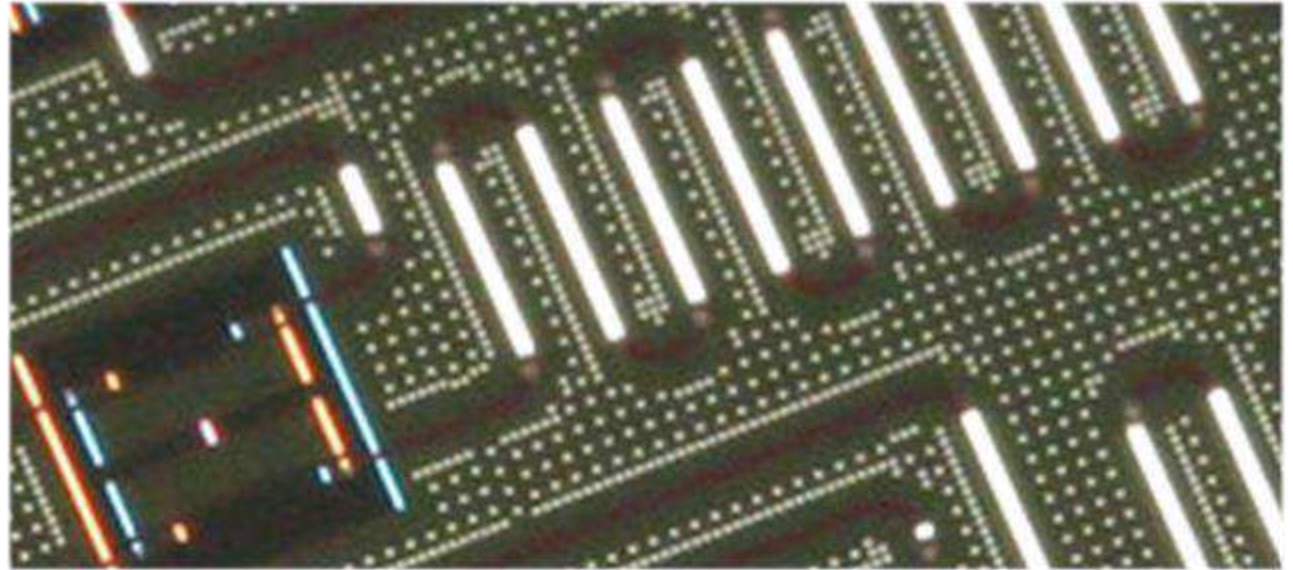
Those who believe all of physics is *oscillators*.

cQED Ingredients

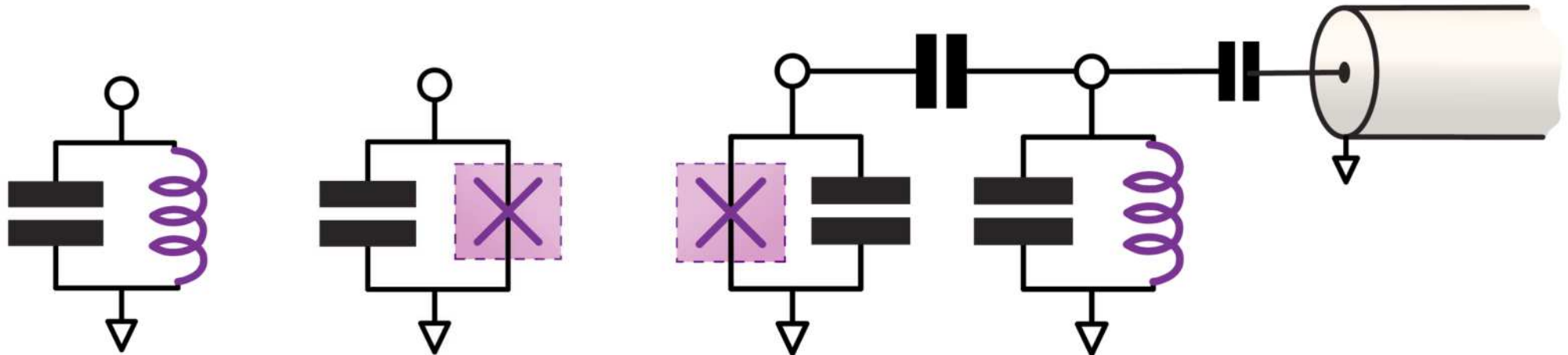
Circuit elements



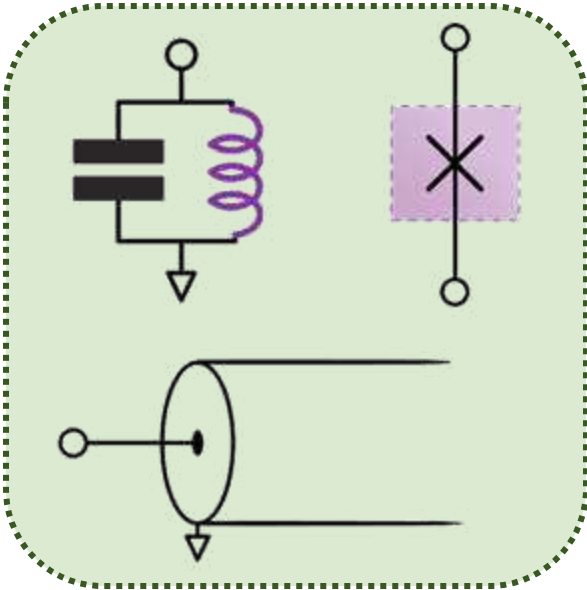
Microwave oscillators



Combine



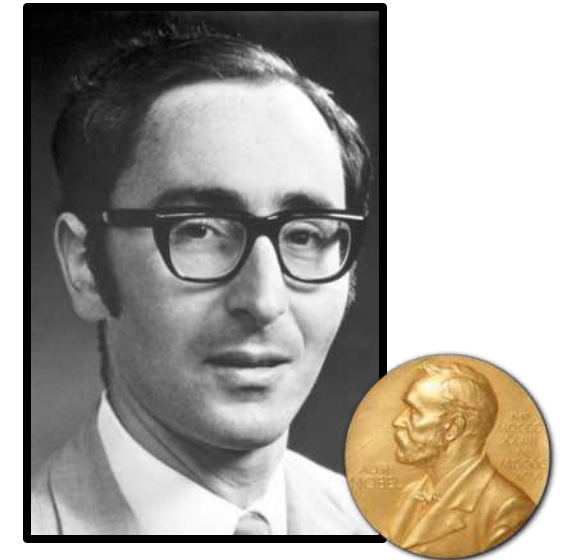
cQED Ingredients

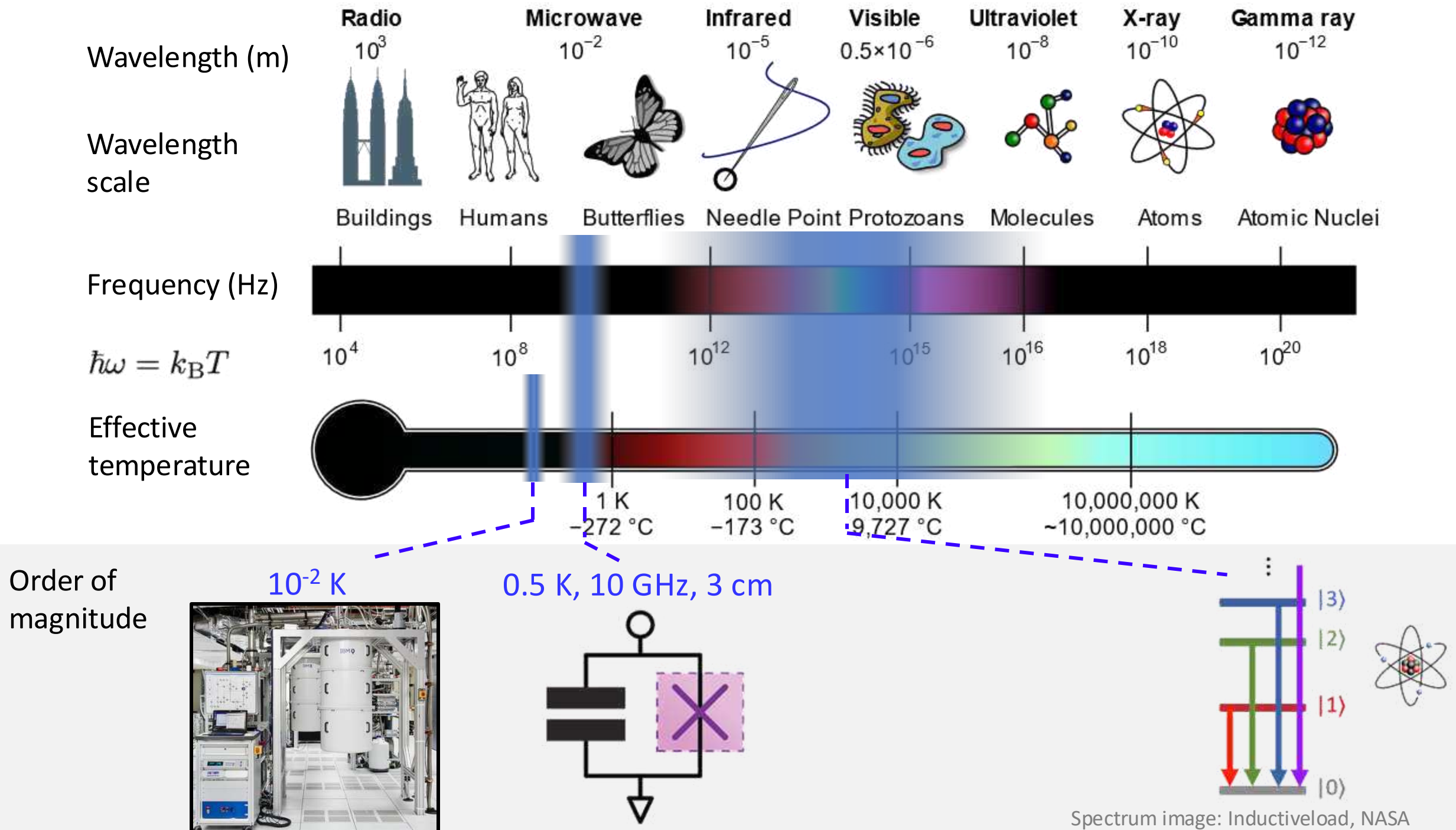


- Small dissipation
- Isolation from environment
- Low temperature
- Nonlinearity
- Large vacuum fluctuations

Superconductivity

- Nominally zero intrinsic dissipation and heat
- Nominal temperature far below energy level splitting
- Non-linear, robust Josephson tunnel junction effect





A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

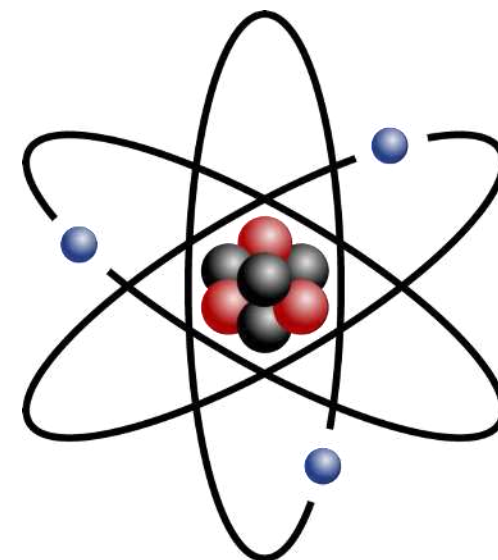
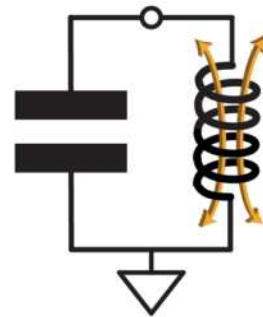
Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

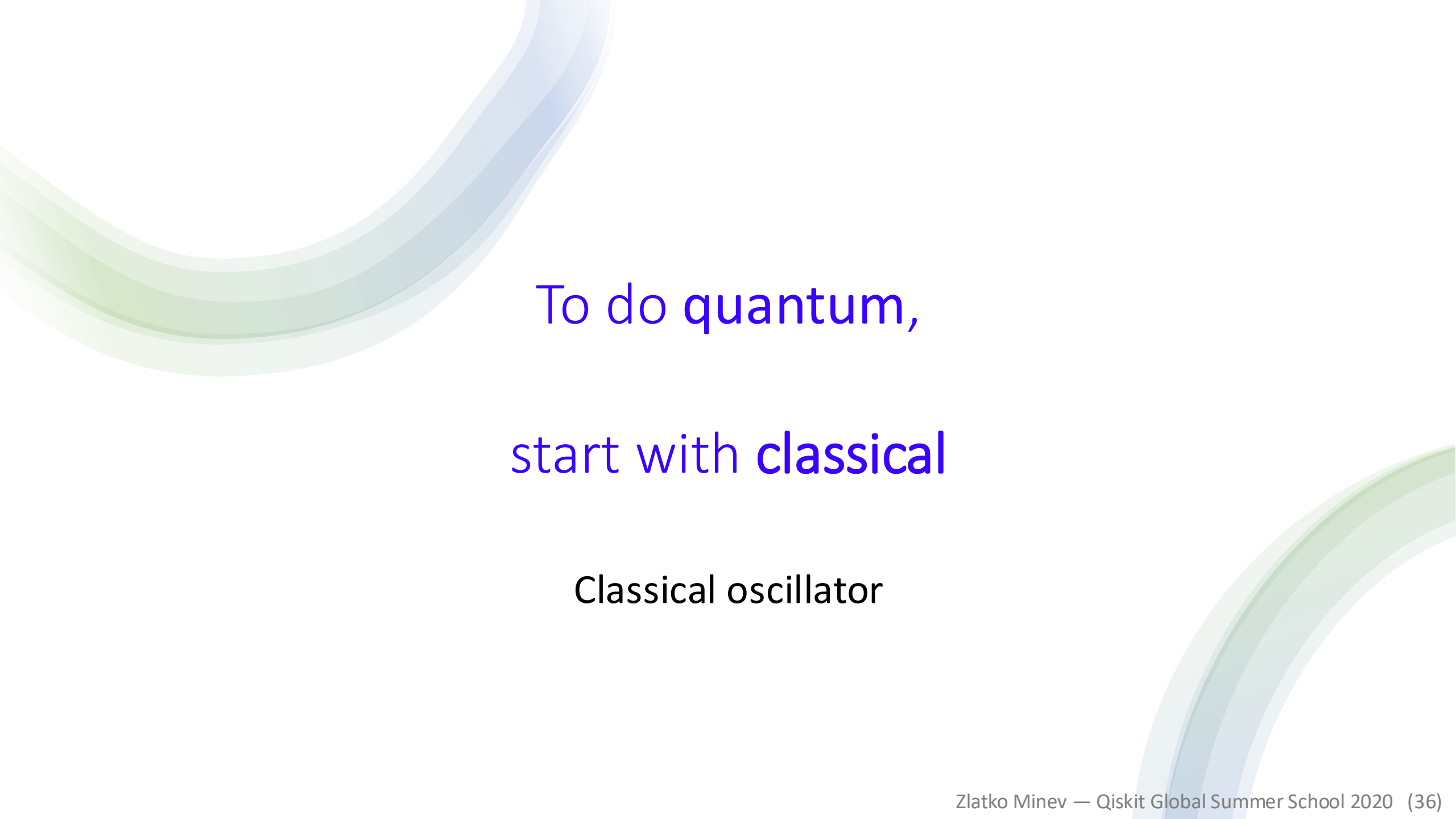
Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...

Circuit Quantum Electrodynamics

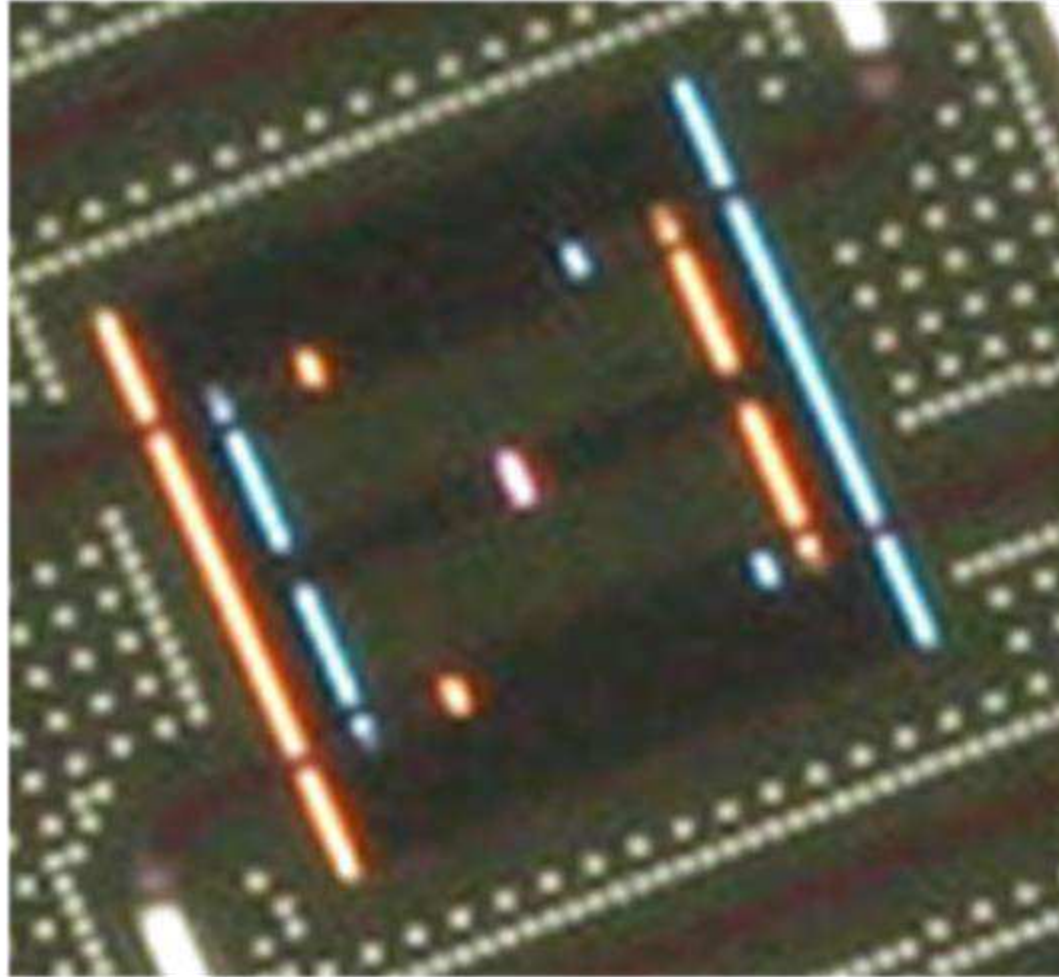




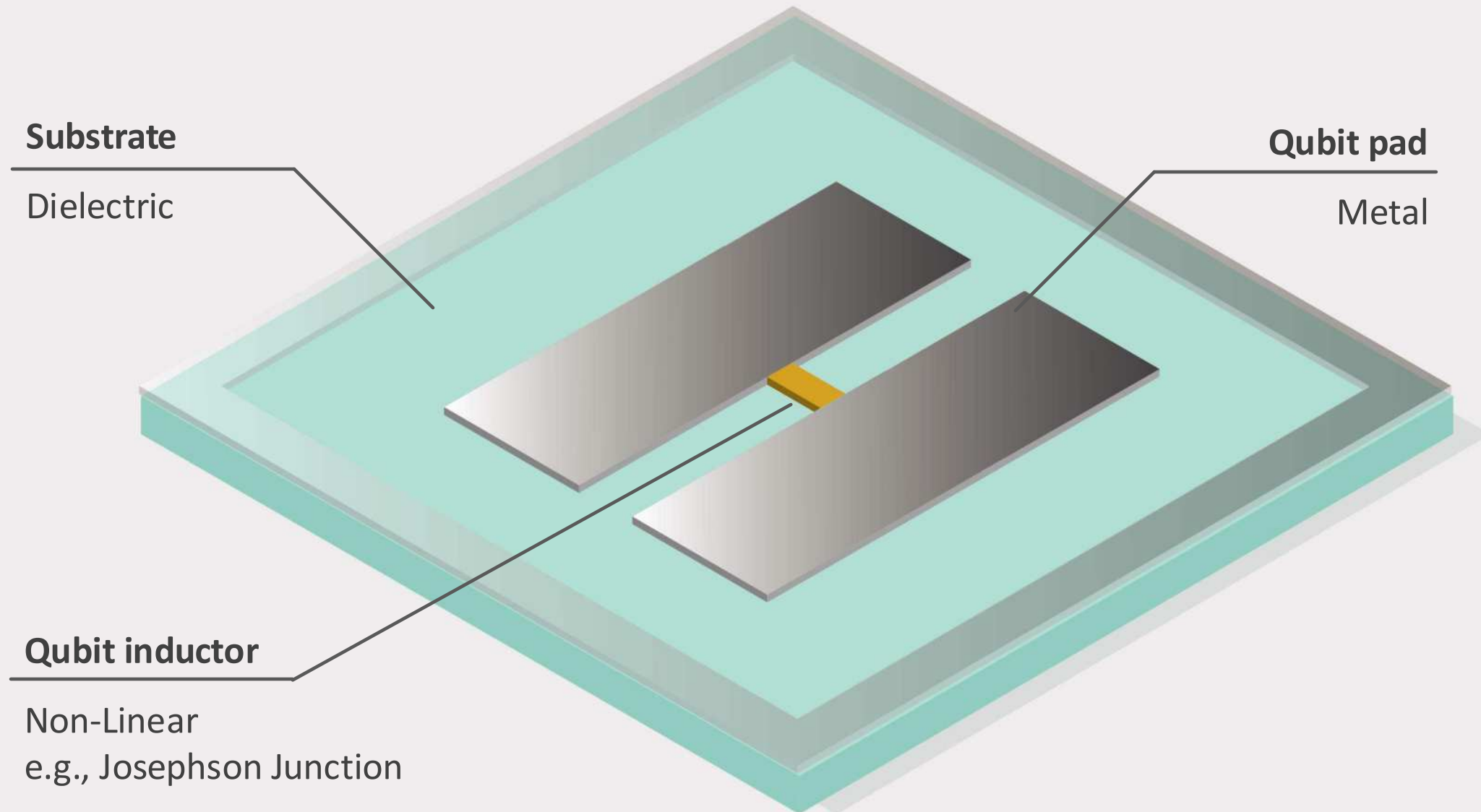
To do quantum,
start with **classical**

Classical oscillator

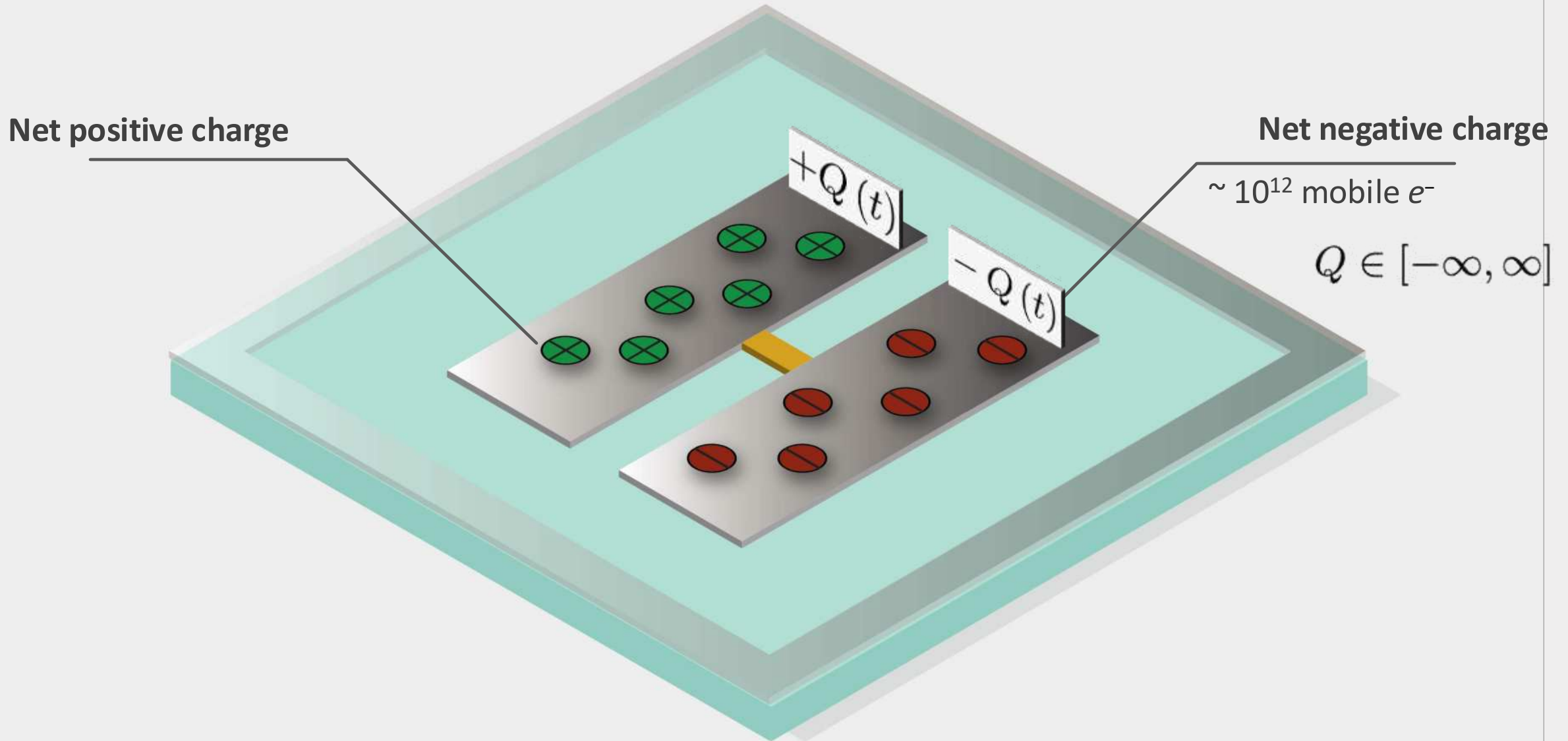
Transmon qubit



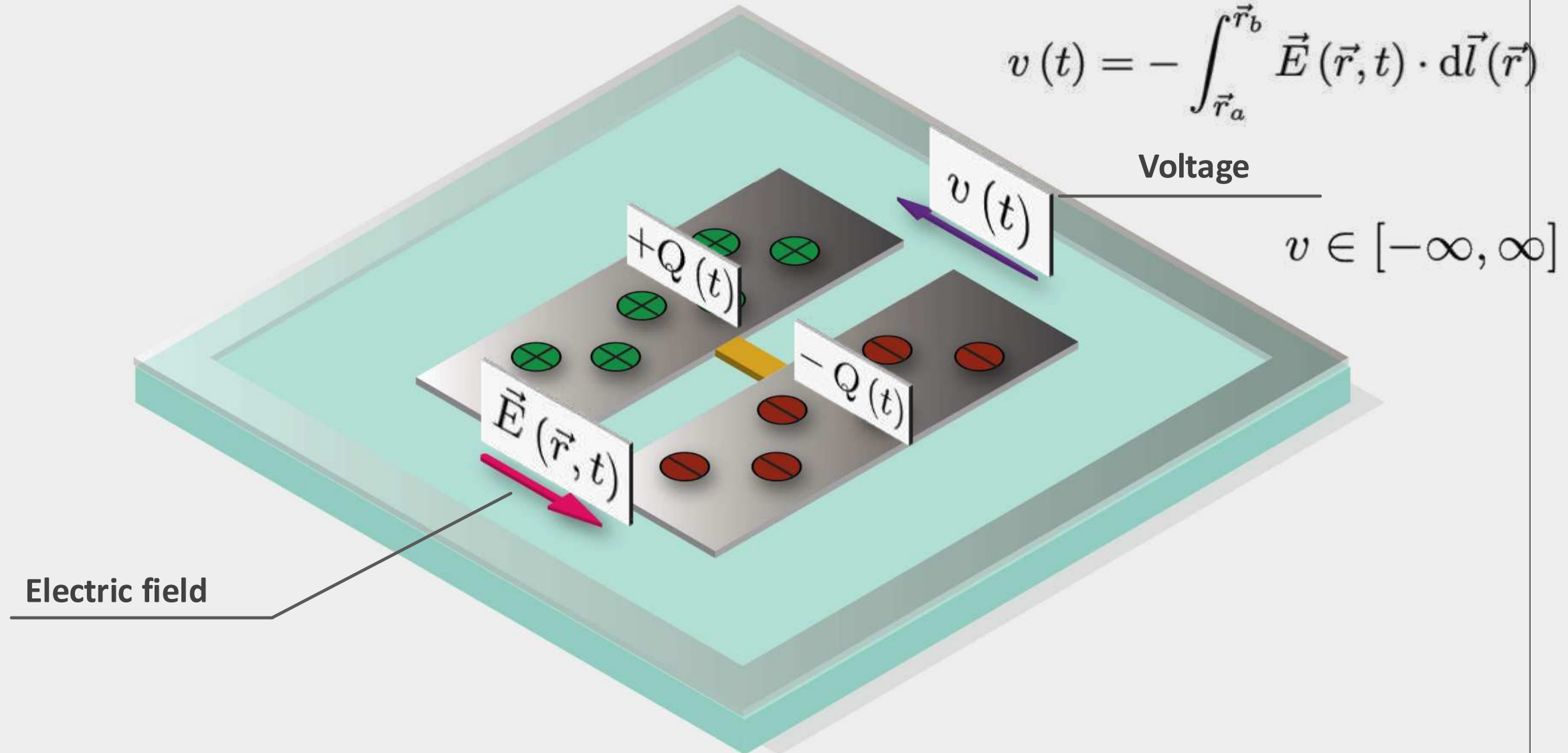
Transmon qubit



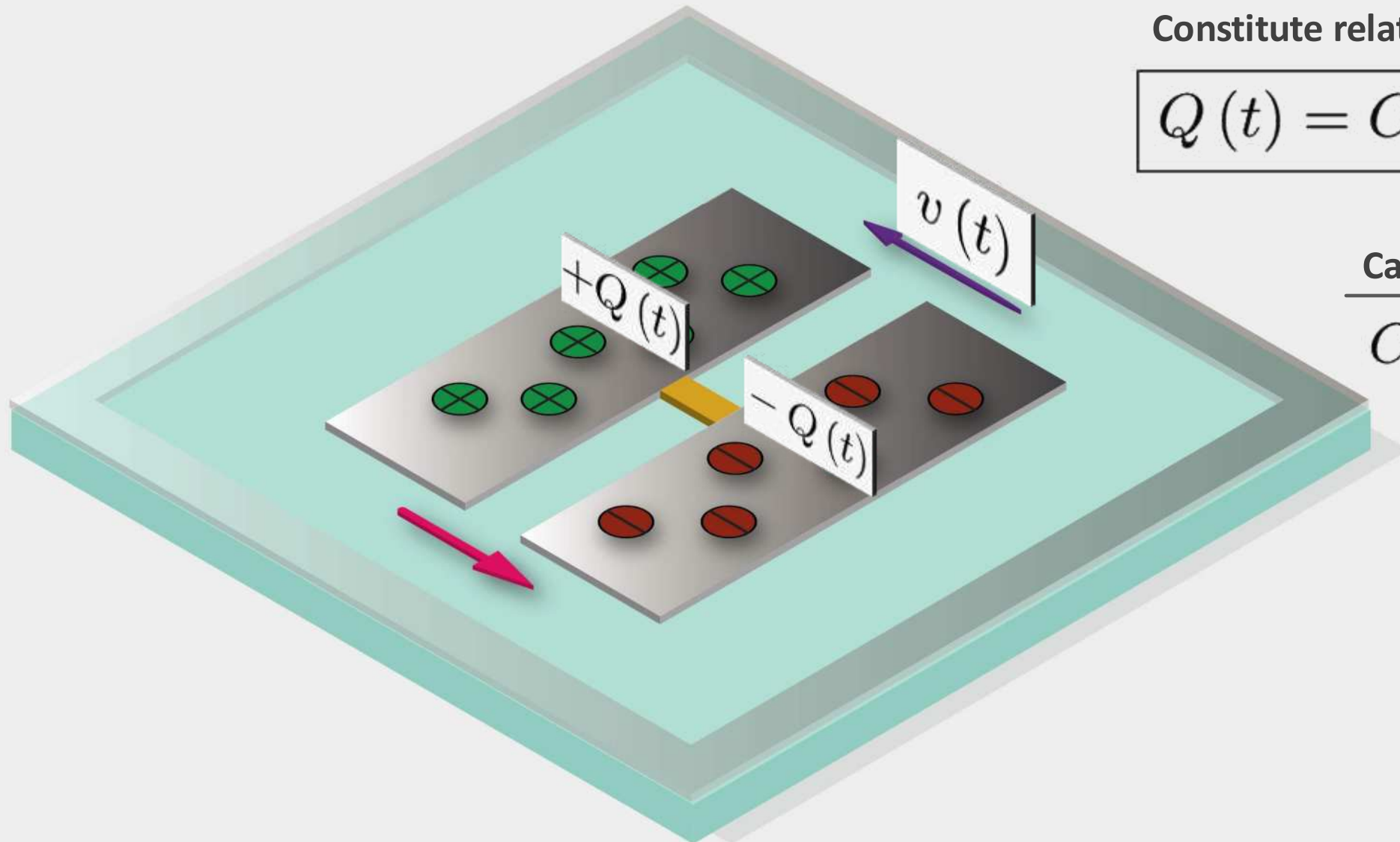
Transmon qubit: charge



Electric field and voltage



Charge and capacitance



Constitute relationship

$$Q(t) = C v(t)$$

Capacitance

$$C \in (0, \infty)$$



Charge and current

Conservation of charge

Universal relationship

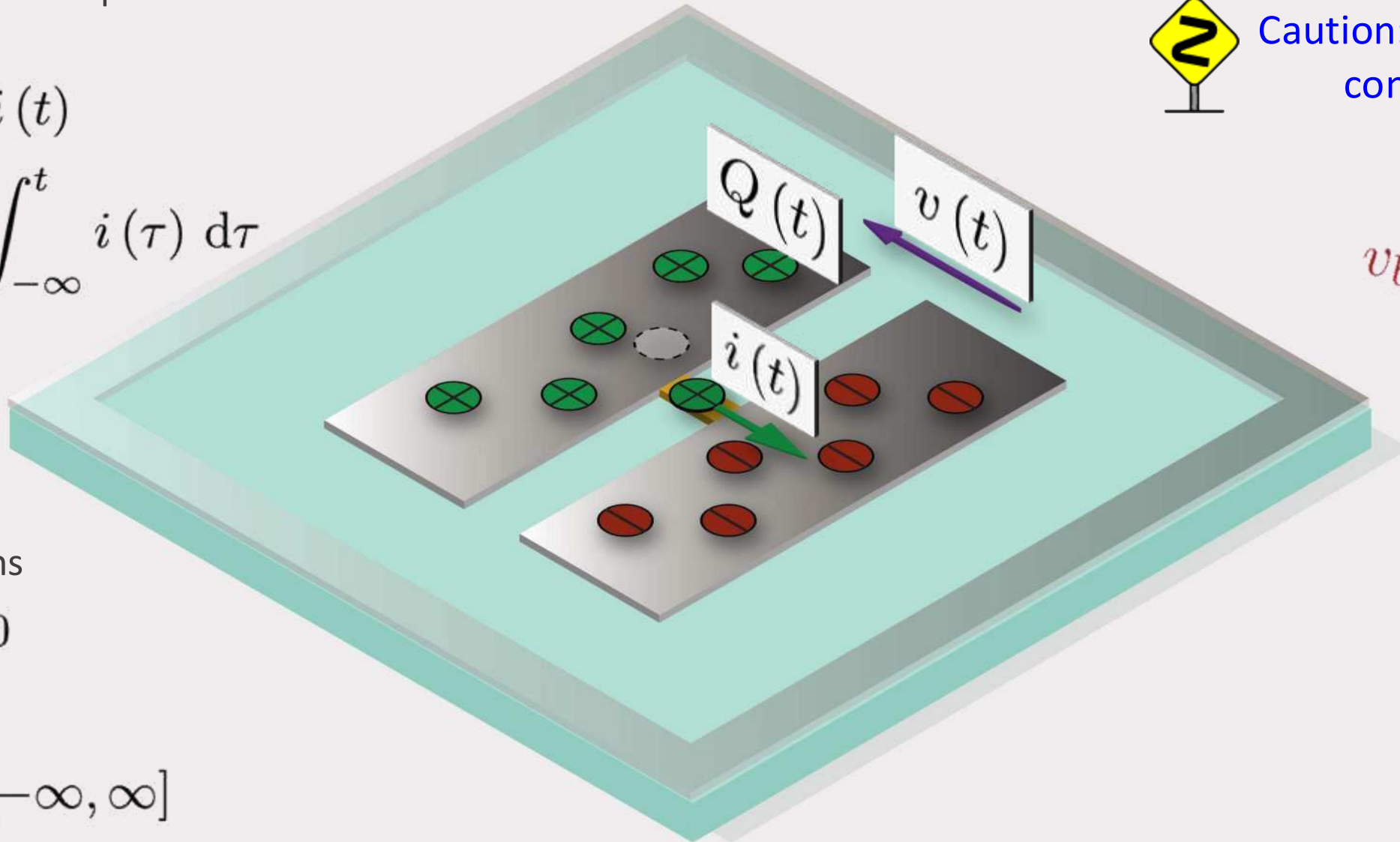
$$\frac{d}{dt}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^t i(\tau) d\tau$$

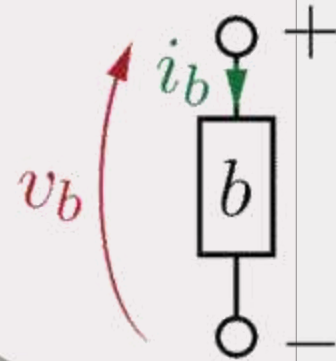
Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$



Caution: Passive sign convention



Magnetic flux and inductance

Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

$$\frac{d}{dt}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

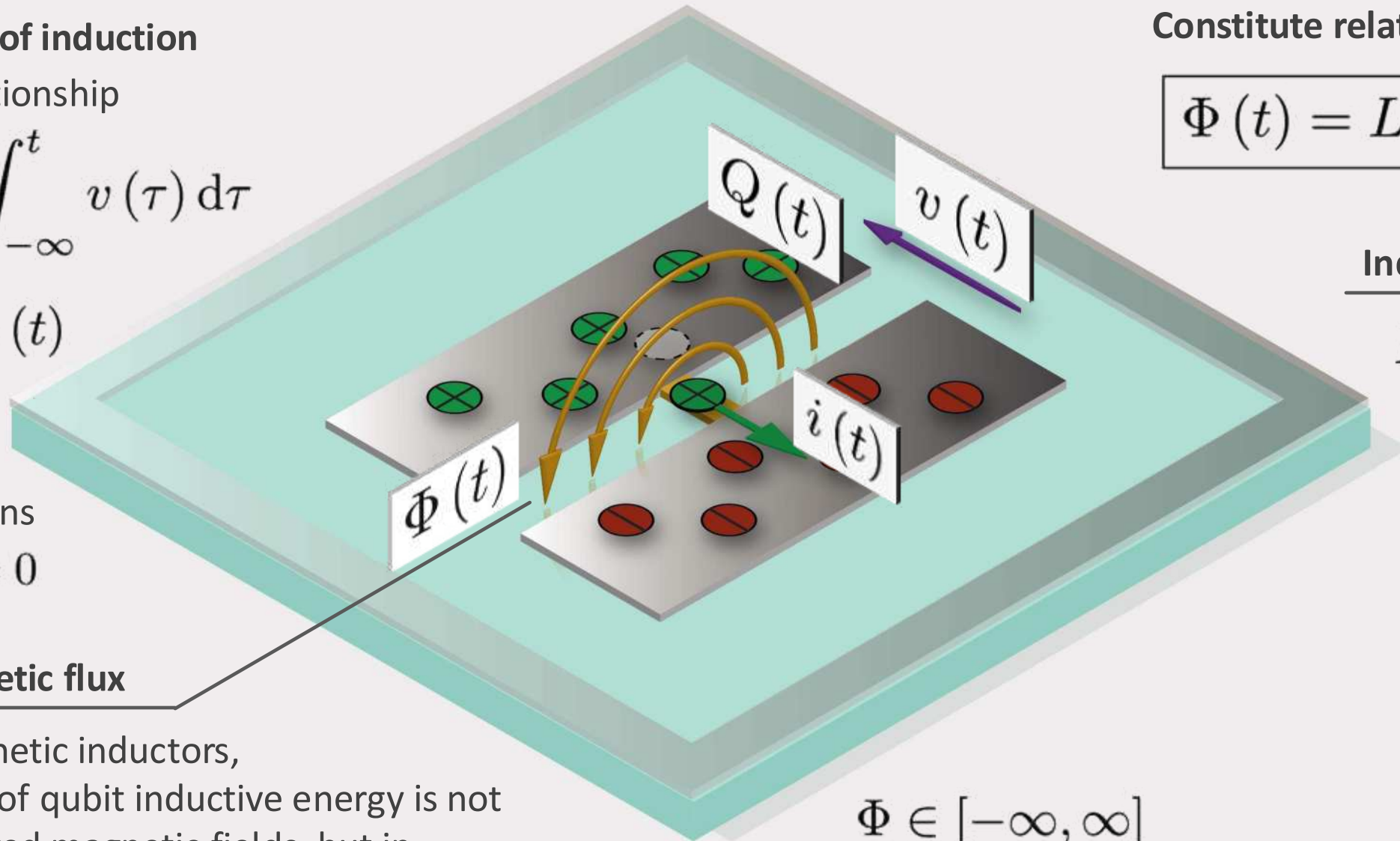
Constitute relationship

$$\Phi(t) = Li(t)$$

Inductance

$$L \in (0, \infty)$$

$$\Phi \in [-\infty, \infty]$$



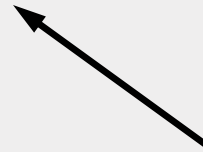
Power and energy

Universal

$$\frac{d}{dt}\mathcal{E}(t) = p(t) \equiv v(t)i(t)$$

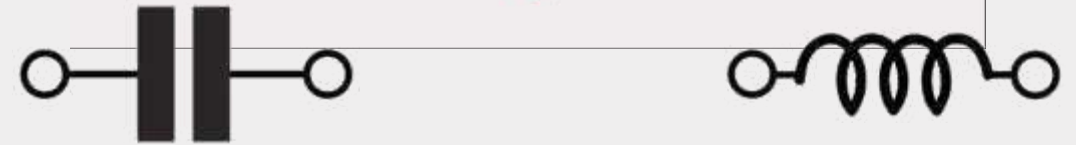
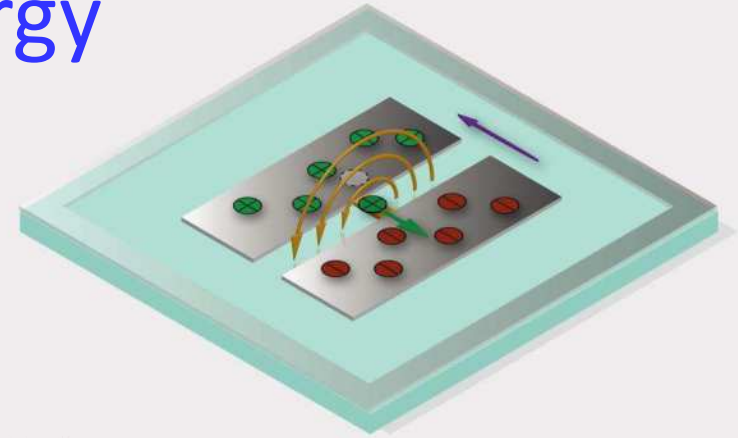


Energy stored *in*
(delivered *to*)
component



Instantaneous
power flowing *to*
component

$$\mathcal{E}(t) = \int_{-\infty}^t p(\tau) d\tau$$



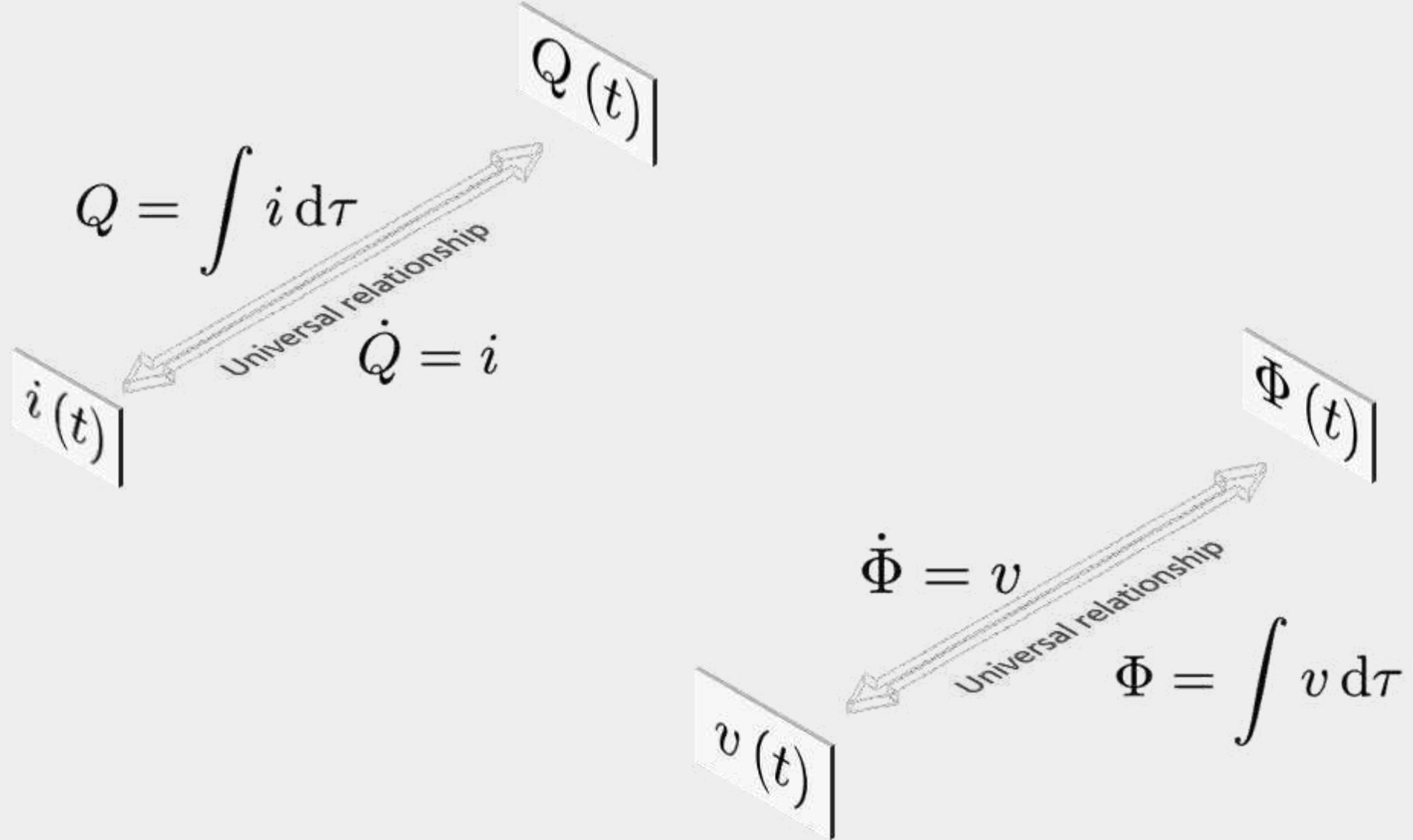
$$\mathcal{E}_{\text{cap}}(\dot{\Phi}) = \frac{1}{2}C\dot{\Phi}^2$$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

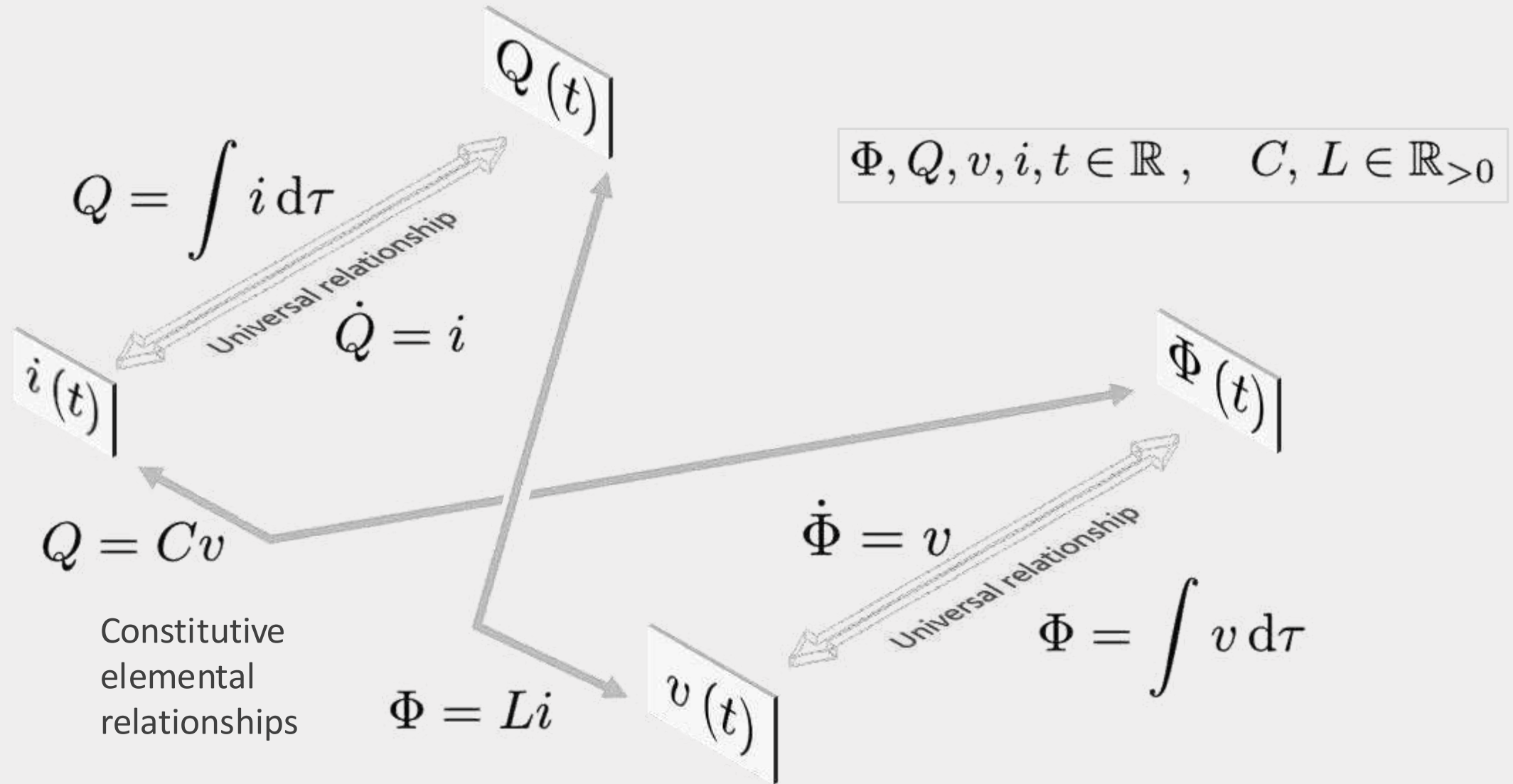
$$\mathcal{E}_{\text{cap}}(Q) = \frac{Q^2}{2C}$$

$$\mathcal{E}_{\text{ind}}(\dot{Q}) = \frac{1}{2}L\dot{Q}^2$$

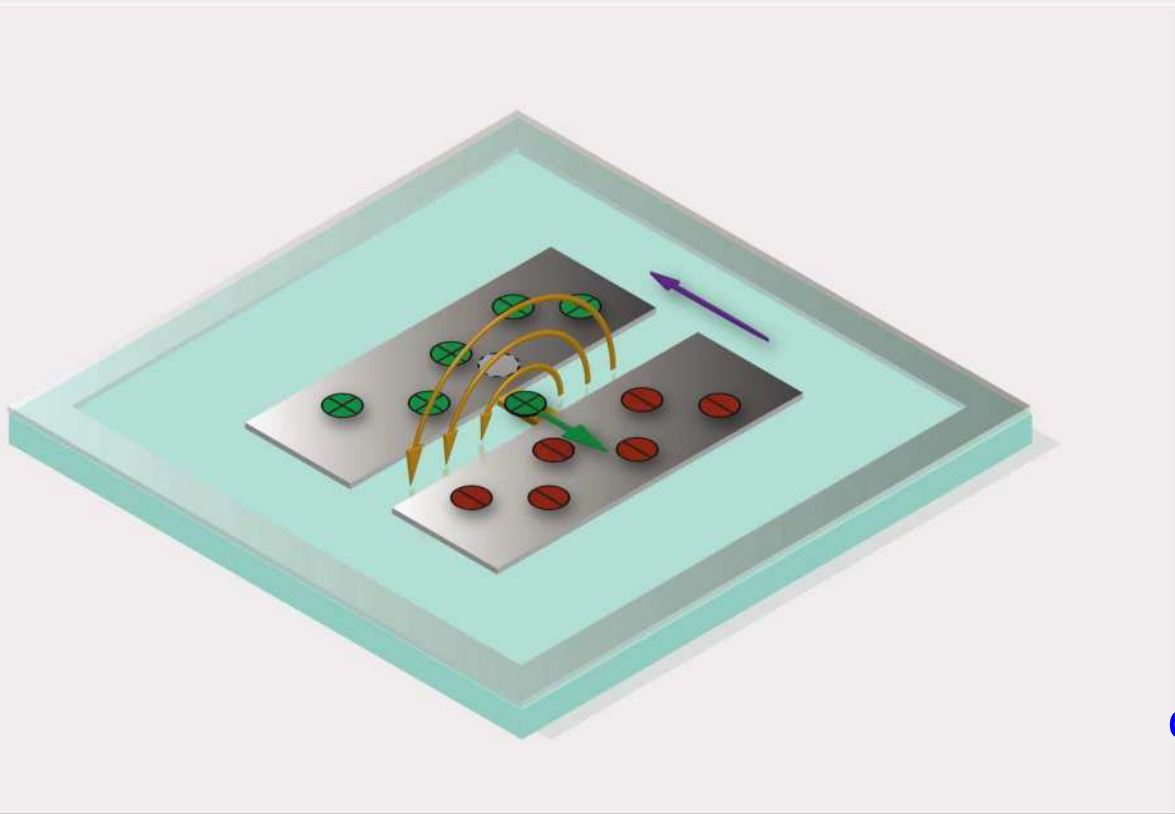
Four fundamental manifestations of electricity



Four fundamental manifestations of electricity



Electromagnetic oscillator



capacitance value

node of the circuit

C

L

inductance
value

inductor

capacitor

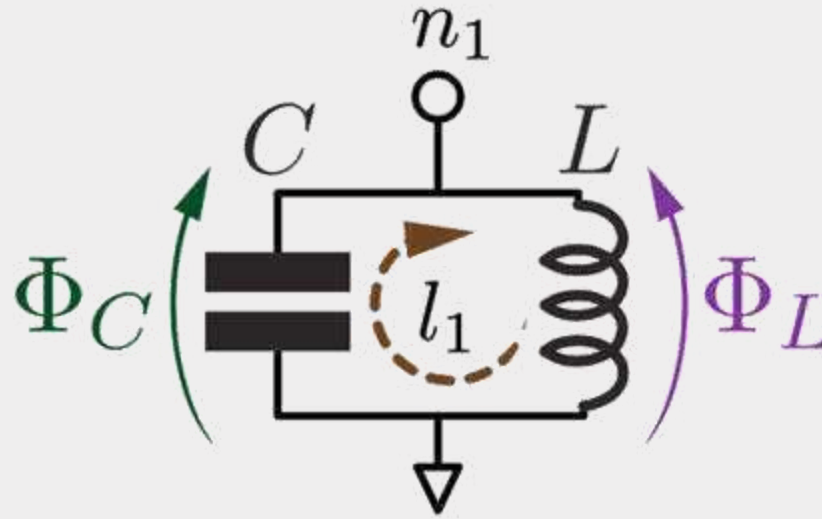
ground

Kirchhoff's network laws*

Conservation of charge

Kirchhoff's current law

$$\sum_{b \in \text{node}} \pm \dot{Q}_b(t) = 0$$



Faraday's law of induction

Kirchhoff's voltage law

$$\sum_{b \in \text{loop}} \pm \dot{\Phi}_b(t) = 0$$

$$n_1 : \quad \dot{Q}_C + \dot{Q}_L = 0$$

$$l_1 : \quad \dot{\Phi}_C - \dot{\Phi}_L = 0$$

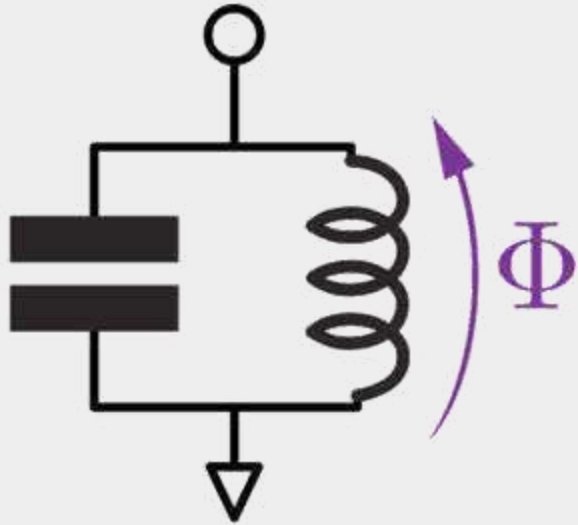
$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\Phi_C = \Phi_L$$



As we will see later, for the Lagrangian description in flux basis, KVL acts as a set of holonomic constraints and KCL as the equations of motion

Oscillator analogy



$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\ddot{\Phi} = -\omega_0^2\Phi, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Resonance
frequency



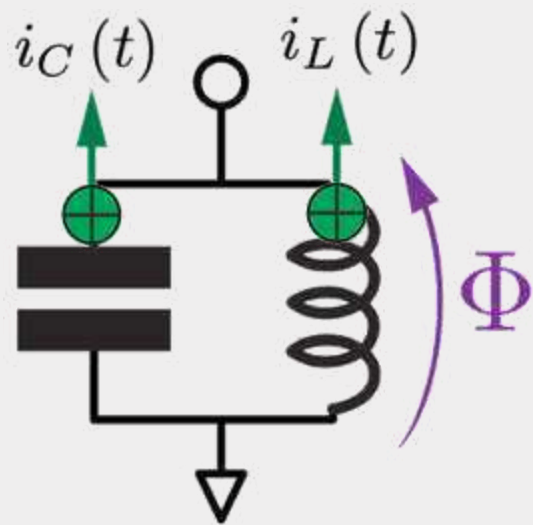
$\dot{\Phi}(t)$ Velocity

$\Phi(t)$ Deviation from equilibrium

Equilibrium position
 $x = 0, \quad \Phi = 0$



Lagrangian and Hamiltonian*



$$\underbrace{C\ddot{\Phi}}_{i_C(t)} + \underbrace{L^{-1}\Phi}_{i_L(t)} = 0$$

$$\longleftrightarrow \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = \frac{\partial \mathcal{L}}{\partial \Phi}$$

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\mathcal{L}(\Phi, \dot{\Phi}) = \mathcal{E}_{\text{cap}}(\dot{\Phi}) - \mathcal{E}_{\text{ind}}(\Phi)$$

$$= \frac{1}{2}C\dot{\Phi}^2 - \frac{\Phi^2}{2L}$$

$$Q = \frac{\partial \mathcal{L}}{\partial \dot{\Phi}} = C\dot{\Phi}$$

Kinetic energy – Potential energy

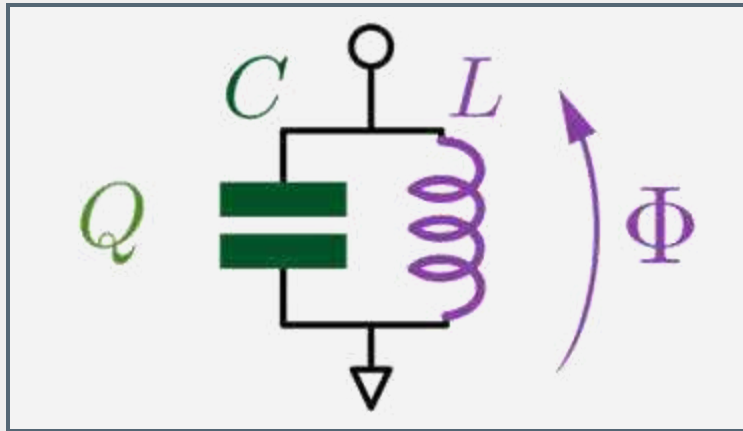
Canonically conjugate variable
(momentum; charge)

Euler-Lagrange equations, $F = ma$

Kinetic energy + Potential energy

* Temporarily going to assume some minimal knowledge of classical mechanics.
Since we eliminated the KVL constraints, this is a Lagrangian of the second kind.

The LC classical harmonic oscillator



Position: $\Phi \mapsto x$

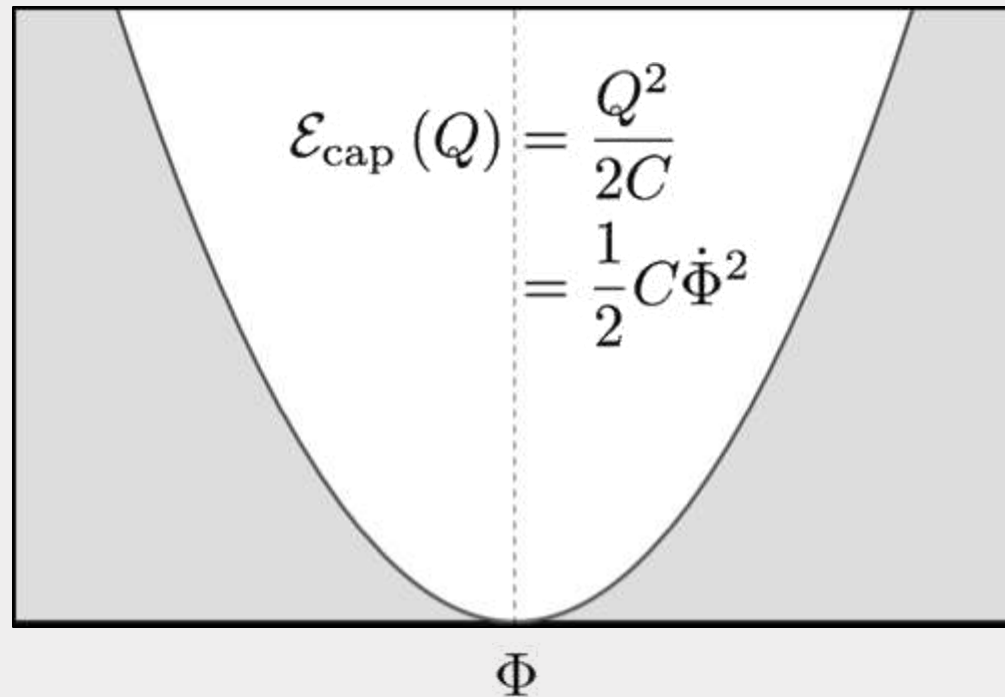
Mass: $C \mapsto m$

Momentum: $Q \mapsto p$

Spring constant: $L^{-1} \mapsto k$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \quad Z_0 = \frac{L}{C}$$

Graph: Minev; Spring-mass: Svjo

“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

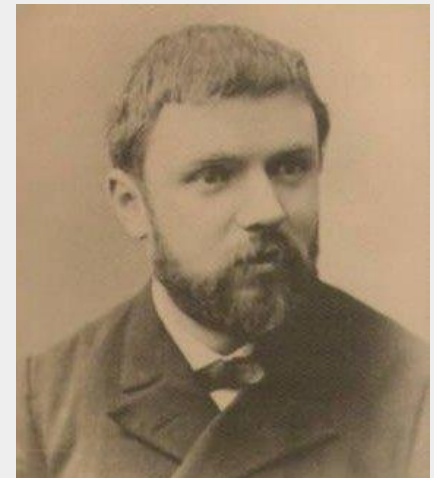


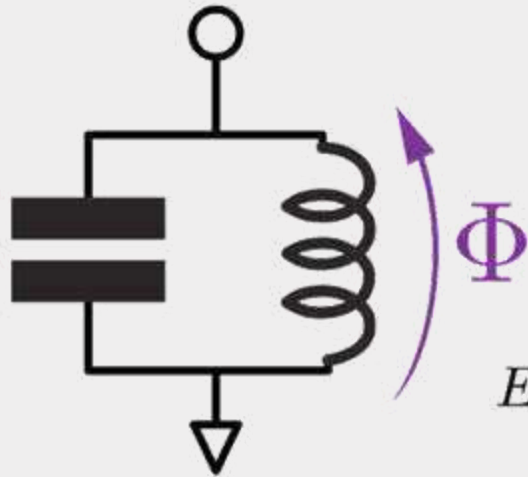
Photo by Eugène Pirou

Hamiltonian dynamics and phase space

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

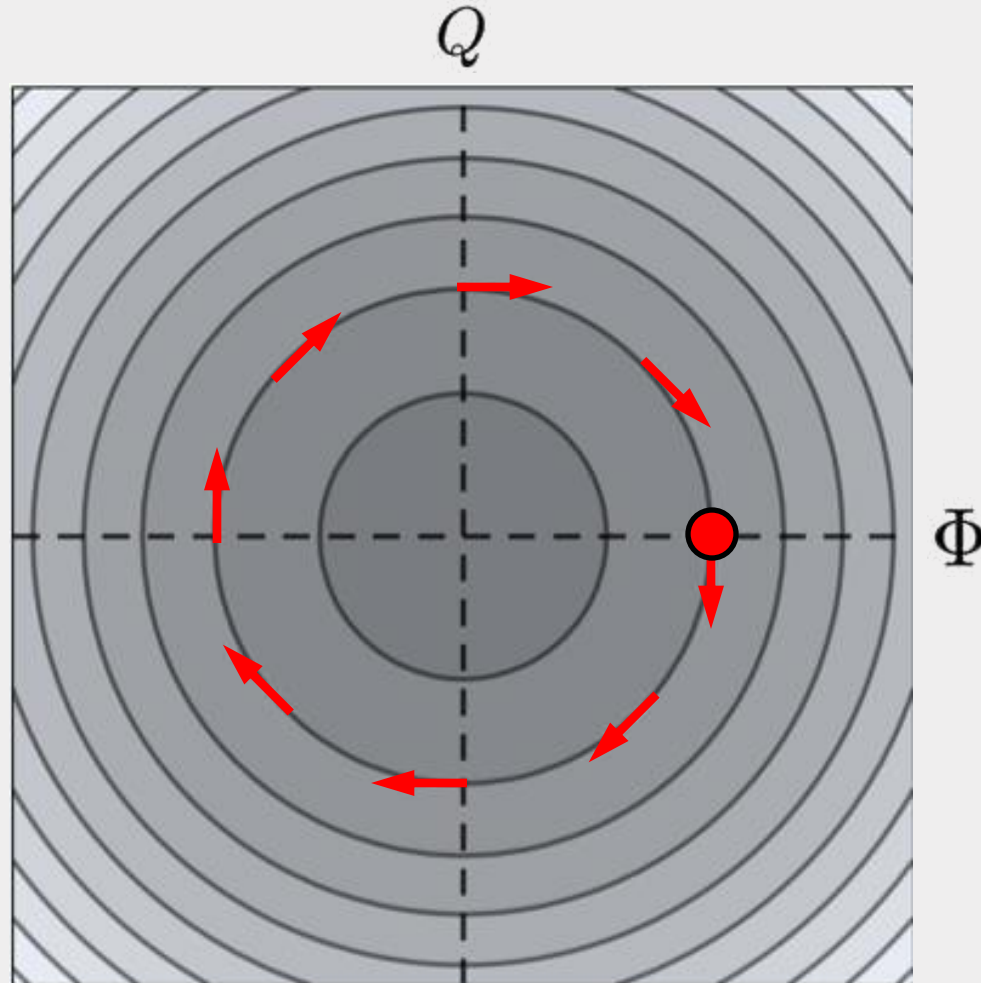
with $L = C = 1$

$$E = \hbar\omega_0 \left(n + \frac{1}{2} \right) = \frac{1}{2} (Q^2 + \Phi^2)$$



$$E = \frac{3\hbar\omega_0}{2}$$

$$E = \frac{\hbar\omega_0}{2}$$



$$\dot{\Phi} = +\frac{\partial \mathcal{H}}{\partial Q} = +Q$$

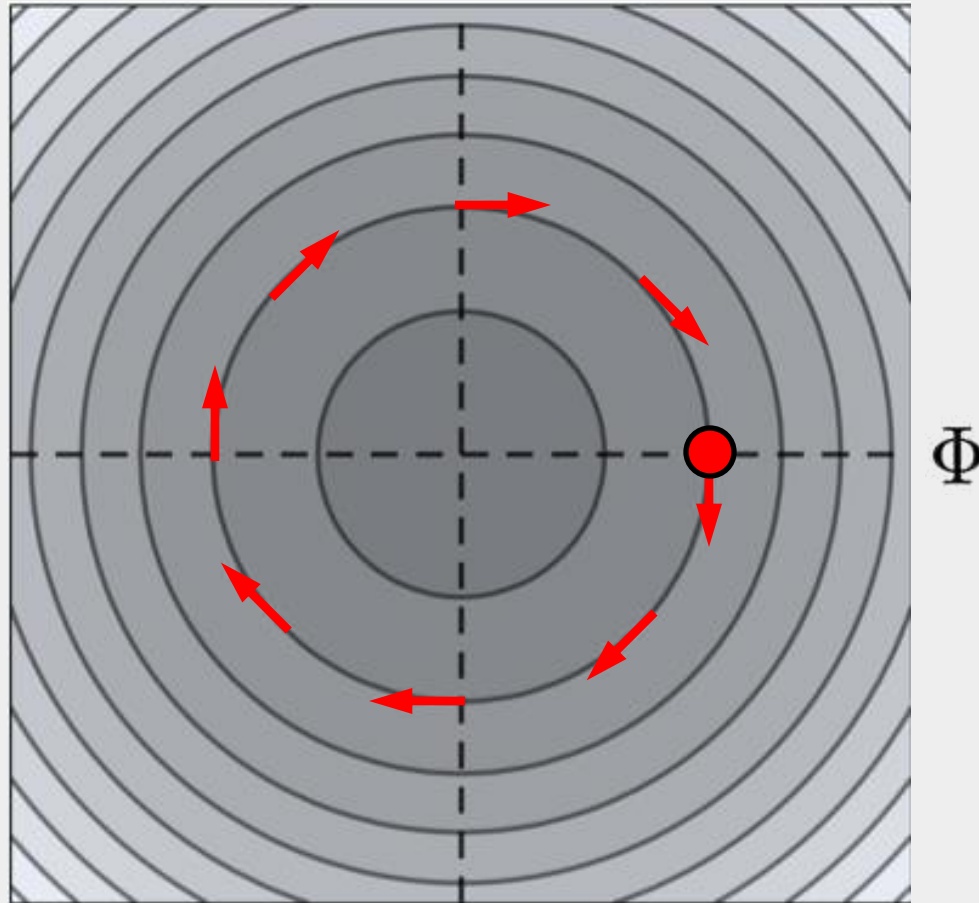
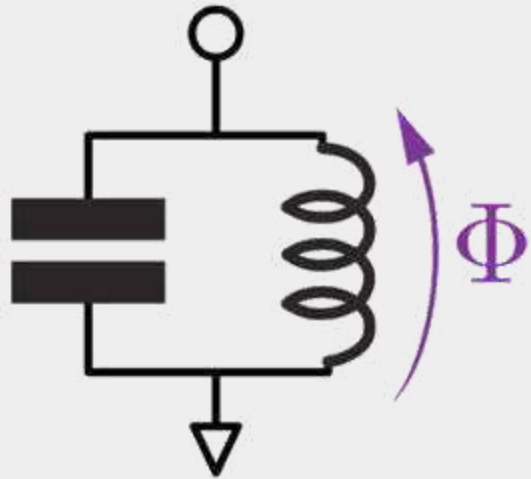
$$\dot{Q} = -\frac{\partial \mathcal{H}}{\partial \Phi} = -\Phi$$

Complex action-angle variable

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} (\alpha^* \alpha + \alpha \alpha^*)$$

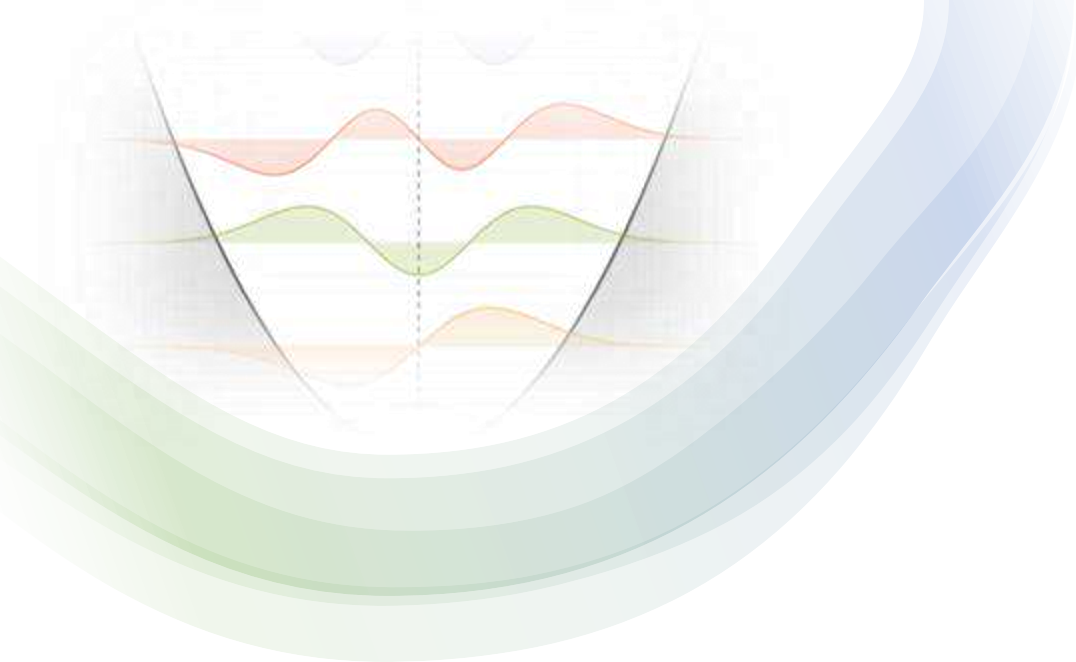
$$E = \hbar \omega_0 \left(n + \frac{1}{2} \right)$$

$$\alpha(t) = \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)]$$



Classical analog of the bosonic ladder operator

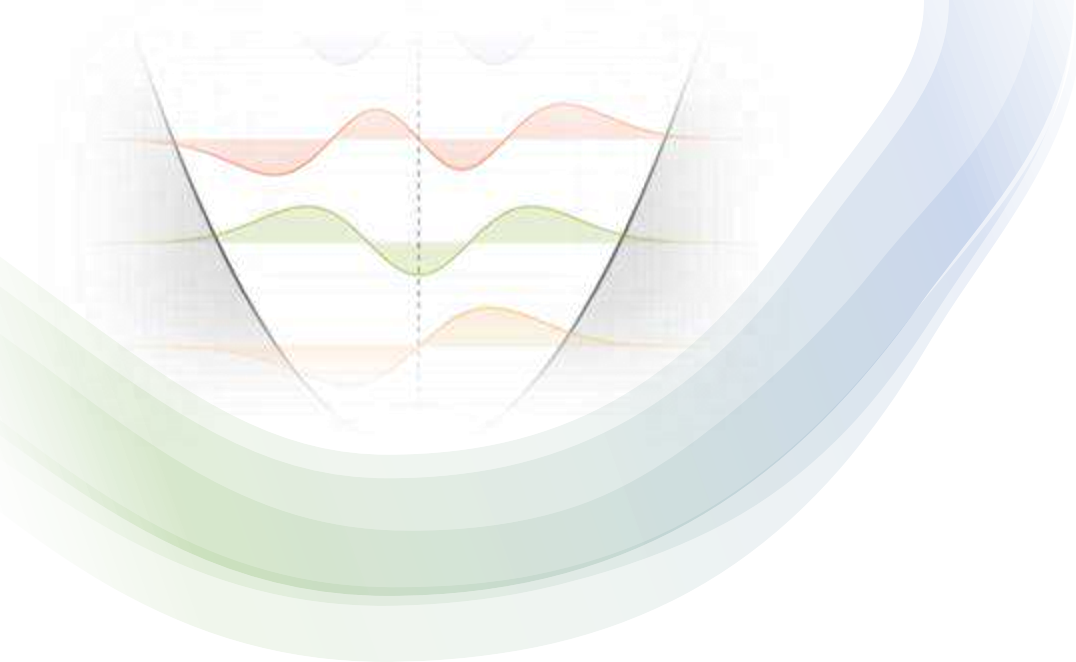
$$\alpha(t) = \alpha(0) e^{-i\omega_0 t}$$



Unveiling the quantum

Quantum harmonic oscillator

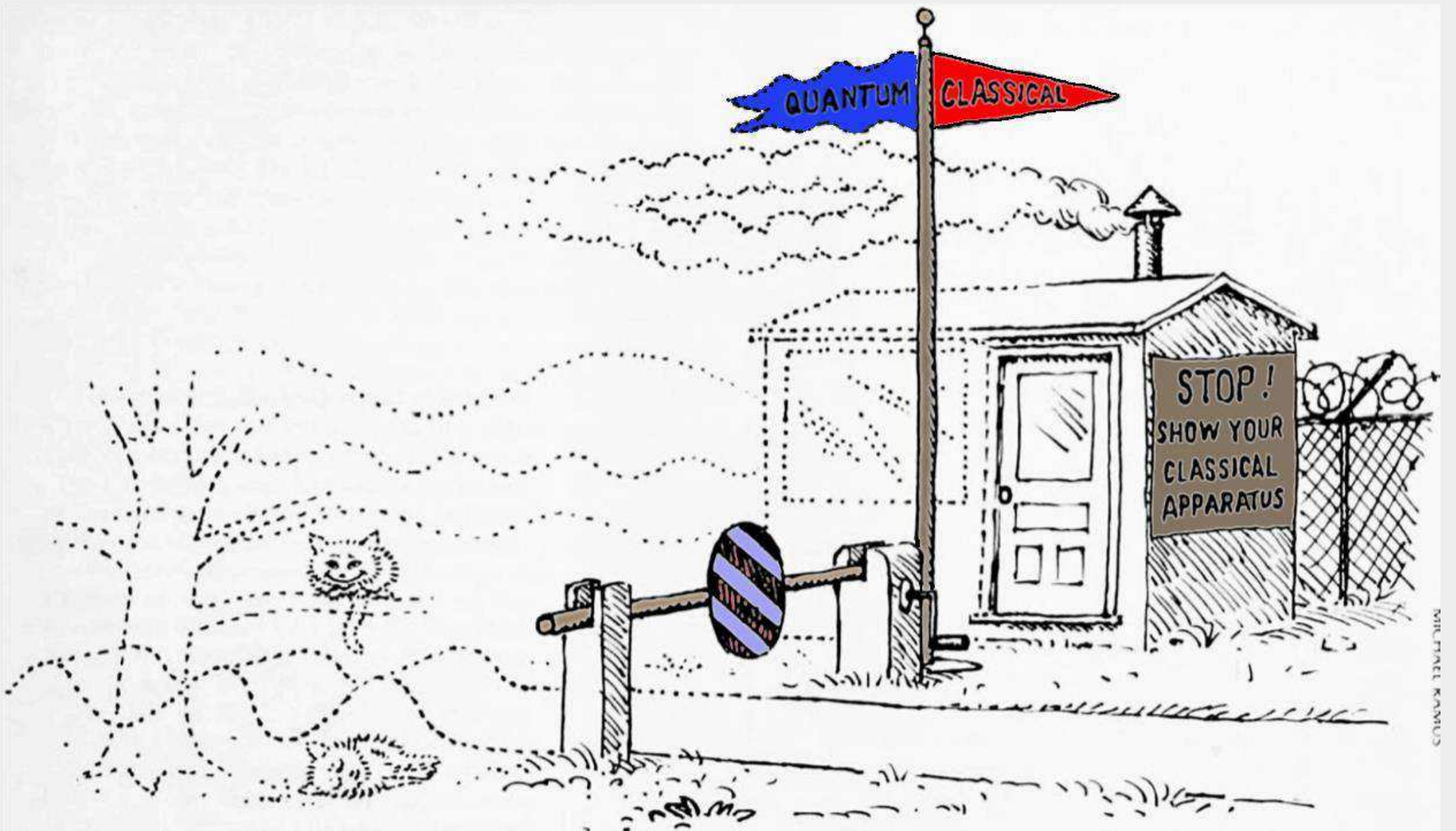




Unveiling the quantum

Dangerous bend ahead
on quantization





Drawing: Zurek, Physics Today (1991)



Dirac's canonical quantization



Source: Cambridge University, Cavendish Laboratory / Wikimedia Commons

Quantum (commutator)

$$-\frac{i}{\hbar} [\hat{x}, \hat{p}] = \hat{1}$$

$$\frac{d}{dt} \hat{O} = -\frac{i}{\hbar} [\hat{O}, \hat{H}]$$

where $\hat{O}(\hat{x}, \hat{p}; t), \hat{H}(\hat{x}, \hat{p}; t)$

These look
a lot alike!

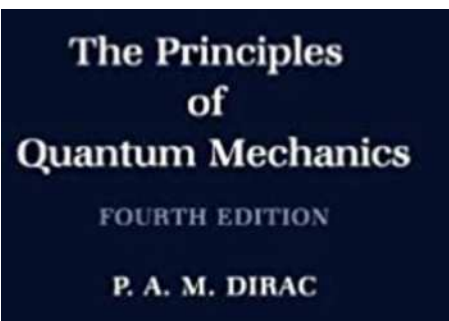
Classical (Poisson bracket)

$$\{x, p\}_P = 1$$

$$\frac{d}{dt} O = \{O, \mathcal{H}\}_P$$

where $O(x, p; t), \mathcal{H}(x, p; t)$

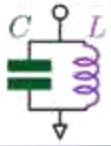
Procedure: Supplant the Poisson brackets by commutators



$$\begin{aligned} \{x, p\} &\longmapsto \frac{1}{i\hbar} [\hat{x}, \hat{p}] \\ \{O, O'\} &\longmapsto \frac{1}{i\hbar} [\hat{O}, \hat{O}'] \end{aligned}$$

“There is, however, a fairly general method of obtaining quantum conditions, applicable to a very large class of dynamical systems. This is the **method of classical analogy**”

P.A.M. Dirac

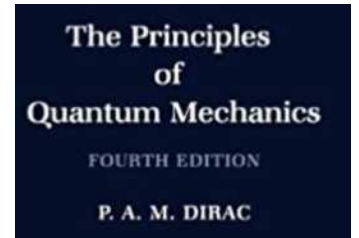


Dirac's canonical quantization: Quick exposure

Supplant classical Poisson bracket

and all quantum algebra follows...

$$\{O, O'\} \mapsto \frac{1}{i\hbar} [\hat{O}, \hat{O}']$$



Derivation: Dirac derives the quantum form of the Poisson bracket — the commutator — from mere assuming that

1. Classical Poisson bracket rules hold (by analogy, the new theory must be consistent with the old!)
2. The dynamical variables do not commute; i.e., $xp \neq px$
3. The Poisson bracket has a single result and single unique meaning



Problems:

- Operator ordering ambiguities.
Consider A and B are polynomials in x and p ; e.g., x^2xp or pxx^2
- Curvilinear coordinate systems (potentially transmon if $\cos(\Phi/\varphi_0)$ considered wrapped)
- Quantum gravity ...



If I knew what I was doing, it
wouldn't be called research.

Albert Einstein
See Hawken *et al.* (2010)

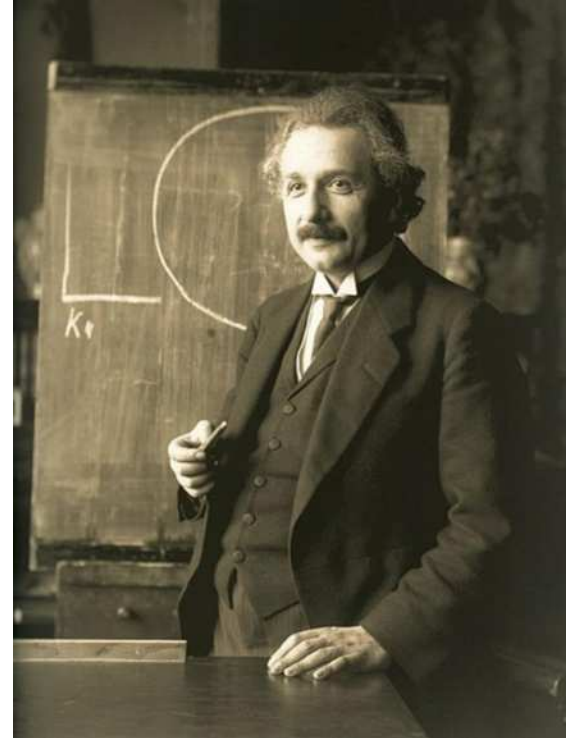
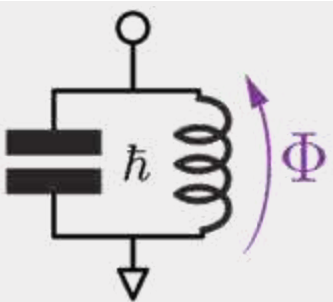


Photo: F. Schmutzer



The classical and quantum oscillator

Classical

Quantum

$$\Phi(t) \mapsto \hat{\Phi}$$

$$Q(t) \mapsto \hat{Q}$$

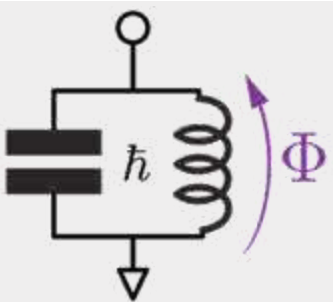
$$\mathcal{H}(\Phi, Q) \mapsto \hat{H}(\hat{\Phi}, \hat{Q})$$

$$\{\Phi, Q\} = 1 \mapsto [\hat{\Phi}, \hat{Q}] = i\hbar \hat{1}$$

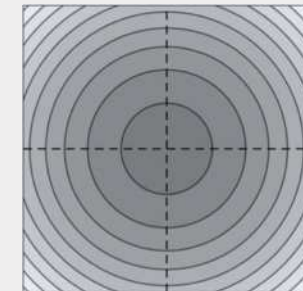
$$\{\alpha, \alpha^*\} = 1/(i\hbar) \mapsto [\hat{a}, \hat{a}^\dagger] = \hat{1}$$

$$\{A, B\} = \frac{\partial A}{\partial \Phi} \frac{\partial B}{\partial Q} - \frac{\partial B}{\partial \Phi} \frac{\partial A}{\partial Q}$$

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$



The classical and quantum oscillator



Classical

Quantum

Hamiltonian

$$\begin{aligned}\mathcal{H} &= \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \\ &= \frac{1}{2} \hbar \omega_0 (\alpha^* \alpha + \alpha \alpha^*)\end{aligned}$$

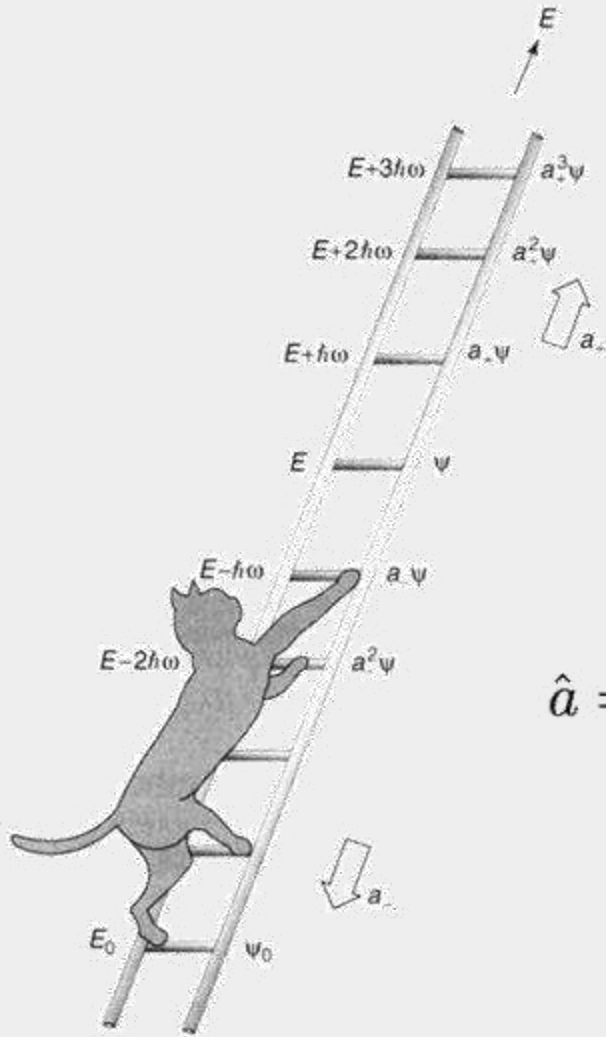
$$\begin{aligned}\hat{H} &= \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \\ &= \hbar \omega_0 (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)\end{aligned}$$

Phase space

$$\begin{aligned}\alpha(t) &= \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)] \\ \alpha(t) &= \alpha(0) e^{-i\omega_0 t} \\ \Phi(t) &= \sqrt{\frac{\hbar Z}{2}} (\alpha^*(t) + \alpha(t)) \\ Q(t) &= i\sqrt{\frac{\hbar}{2Z}} (\alpha^*(t) - \alpha(t))\end{aligned}$$

$$\begin{aligned}\hat{a} &= \sqrt{\frac{1}{2\hbar Z}} (\hat{\Phi} + iZ\hat{Q}) \\ \hat{a}(t) &= \hat{a}(0) e^{-i\omega_0 t} \quad (\text{Heisenberg picture}) \\ \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a})\end{aligned}$$

Ladder operators and matrix representation



annihilation

$$\hat{a} |0\rangle = 0$$

$$\hat{a} |1\rangle = \sqrt{1} |0\rangle$$

creation

$$\hat{a}^\dagger |0\rangle = \sqrt{1} |1\rangle$$

$$\hat{a}^\dagger |1\rangle = \sqrt{2} |2\rangle$$

general hopping

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{3} & \ddots \end{pmatrix}$$

$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

Hand-written notes

Hamiltonian and energy

$$\begin{aligned}\hat{H} &= \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \\ &= \hbar\omega_0 (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)\end{aligned}$$

Mean, variance, and RMS fluctuations

$$\langle 0 | \hat{\Phi}^2 | 0 \rangle = \Phi_{\text{ZPF}}^2$$

$$\langle 0 | \hat{Q}^2 | 0 \rangle = Q_{\text{ZPF}}^2$$

Calculations of the energy

For notational simplicity, I will drop the hats on the operators temporarily and the 0 from ω_0 .

$$\hat{H} = \frac{\hbar\omega}{2} (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)$$

$$\hat{a}^\dagger \hat{a} |n\rangle = n |n\rangle \quad \leftarrow \text{Fock}$$

$$\hat{a}^\dagger (\hat{a} |n\rangle)$$

$$\hat{a}^\dagger \sqrt{n} |n-1\rangle$$

$$\text{using } \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\downarrow \sqrt{n+1-1} \sqrt{n} |n-1+1\rangle$$

$\downarrow$

$$= n |n\rangle$$

$\hat{a}^\dagger a = \text{photon number } n$

$$[a, a^\dagger] = 1$$

$$a a^\dagger - a^\dagger a = 1$$

$$a a^\dagger = 1 + a^\dagger a$$

$$\hat{H} = \frac{\hbar\omega}{2} (a^\dagger a + a a^\dagger)$$

$$= \frac{\hbar\omega}{2} (a^\dagger a + a^\dagger a + 1)$$

$$= \frac{\hbar\omega}{2} (2a^\dagger a + 1)$$

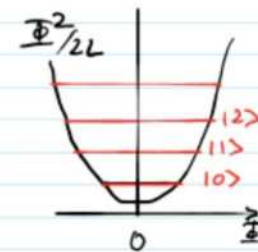
$$\hat{H} = \hbar\omega_0 \left(\underbrace{a^\dagger a}_n + \underbrace{1/2}_{\text{zero-point energy}} \right)$$

n zero-point energy

$$\hat{H} |n\rangle = E_n |n\rangle$$

$$= \hbar\omega_0 (n + 1/2)$$

$$n = 0, 1, 2, \dots$$



Expectation value of magnetic flux and charge

$\langle \hat{\Phi} \rangle = \langle n | \Phi_{\text{ZPF}} (\hat{a} + \hat{a}^\dagger) | n \rangle$

$\leftarrow \text{some \#} = \sqrt{\frac{\hbar Z}{2}} \quad Z = \sqrt{L/C}$

$\langle n | \hat{a} | n \rangle = \sqrt{n} \langle n | n-1 \rangle = 0$

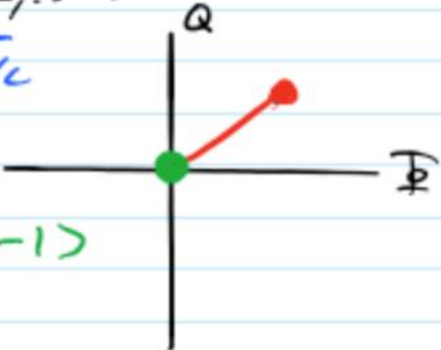
$\langle n | \hat{a}^\dagger | n \rangle = \sqrt{n+1} \langle n | n+1 \rangle = 0$

$\langle \hat{\Phi} \rangle = \Phi_{\text{ZPF}} (\langle \hat{a} \rangle + \langle \hat{a}^\dagger \rangle) = 0$

$\langle \hat{Q} \rangle = \langle n | -i Q_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) | n \rangle$

$= -i Q_{\text{ZPF}} (\langle \hat{a}^\dagger \rangle - \langle \hat{a} \rangle) = 0$

$Q_{\text{ZPF}} = \sqrt{\frac{\hbar}{2Z}}$



Fluctuations of flux

$$\begin{aligned}\text{Var}(\hat{\Phi}) &= \langle \hat{\Phi}^2 \rangle - \langle \hat{\Phi} \rangle^2 \\&= \langle \hat{\Phi}^2 \rangle \\&= \langle n | (a^\dagger + a)^2 | n \rangle \times \Phi_{\text{ZPF}}^2 \\&= \langle n | a^{\dagger 2} + a^2 + a^\dagger a + a a^\dagger | n \rangle \times \dots \\&= \underbrace{\langle n | a^{\dagger 2} | n \rangle}_{\langle n | n+2 \rangle} + \underbrace{\langle a^2 \rangle}_{\rightarrow 0} + \langle a^\dagger a + a a^\dagger \rangle \times \Phi_{\text{ZPF}}^2 \\&= \Phi_{\text{ZPF}}^2 \langle a^\dagger a + a a^\dagger \rangle \\&= \Phi_{\text{ZPF}}^2 \langle a^\dagger a + a^\dagger a + 1 \rangle\end{aligned}$$

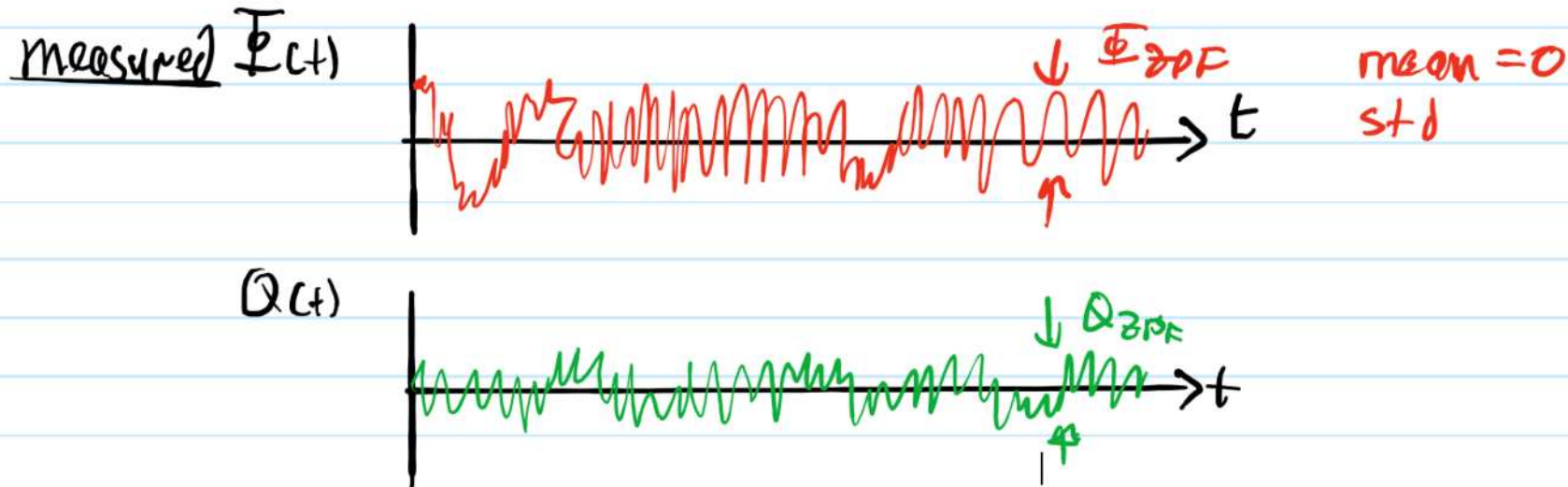
Fluctuations of flux and charge

$$\text{Var}(\hat{\Phi}) = \langle \hat{\Phi}^2 \rangle = (2n+1) \Phi_{\text{ZPF}}^2$$

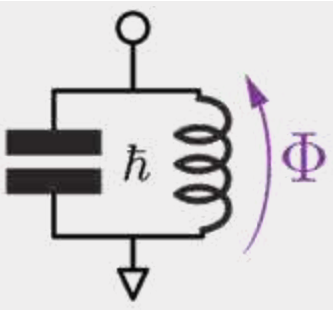
$$\text{For } |n\rangle = 0 : \quad \langle \hat{\Phi}^2 \rangle = \Phi_{\text{ZPF}}^2$$

i.e., RMS fluctuation of $\hat{\Phi}$: $|n \text{ ground } \rangle$

$$\sigma_{\hat{\Phi}} = \Phi_{\text{ZPF}}$$

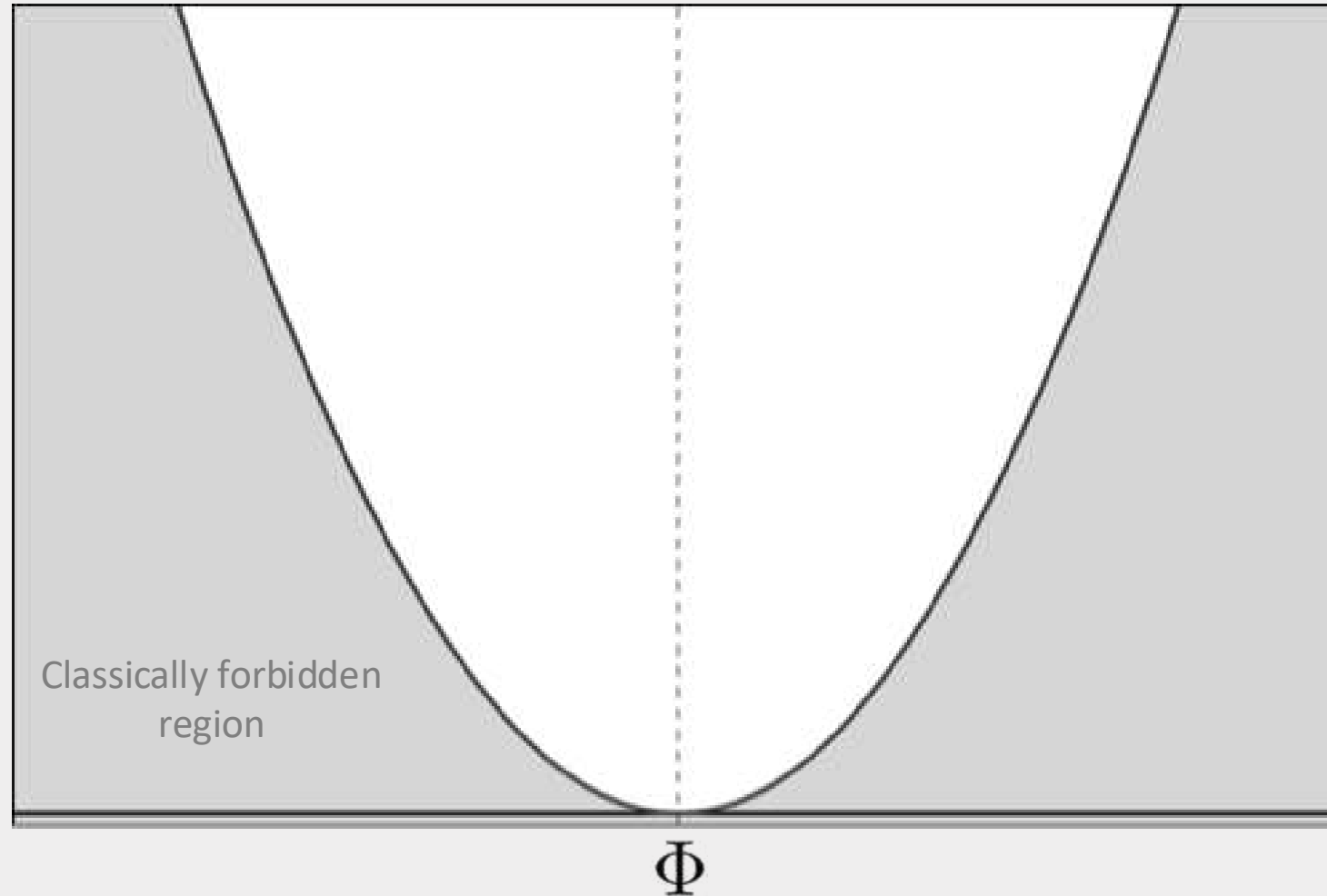


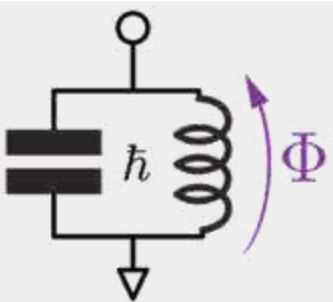
Wavefunctions of the quantum oscillator



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



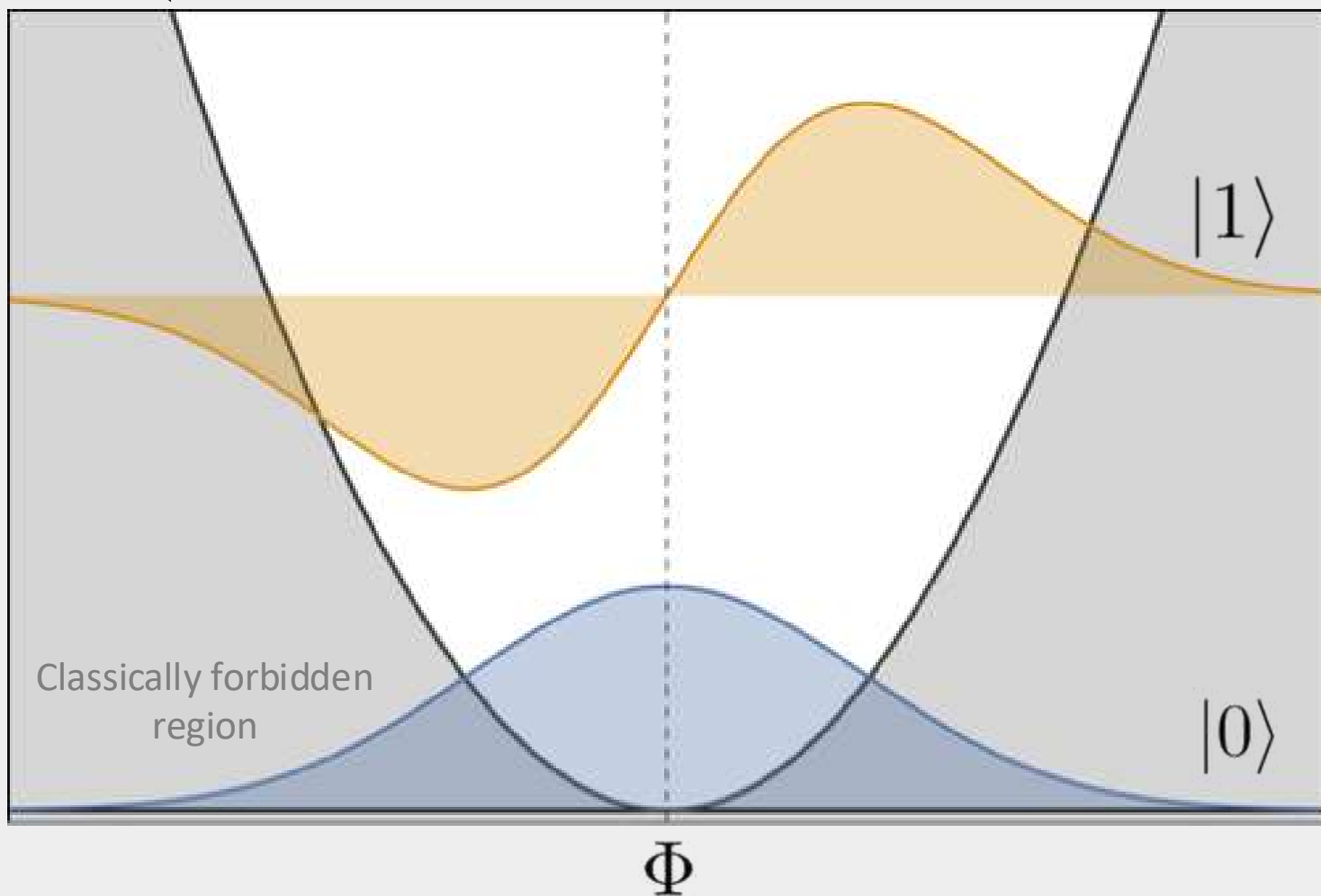


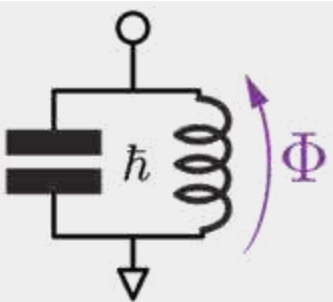
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



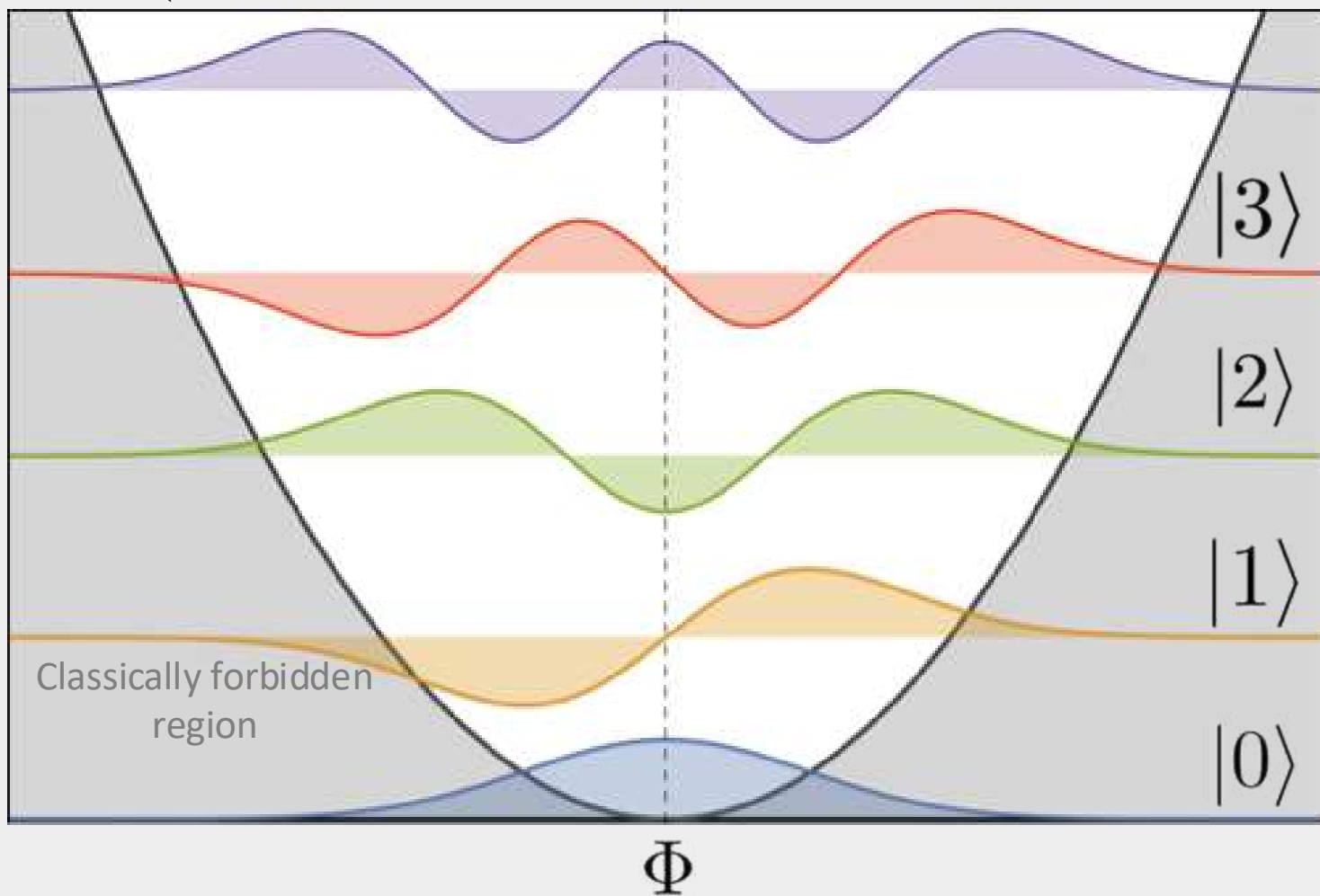


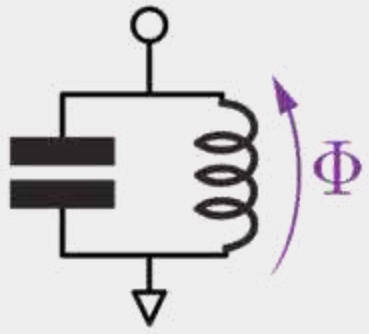
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

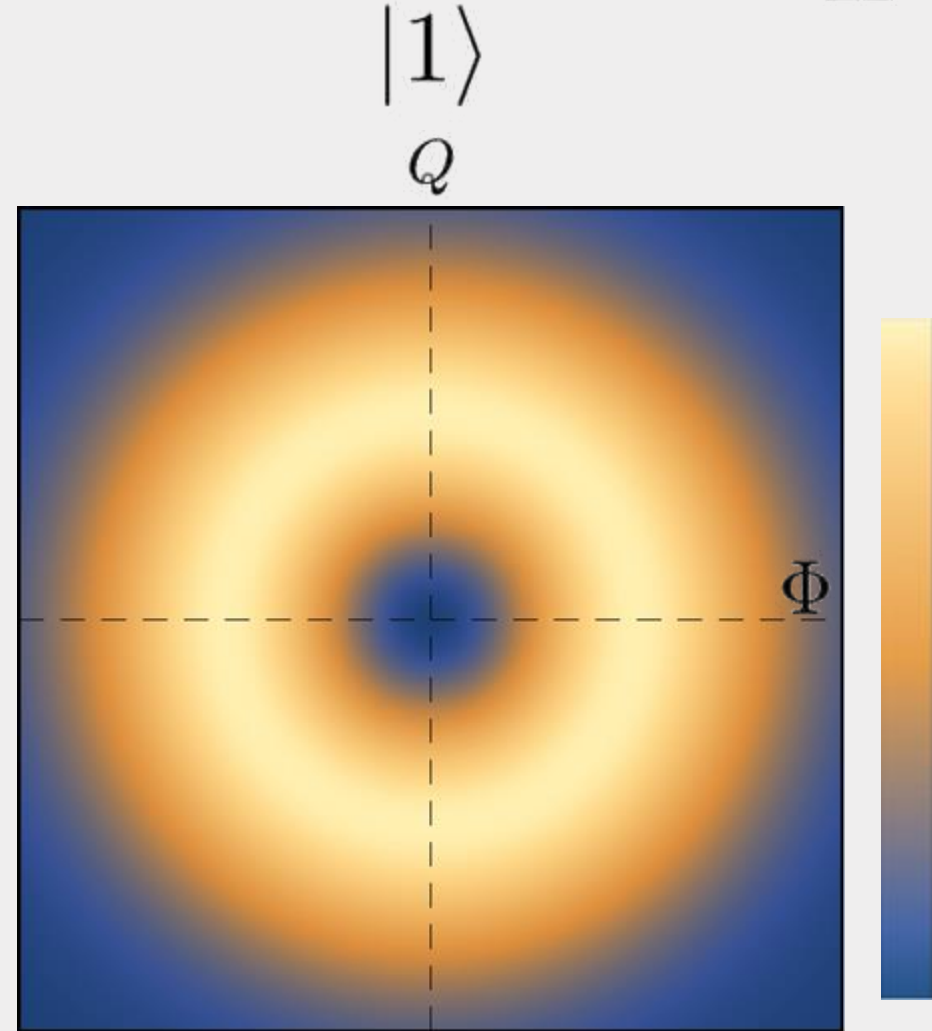
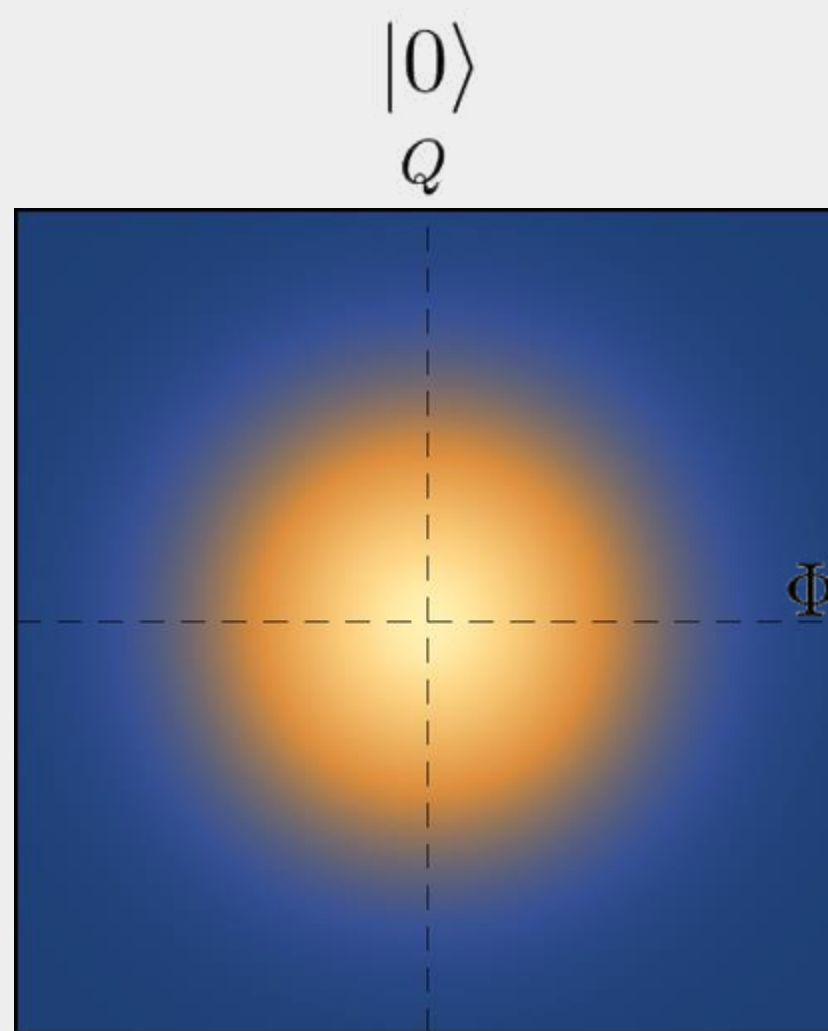
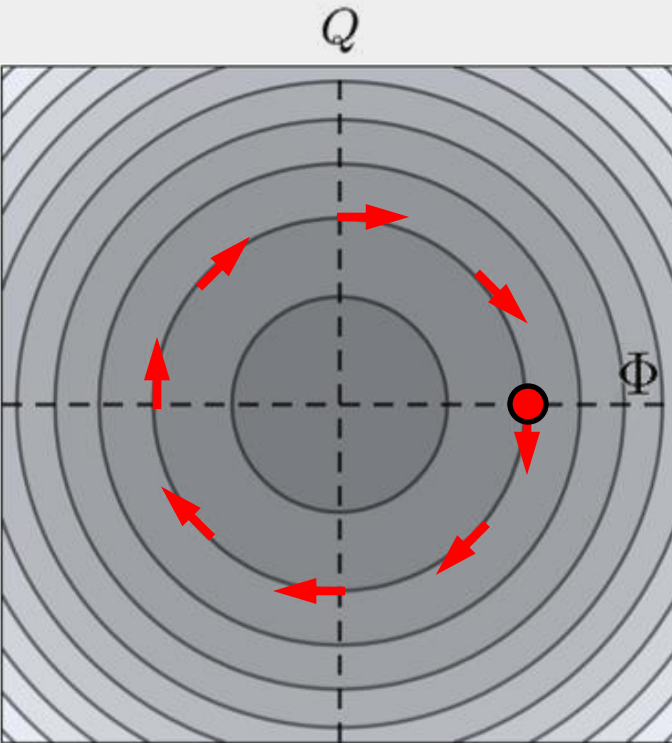
$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



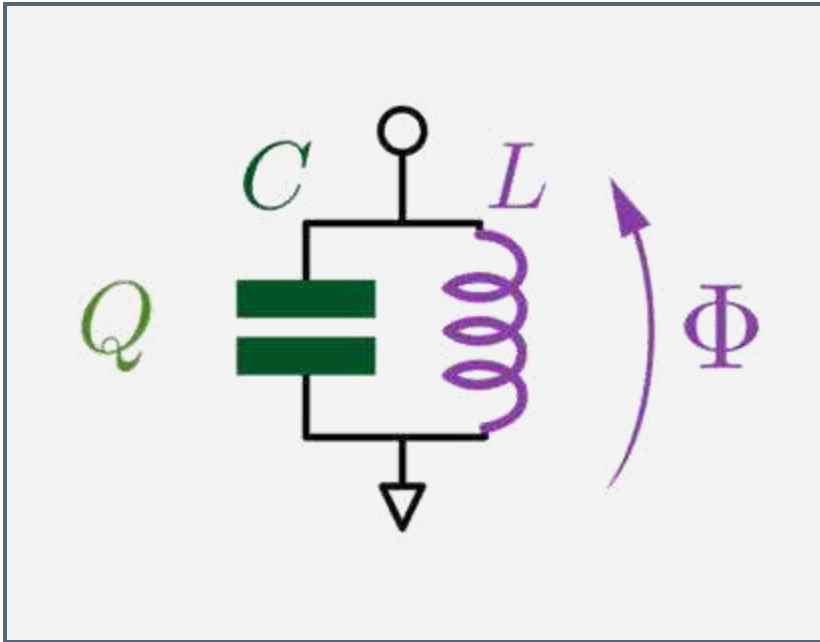


Phase-space (Husimi Q function)



$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \hat{\rho} | \alpha \rangle$$

Pop-up question



The flux and charge operators are Hermitian observables.

How can some expectations, such as

$$\langle 0 | \hat{\Phi} \hat{Q} | 0 \rangle = \frac{1}{4} i ,$$

be imaginary?

Or, others be negative...?

Advanced questions



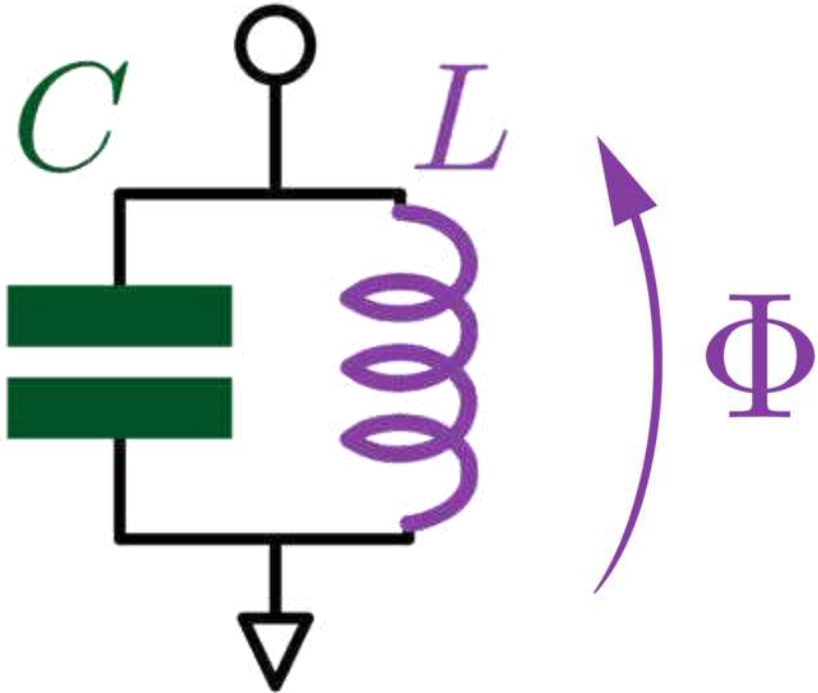
1. What is $\hat{\Phi}(t)$ in terms of $\hat{a}$ in the Heisenberg picture?
2. **Autocorrelation.** Find the autocorrelation operator $\hat{C}_{\Phi\Phi}(t) = \hat{\Phi}(t) \hat{\Phi}(0)$. What frequency components does it have?
 - (a) What is the flux autocorrelation expectation value $C_{\Phi\Phi}(t) = \langle n | \hat{\Phi}(t) \hat{\Phi}(0) | n \rangle$ for the Fock state $|n\rangle$? Is it real? Why not? What is the frequency spectrum?
 - (b) Repeat for a coherent state $|\alpha\rangle$.



3. **Thermal state.** The thermal state is $\hat{\rho}_{\text{th}} = \exp[-\beta \hat{a}^\dagger \hat{a}] / \text{Tr}[\exp(-\beta \hat{a}^\dagger \hat{a})]$, where $\beta = \hbar\omega_0/k_B T$, and T is the temperature of the oscillator.
 - (a) What is the mean and variance of the flux $\hat{\Phi}$ and charge $\hat{Q}$ operators?
 - (b) How does the frequency spectrum of the autocorrelation $\langle \hat{C}_{\Phi\Phi}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{\Phi\Phi}(t)]$ change when with that for the state $|n\rangle$ and $|\alpha\rangle$?
 - (c) The spectrum is not symmetric in frequency. How can you interpret that positive and negative frequencies have different weights? How is this related to absorption and emission of the oscillator? (Consider Fermi's golden rule).
 - (d) What happens to the spectrum in the limit of high temperature, $k_B T \gg \hbar\omega_0$? How about low temperature, $k_B T \ll \hbar\omega_0$?
 - (e) How do the above conclusions change for charge; i.e., for $\langle \hat{C}_{QQ}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{QQ}(t)]$?

Will discuss
answers on my
blog sometime
soon

Quantum Harmonic Oscillator Applets

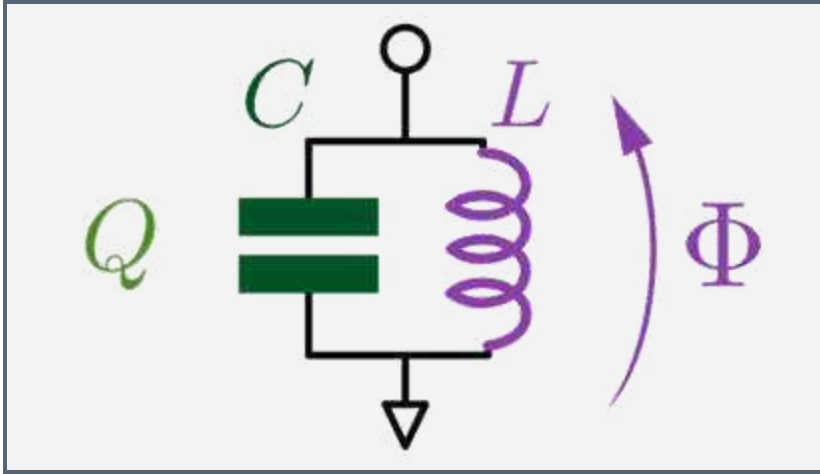


[Energy levels](https://www.st-andrews.ac.uk/physics/quvis) of SHO applet from
<https://www.st-andrews.ac.uk/physics/quvis>

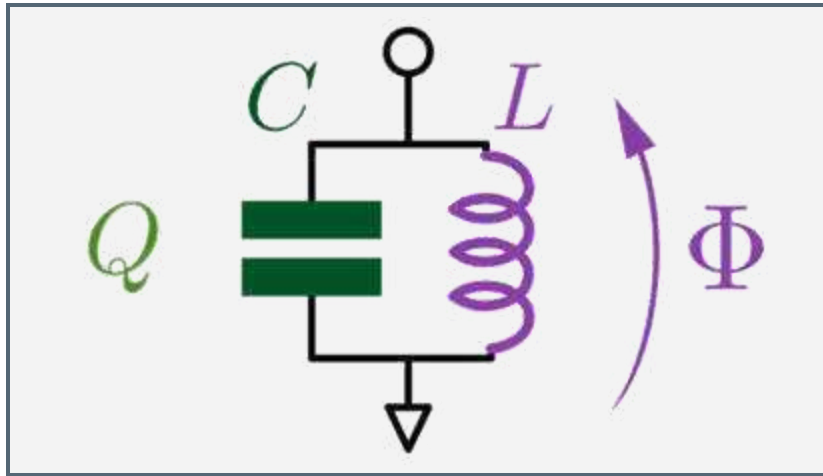
Wigner function
<https://demonstrations.wolfram.com/WignerFunctionOfHarmonicOscillator/>

Coherent states
<https://demonstrations.wolfram.com/CoherentStatesOfTheHarmonicOscillator/>

Linear harmonic oscillator summary



The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

$$\hat{H} |n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0 = \Phi_0 \sqrt{\frac{z_0}{2\pi}}, \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0^{-1} = (2e) \sqrt{\frac{1}{2\pi z_0}}, \end{aligned}$$

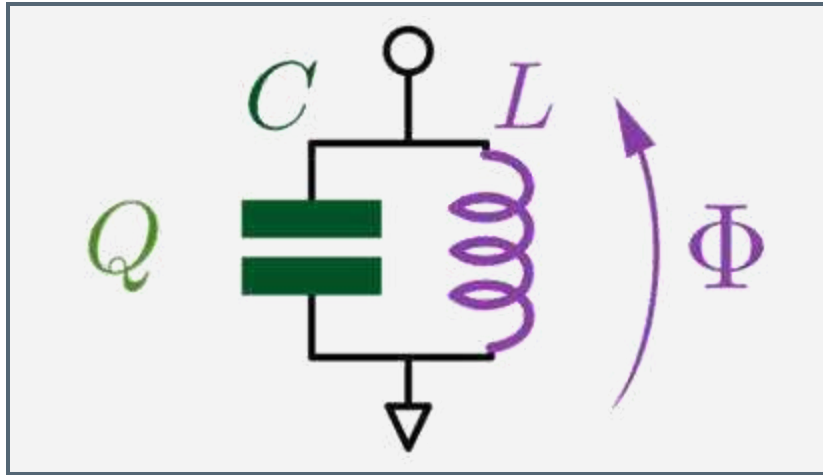
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\begin{aligned} \mathcal{E}_{\text{cap}}(Q) &= \frac{Q^2}{2C} \\ &= \frac{1}{2} C \dot{\Phi}^2 \end{aligned}$$

Energy

Φ/ϕ_0

The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

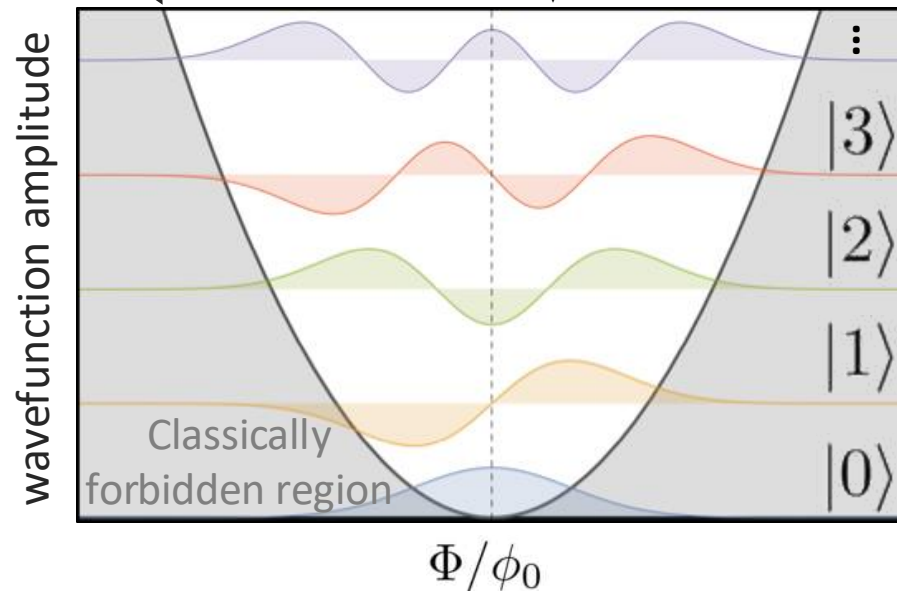
Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad \Phi_{\text{ZPF}} Q_{\text{ZPF}} = \frac{\hbar}{2} \quad Z_0 = \frac{L}{C}$$

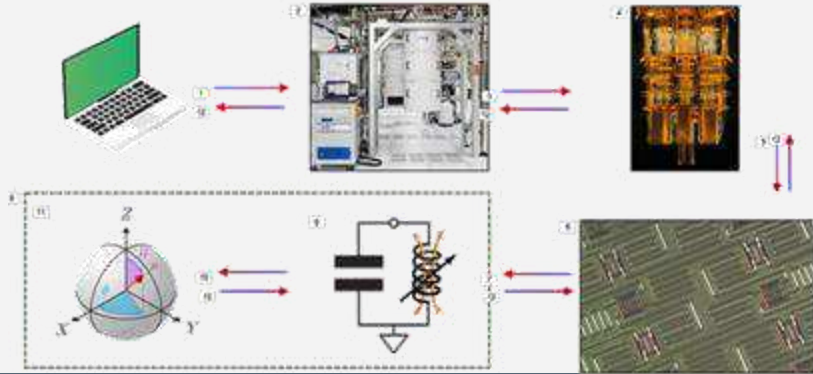
$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0} & \langle 0 | \hat{\Phi}^2 | 0 \rangle &= \Phi_{\text{ZPF}}^2 \\ \hat{Q} &= i Q_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0^{-1}} & \langle 0 | \hat{Q}^2 | 0 \rangle &= Q_{\text{ZPF}}^2 \end{aligned}$$

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L} \quad \psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

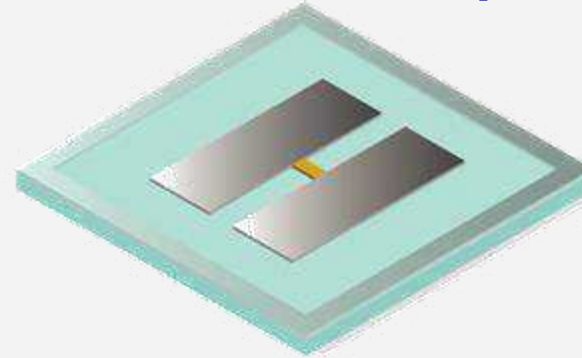


The road behind and ahead

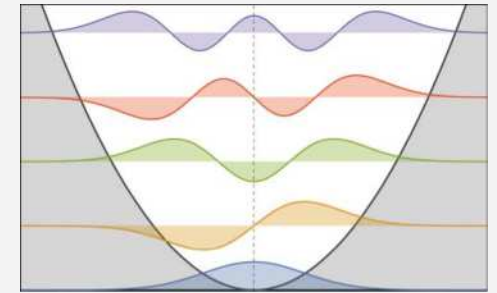
Qubit in the cloud



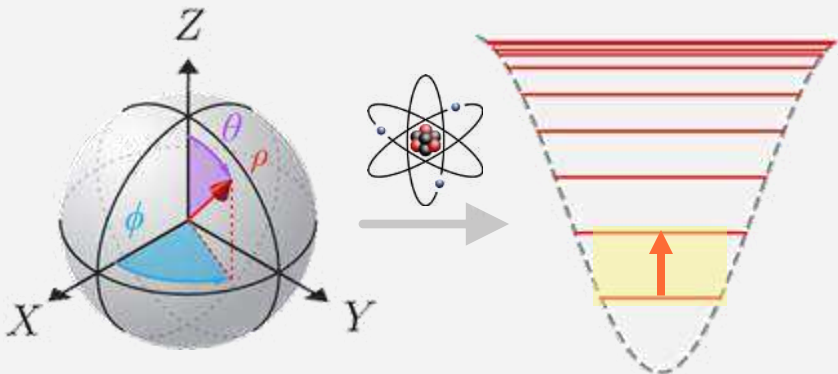
cQED: Transmon qubit



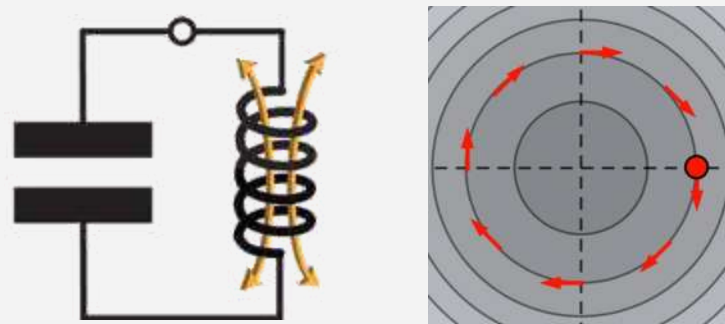
Unveiling the quantum oscillator



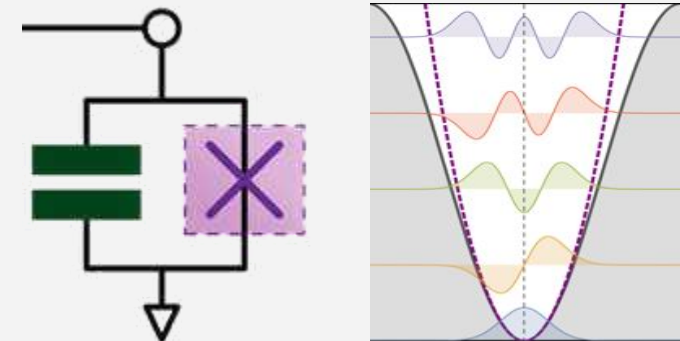
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



Next steps

Tightly integrated lab work with Dr. Nick Bronn and Co.!

More depth on qubit control

Run experiments on real devices

Check out references, problems given in the lecture,
dangerous bends

Break away from the rules of today



Thank you!

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IBM Quantum